Interlacing and Additive Inapproximability of Deterministic Communication Complexity

Abstract

Given the full truth table of a Boolean function, how accurately can one estimate its deterministic communication complexity? Earlier versions of this paper established exact NP-hardness and ETH-based additive inapproximability. We prove that approximation within every fixed additive error is NP-hard under deterministic polynomial-time reductions, both for unrestricted protocols and for Yao's strictly alternating model, including receiver output. For unrestricted protocols, randomised sparse ETH rules out bounded-error randomised polynomial-time estimation within $c \log T / \log \log T$ for every fixed $0 \leq c < 1/8$, where T is the number of table entries. Deterministic ETH rules out deterministic polynomial-time estimation within $c_* \log \log T$ for an effective absolute constant $c_* > 0$.

These results bring together the interlacing program, from its origins in a direct-sum counterexample through relaxed interlacing and exact hardness to repeatable gap amplification and sparsification. The central obstacle is to enlarge a communication gap while keeping the table small enough to print. We develop an interlacing step whose extra bit survives input deletions and whose lower-bound guarantees are restored for the next step. Short explicit input lists make every fixed gap polynomial in size; keeping the full products implicit and sampling once gives the stronger randomised bound. The same size-control principle extends the program to strict alternation: an explicit array list preserves a different lookup lower bound that exploits the fixed speaking schedule.

1 Introduction

Alice knows a row of a Boolean matrix and Bob knows a column. They must determine the entry where the two meet, communicating as few bits as possible. Their local computation is free, and their protocol must answer every input correctly. The deterministic communication complexity $\mathrm{D}(M)$ is the smallest worst-case cost of such a protocol. We study the algorithmic problem in which the *whole table* is given: how accurately can this optimum be estimated? The answers are already in the input; the difficulty is finding how little communication suffices to produce them.

Yao asked this question in 1979, when he introduced communication complexity [20]. Exact NP-hardness alone says little about approximation. Costs B and $B+1$ can receive the same estimate if an error of one bit is allowed. Moreover, Yao required the players to alternate after every bit, whereas unrestricted protocols choose the speaker at each node. These are different optimisation problems. We prove that approximation within every fixed additive error is NP-hard in both models, and obtain growing-error hardness in the unrestricted model.

The interlacing program. This paper is the culmination of a line of work that connects the structure of optimal protocols to the difficulty of finding their cost. Its organising operation is *interlacing*: Alice receives a base row and a selector, Bob receives a tuple of base columns, and they must compute the entry at the selected column. Mackenzie and Saffidine introduced this framework

in their direct-sum counterexample, first circulated in 2024 [17]. Interlacing made it possible to prove that solving several independent instances together can cost less than solving them separately.

The next obstacle was the size of the construction. Interlacing with all column tuples creates an exponential input family. *Relaxed interlacing* replaces that family by a small pseudorandom one while preserving the lower-bound argument. Earlier versions of this paper, first circulated in 2025 [5], used this method to establish exact NP-hardness, concurrently and independently with Hirahara, Ilango and Loff [9]. The reduction turns a violated combinatorial constraint into one unavoidable extra bit of communication.

Approximation required that extra bit to accumulate. This led to *repeatable gap amplification*: make the lower bound survive the input restrictions created by a protocol, and restore the guarantees needed for another round. The earlier ETH stage of this paper [6], prepared as version 5 of arXiv:2508.05597, rules out every fixed additive error under ETH and gives a computable unbounded error allowance. Its fixed-gap constructions take subexponential time in the source length; they do not give the polynomial-time reductions needed for fixed-error NP-hardness.

What this paper adds. The present work makes the amplified gap survive in a small enough table for every fixed gap to be produced in deterministic polynomial time. Thus fixed-error hardness now follows from NP-hardness itself. We also obtain explicit growing-error rates under deterministic and randomised ETH, replacing the earlier unspecified unbounded allowance by quantitative bounds. The key advance is preserving the properties needed by the lower-bound proof while reducing its input families. One construction keeps short explicit lists at every round; the other keeps full products implicit and samples only after amplification. Interlacing supplies the unrestricted lower bounds in both constructions, and restoration makes the gap repeatable. Sparsification controls the size of the table that must ultimately be printed.

This manuscript continues the same paper under arXiv:2508.05597. Our version-specific references identify earlier stages of that work; the approximation and strict-alternation results are later contributions. The direct-sum paper is a separate precursor to this continuing program.

From sparsification to strict alternation. Separating the lower-bound structure from the size of its input family also opens a route to Yao's model. The short protocols in the unrestricted construction use blocks of consecutive bits from one player; inserting the missing turns can consume their additive advantage. For strict alternation, we build a lookup table whose lower bound uses the fixed speaking schedule. Allowing Bob every Boolean array initially gives an exponential family of columns. The proof only needs a bound on how often an array matches a prescribed pattern of bits. A short explicit array list preserves that bound and makes the table polynomial in size for every fixed gap. The shared principle is sparsification; the alternating proof counts response strategies and does not invoke the unrestricted amplifier. This construction gives fixed-error hardness, with no corresponding growing-error bound established here.

Write T for the number of table entries; polynomial time means polynomial in T. Additive error measures accuracy in bits, the natural scale here: on an $n \times n$ table, Alice can send her row index and Bob the answer, so $\mathrm{D}(M) \le \lceil \log n \rceil + 1$. An estimator returns a number, not necessarily a protocol. A bounded-error randomised estimator must meet its error guarantee with probability at least $2/3$ on every input. The protocols whose costs we estimate are always deterministic and exact.

We write $D_A^{\mathrm{alt}}(M)$ for the strictly alternating public-output cost with Alice starting. Yao's receiver-output cost $Y_{\mathrm{free}}(M)$ allows an input-independent choice of starter and lets the receiver of the last bit determine the answer. Stopping remains public. Section 2 fixes these conventions, including depth zero. The growing-error results use sparse ETH: 3-SAT on N variables and a

fixed linear number of clauses has no $2^{o(N)}$-time algorithm. Its deterministic and bounded-error randomised forms are distinguished in Definition 2.3.

1.1 Results

The central conclusion is NP-hardness for every fixed additive error in both speaking models. The two quantitative growing-error bounds concern unrestricted protocols and use the stated forms of ETH. The table separates these conclusions and their assumptions.

Protocol model	Additive error	Basis of hardness	Estimator excluded
Unrestricted	$c \log T / \log\log T$, $c < 1/8$	Randomised sparse ETH	Bounded-error randomised
Unrestricted	$c_* \log\log T$	Deterministic ETH	Deterministic
Unrestricted	Every fixed error	NP-hard by deterministic reduction	See the consequences below
Strict alternation	Every fixed error	NP-hard by deterministic reduction	See the consequences below

Theorem 1.1 (Randomised growing-error hardness)**.** *Assume randomised sparse ETH and fix $0 \le c < 1/8$. No bounded-error randomised estimator running in time polynomial in T estimates $\mathrm{D}(M)$ within additive error $c \log T / \log\log T$ on every sufficiently large explicit total Boolean table. This remains true for square tables whose side length is a power of two.*

Theorem 1.2 (Deterministic growing-error hardness)**.** *There is an effective absolute constant $c_* > 0$ such that deterministic ETH rules out every deterministic polynomial-time estimator that estimates $\mathrm{D}(M)$ within additive error $c_* \log\log T$ on every sufficiently large explicit total Boolean table.*

For fixed gaps, no ETH assumption is needed. In each reduction below the threshold B is computed from the formula and requested gap, without deciding satisfiability.

Theorem 1.3 (Fixed gaps for unrestricted protocols)**.** *For every fixed integer $k \ge 1$, a deterministic polynomial-time algorithm maps a 3-CNF formula Φ to an explicit total Boolean table M and an integer B such that*

$$\Phi \in \mathsf{SAT} \implies \mathrm{D}(M) \le B, \qquad \Phi \notin \mathsf{SAT} \implies \mathrm{D}(M) \ge B + k.$$

The polynomial may depend on k. The output may be square with power-of-two side length.

Theorem 1.4 (Fixed gaps under strict alternation)**.** *For every fixed integer $k \ge 1$, a deterministic polynomial-time algorithm maps a 3-CNF formula Φ to an explicit total Boolean table M and an integer B such that*

$$\Phi \in \mathsf{SAT} \implies D_A^{\mathrm{alt}}(M) \le B, \qquad \Phi \notin \mathsf{SAT} \implies D_A^{\mathrm{alt}}(M) \ge B + k.$$

The polynomial bounds on output size and running time may depend on k.

Observation 1.5. *A reduction with YES cost at most B and NO cost at least $B + k$ defeats additive error a whenever $k > 2a$. Thus, for every fixed error, estimating either cost in the preceding two theorems in deterministic polynomial time implies $\mathsf{P} = \mathsf{NP}$; bounded-error randomised estimation implies $\mathsf{NP} \subseteq \mathsf{BPP}$.*

Proof. An accurate estimate is at most $B + a$ in the YES case and at least $B + k - a$ in the NO case. Since $B + a < B + k/2 < B + k - a$, comparison with the public midpoint decides satisfiability whenever the estimate is accurate. For fixed a, choose a fixed integer $k > 2a$ and compose this comparison with the corresponding deterministic reduction. □

For example, a gap of three bits separates estimates with error one. Changing the starter or announcing a receiver's answer costs at most one bit each (Lemma 8.1), so increasing the fixed gap also gives Yao's output convention.

Corollary 1.6. *For every fixed nonnegative additive error, a deterministic polynomial-time estimator of $Y_{\mathrm{free}}(M)$ on all explicit total Boolean tables implies $\mathsf{P} = \mathsf{NP}$. A bounded-error randomised estimator implies $\mathsf{NP} \subseteq \mathsf{BPP}$. The same conclusions hold for public-output strict alternation with Alice first, Bob first, or an input-independent choice of starter.*

The strict-alternation construction has polynomial size for each fixed gap, with a degree that may depend on that gap. It gives no growing-error bound. The stronger unrestricted rate uses both a randomised ETH assumption and randomised estimators; under deterministic ETH our deterministic obstruction is $c_* \log \log T$.

1.2 Proof overview

The two unrestricted reductions combine repeatable gap amplification with sparsification. Recall the interlacing operation: Alice receives a base row and a selector i, Bob receives a tuple of base columns, and they must answer the i-th question. Repetition enlarges the gap but also creates large products of input sets. Deleting inputs shrinks the table, yet can erase the selector's effect: if Bob's questions are all equal, the selector no longer matters. Figure 2 gives the smallest example. A protocol itself creates deletions as messages remove incompatible rows or columns.

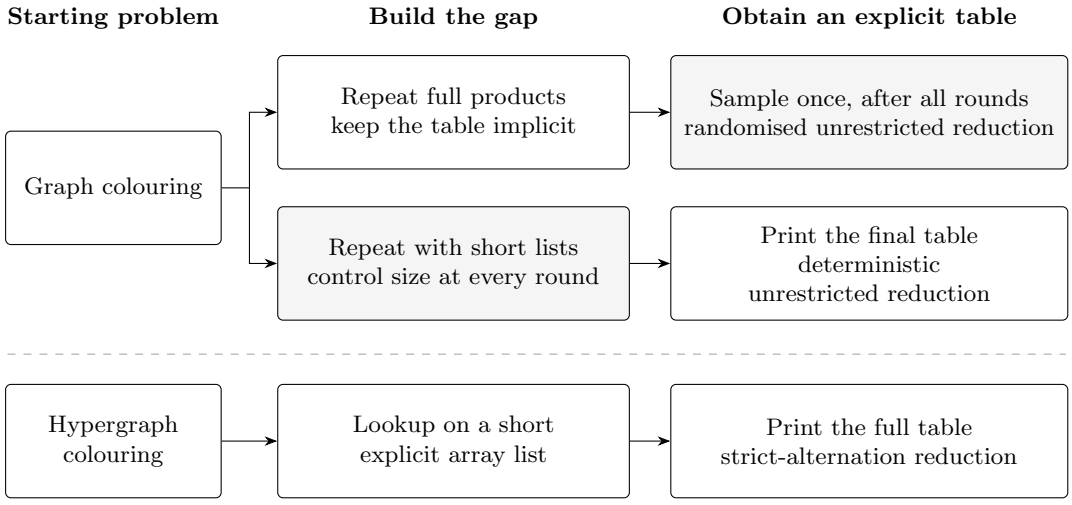


Figure 1: Where size is controlled (shaded boxes). The two unrestricted routes share the internal-interlace argument: one samples only after growing the gap, while the other keeps the input lists short throughout. Strict alternation uses a separate lookup construction. All sources come from 3-SAT.

A repeatable extra bit. The graph supplies a simple obstruction to a short protocol. At each edge coordinate, its two endpoints and a few auxiliary items are active. A failed colouring forces two active items into a bin that also contains an inactive item. These items request three distinct lookup positions. First forcing the protocol to fix an assignment to bins makes the same obstruction available after inputs are deleted.

To repeat the construction, we track two deletion thresholds. At a fixed minimum row mass, retaining 3/5 of the columns preserves the current lower bound; retaining 3/10 may lose one bit. The lower bound then follows by tracking where the protocol sends its inputs. Most copies retain a large unsplit share at a distinct continuation; we call that continuation the copy's *owner*. Special diagonal inputs expose the collision at an owner, forcing some of these protected inputs to go elsewhere. Neither other owners nor the few exceptional continuations can receive all this displaced mass. This contradiction handles arbitrary speaking order. One final interlace restores the bounds after deletion that the next round needs.

Both unrestricted constructions use the same argument for this internal interlace. It bounds how much restricting the tuple inputs can change the law of a few coordinates. The inner source estimates and final restoration remain separate: their smaller retained events require stronger estimates for full products.

One final sample. For the randomised reduction, we keep every product implicit through all k rounds. Rank bounds the number of column sets on which a short protocol can succeed, uniformly over retained row sets. This lets a union bound protect a column sample, followed by an independent row sample. The samples have the same power-of-two size, so the output is already square. Recursive entry evaluation prints it without expanding the products. With our choice of k, the size is $\log T = (4 + o(1))k \log N$ for a sparse formula on N variables, which yields the coefficient $1/8$ after allowing error on both sides of the gap. The sample need not support any further amplification.

Explicit lists at every round. For the deterministic reduction, short weighted lists replace the products. They agree with the product law on every small set of coordinates. We build a list one coordinate at a time and remove redundant support points while preserving those marginals. Conditioning on a transcript can destroy this agreement: Bob can announce whether two independent questions agree, making them identical on that branch. A moment bound shows that only a bounded set of coordinates has badly changed small marginals. This supplies the common internal argument and, at larger widths, the list-specific source and restoration bounds. The list sizes compound over the rounds, giving at most $N^{2^{O(k)}}$ entries and the $\log \log T$ error scale.

Strict alternation. Here we sparsify a lookup table that encodes a hypergraph colouring instance. If Bob could hold every Boolean array, the column family would be exponential, but a uniform array would match a prescribed pattern on r positions with probability exactly 2^{-r}. A short explicit list retains this bound up to a small additive error. This is the property needed by the following lower-bound argument.

A hypergraph-colouring argument forces many distinct lookup requests to share Alice's early response strategy: she responds identically for *every* possible early history of Bob's bits. The remaining turns allow too few output patterns to cover the surviving arrays, even with the small error allowed by the list. The fixed speaking schedule makes this counting argument possible. We print the whole list, obtaining a deterministic reduction for each fixed gap.

1.3 Related work

Yao introduced the strictly alternating model and asked how hard its optimum is to compute from a truth table [20]. Kushilevitz and Weinreb proved approximation hardness under cryptographic assumptions [15]. Ilango, Loff and Oliveira proved NP-hardness and strong inapproximability for partial matrices [11]. For total matrices, Hirahara, Ilango and Loff proved additive bit-cost hardness for three-message protocols and leaf-count hardness for every fixed number of messages at least three [8]. Messages may contain many bits, so these results cover neither bit-by-bit alternation nor the unrestricted optimum. Earlier versions of this paper [5] established exact unrestricted NP-hardness, concurrently and independently with Hirahara, Ilango and Loff [9]. The present fixed-gap results address the additive-one and alternating questions left open by those exact-hardness results.

The amplifier uses the interlacing operation of Mackenzie and Saffidine [17]. Its graph source combines Dinur's gap amplification with the bounded-degree colouring reduction of Bogdanov, Obata and Trevisan [2, 4]. The alternating source is the balanced rainbow-colouring hardness of Guruswami and Lee [7], combined with an explicit finite-field array construction based on the small-bias construction of Alon et al. [1]. The deterministic lists use sequential extension and support compression in the tradition of Koller and Megiddo [14]. Rank-based counts have antecedents in rankwidth and clique-width algorithms [3, 18] and hard-submatrix arguments [10]. In particular, the row-sum method of Oum, Sæther and Vatshelle [18] motivates our short leaf-condition count. Mackenzie's lossless hardness condensation [16] also counts inputs served by short protocols; it preserves hardness relative to the new input length, whereas our final sample preserves an absolute threshold. We prove the sampling theorem needed here directly.

1.4 Formal verification

Formal verification accompanies this program. The independent Beyond the Library team [19] has formalised the corrected Mackenzie–Saffidine direct-sum result [17] and its supporting arguments in Lean. Their public proof inspector gives the statements and formal proofs.

A separate companion formalisation of the present paper is being developed for its sparsified constructions and the strictly alternating model. The final artifact will specify the verified theorem coverage, source version, assumptions and correspondence with the statements here. **[Companion formalisation link to be inserted before submission.]** The direct-sum formalisation and this companion development have distinct theorem scopes; the latter remains in progress.

Reading guide. The proof order follows the three constructions rather than their chronology. After the basic definitions, we state the repeatable amplifier and derive the randomised result by one final sample. We then prove the amplifier's lower bounds, replace its products by explicit lists for the deterministic results, and give the alternating lookup construction. The appendices supply the interlacing estimates, source conversions, implementation details and effective parameter choices.

2 Preliminaries

2.1 Communication and estimation

An estimator sees the whole table; a communication protocol sees one row and one column, held by different players.

Definition 2.1. A table $M : X \times Y \to \{0,1\}$ is total and its indexed input sets are finite and nonempty. Equal rows or columns may have different indices. Its size is $T = |X||Y|$. A deterministic public-output protocol is a binary tree whose internal nodes specify a speaker and a bit-valued function of that player's input and the preceding transcript. Each leaf has a fixed output, and the protocol must compute every entry correctly. Its cost is its maximum depth, charging every bit, including fixed bits. Local computation is free. Write $\mathrm{D}(M)$ for the least cost with unrestricted speaking order.

An additive-$a(T)$ estimator returns $\widehat{D}(M)$ satisfying $|\widehat{D}(M) - \mathrm{D}(M)| \le a(T)$; its output must admit polynomial-time comparison with polynomial-length binary rationals. A bounded-error randomised estimator satisfies this guarantee with probability at least $2/3$ on every input. Polynomial time means worst-case time polynomial in T. The same convention applies to the alternating costs below.

Definition 2.2. In strict alternation the players speak one bit at a time, in turn. Let D_A^{alt} and D_B^{alt} be the minimum public-output costs when Alice or Bob starts, respectively, and let $D_{\mathrm{free}}^{\mathrm{alt}} = \min\{D_A^{\mathrm{alt}}, D_B^{\mathrm{alt}}\}$. The starter is chosen independently of the inputs. Every bit is charged, and stopping is public: the transcript determines whether communication has ended.

Yao's receiver-output cost Y_{free} permits the same choice of starter, but the receiver of the last bit determines the answer from that player's input and the transcript. At depth zero, the protocol designates one player whose input determines the answer. Stopping is still public and correctness is required on every input.

Definition 2.3. Randomised sparse ETH asserts that, for some fixed constant c_s, no bounded-error randomised algorithm decides 3-SAT on N variables and at most $c_s N$ clauses in worst-case time $2^{o(N)}$. Deterministic sparse ETH makes the corresponding assertion for deterministic algorithms. We use this deterministic form, related to the usual formulation by sparsification [12, 13].

All communication protocols in this paper are deterministic and exact. Randomness belongs only to an estimator or a reduction. All logarithms are binary unless written ln. Write $[q] = \{1, \dots, q\}$ and $\mathrm{pow}_2(a) = 2^{\lceil \log a \rceil}$ for the least power of two at least $a \ge 1$.

Observation 2.4. *For every $n \times n$ table, $\mathrm{D}(M) \le \lceil \log n \rceil + 1$: Alice sends her row index and Bob sends the entry.*

2.2 Relabelling and probability laws

We attach positive rational probability laws to finite input sets to measure how much of each set survives a restriction. These weights do not change protocol cost: correctness is still required at every entry. Repeated labels have their own weights even when their rows or columns coincide.

Observation 2.5 (Local decoding)**.** *Suppose $F'(x', y') = F(f(x'), g(y'))$ for maps $f : X' \to X$ and $g : Y' \to Y$. In every communication model above, the cost of F' is at most that of F; it is equal when both maps are onto. Also $\mathrm{rank}(F') \le \mathrm{rank}(F)$, with equality when both maps are onto. Transposition preserves rank and unrestricted cost, and swaps the starters in the alternating model.*

Proof. Each player decodes their own input and follows the original protocol, with the same bits, speakers, stopping rule and output convention. For onto maps, choosing one preimage of each original label gives the reverse simulation. Selecting or repeating rows and columns cannot increase rank, and the same sections give equality in the onto case. Exchanging the players proves the transposition assertions. □

Corollary 2.6 (Square padding)**.** *Let $M : X \times Y \to \{0,1\}$ have finite nonempty indexed input sets, and let $P \geq 1$ be an integer. Put*

$$S = \mathrm{pow}_2(\max\{|X|, |Y|, P\}).$$

Repeating input labels gives an $S \times S$ table with the same cost in every model above. If ℓ is the complete explicit encoding length of M, including any input laws, the padded table can be printed in time polynomial in $\ell + P$.

Proof. Choose surjections $f : [S] \to X$ and $g : [S] \to Y$ and use entries $M(f(i), g(j))$. Local decoding preserves every cost. Since $|X|, |Y| \leq \ell$ and $S < 2(\ell + P)$, printing the table takes a fixed polynomial in $\ell + P$. The internal weights do not affect its entries. □

Under a decoder $f : X' \to X$, the *pushforward law* assigns a set $A \subseteq X$ the probability $\mu'(f^{-1}(A))$. Thus every retained set $A' \subseteq X'$ has image of pushforward mass at least $\mu'(A')$. Choosing one surviving label per image preserves the corresponding matrix entries; it need not preserve those representatives' original weights. When a projection is assigned its original product law below, this is an explicit choice of law on the projected inputs.

Counting surviving blocks. We will repeatedly partition an input set into q blocks of equal weight. If a restriction retains total mass β, the sum of its within-block masses is $q\beta$. Suppose at most e blocks are exceptional and at most r other blocks retain mass at least t. Bounding these $r + e$ masses by one and every remaining mass by t gives

$$q\beta \leq r + e + qt.$$

Thus a larger total mass forces more than r nonexceptional blocks above the threshold. If the blocks instead form one component of a mixture, of total weight λ, that component retains fraction at least $(\beta - (1 - \lambda))/\lambda$: the other component can contribute at most $1 - \lambda$.

2.3 Lookup and interlacing

Alice holds two bits $x = x_1x_2$ and Bob holds a question $y \in \{1, 2\}$; the lookup table has entry $F(x, y) = x_y$. Give Alice a selector $i \in \{1, 2\}$ and Bob two questions (y_1, y_2), and ask for $F(x, y_i)$. The selector can matter, but deleting questions can make it irrelevant. The following restriction to $x \in \{01, 10\}$ shows both effects.

Entry x_{y_i}: selector can matter

x	i	(1,1)	(1,2)	(2,1)	(2,2)
01	1	0	0	1	1
01	2	0	1	0	1
10	1	1	1	0	0
10	2	1	0	1	0

Shaded columns are deleted.

keep $y_1 = y_2$ →

Selector no longer matters

x	i	(1,1)	(2,2)
01	1	0	1
01	2	0	1
10	1	1	0
10	2	1	0

equal rows within each fixed x

Figure 2: Deleting the unequal-question columns erases the effect of Alice's selector i. In each outlined pair, changing i leaves every retained entry unchanged. Column labels are Bob's question pairs (y_1, y_2).

Observation 2.7. *In the equal-question columns of Figure 2, the answer is independent of Alice's selector, since $F(x, y_1) = F(x, y_2)$ whenever $y_1 = y_2$.*

The general selector construction is the interlacing operation of Mackenzie and Saffidine [17].

Definition 2.8 (Interlacing)**.** For a Boolean matrix $R : X \times Y \to \{0, 1\}$ with specified input laws, the full q-fold interlace $\mathsf{I}_q(R)$ has rows $(i, u) \in [q] \times X$, columns $(y_1, \ldots, y_q) \in Y^q$, and entry

$$\mathsf{I}_q(R)((i, u), (y_1, \ldots, y_q)) = R(u, y_i).$$

The row law is uniform on $[q]$ times the base row law; the column law is the product of the base column law.

Definition 2.9 (Exact tuple lists)**.** Fix integers $q, D \geq 1$. For a finite alphabet Y with a prescribed rational law μ, an indexed tuple list consists of a finite nonempty label set $\mathcal{L}$, a positive rational law ν on its labels, and a decoder $z : \mathcal{L} \to Y^q$. It is *exactly D-wise independent* if every set $J \subseteq [q]$ of at most D decoded coordinates has law μ^J. Exactness refers to equality of these small marginals; higher-order independence is not required. Events are sets of labels and may distinguish labels with the same decoded tuple.

For a base matrix $R : X \times Y \to \{0, 1\}$ with column law μ, interlacing on this list gives column ℓ the entry $R(u, z_i(\ell))$ at row (i, u), with the row law of Definition 2.8 and column law ν. Within-block row mass is measured in the base row law, without division by q. Distinct labels remain distinct inputs even if their decoded tuples agree.

3 Amplification and the randomized result

The main construction turns a one-bit gap into an arbitrary gap while keeping it stable under deletion of inputs. We state its interface first and derive the randomized hardness result; the construction and its proof follow in Section 5.

3.1 Lower bounds that survive restrictions

The restriction profile records how much of a lower bound remains after inputs are removed. We need only two column thresholds: one for the current restriction and one after a single bit halves its column mass.

Definition 3.1 (Restriction profile)**.** Let $F : X \times Y \to \{0, 1\}$ have positive rational input laws. For $0 < \rho \leq 1$ and an integer d, the restriction profile $\mathcal{P}(F; \rho, d)$ holds if, for every $j \in \{0, 1\}$ and all $U \subseteq X$, $V \subseteq Y$,

$$\mu_X(U) \geq \rho \text{ and } \mu_Y(V) \geq (3/5)2^{-j} \quad \Longrightarrow \quad \mathrm{D}(F[U, V]) \geq d - j.$$

The two column thresholds are $3/5$ and $3/10$: halving the retained column mass may lose one bit of the lower bound. No matching upper bound is required.

We next state the bounded-degree colouring source used by the construction. It supplies the activity pattern, while the restriction profile carries the communication lower bound to the next round. We use the bounded-degree colouring promise below; its conversion from sparse 3-SAT, including the size accounting, is proved in Section G and uses the standard sparse gap and bounded-degree reductions [2, 4].

Theorem 3.2 (Bounded-degree colouring source)**.** *There are constants $\Delta \geq 1$ and rational $\gamma > 0$, and a deterministic polynomial-time reduction from 3-SAT to nonempty ordered loopless multigraphs $G = (V, E)$ with no isolated vertices and maximum degree at most Δ, such that:*

1. *a satisfiable formula produces a three-colourable graph;*
2. *an unsatisfiable formula produces a graph for which every three-colouring has at least $\gamma|E|$ monochromatic edge occurrences.*

For sparse formulas on $N \geq 1$ variables, the reduction can use a fixed positive integer p such that, for $n = |V|$,

$$N \leq |E|, \qquad n, |E| \leq N(\log(N+2))^p, \qquad n/2 \leq |E| \leq \Delta n/2.$$

The amplifier combines this graph promise with a restriction profile; when the base cost is at least three, it restores that profile while charging one additional bit on a NO instance.

Theorem 3.3 (Deletion-robust amplification)**.** *Fix the graph promise of Theorem 3.2 and write $L = \lceil \log n \rceil$. For every source larger than an effectively computable constant, there is a public integer $h = L + O_{\Delta,\gamma}(\log L)$ and a deterministic, implicitly represented construction $R \mapsto R^+$ with the following properties:*

(i) For every source,

$$\mathrm{D}(R^+) \leq \mathrm{D}(R) + h + 1,$$

and, if G is YES, the stronger bound $\mathrm{D}(R^+) \leq \mathrm{D}(R) + h$ holds.

(ii) If G is NO, $\mathrm{D}(R) = d \geq 3$, and $\mathcal{P}(R; 2^{-L}, d)$ holds, then

$$\mathrm{D}(R^+) = d + h + 1, \qquad \mathcal{P}(R^+; 2^{-L}, d + h + 1).$$

(iii) On either source promise, $\mathrm{rank}(R^+) \leq 2^h \,\mathrm{rank}(R)$.

All intermediate products are complete. When iterated from the lookup seed below, the construction has the recursive input generators and entry evaluator, with their uniform resource bounds, proved in Lemma 4.5. Its parameter choices and admissibility test are those of Lemma F.1; they are independent of the base matrix and of the number of rounds.

The guarantee is conditional on the graph promise and restriction profile, so it does not apply to an arbitrary base matrix. The deletion argument, including the part that handles whichever player speaks first, is given in Section 5; the interlacing estimates used there are proved in Section A.

3.2 Iteration

One round adds h bits to the YES upper bound and $h + 1$ bits to the NO lower bound. Starting from a lookup table makes the accumulated gap explicit.

Lemma 3.4 (Lookup seed)**.** *For $L \geq 1$, let $p_0 = \mathrm{pow}_2(64(L+1))$ and $R_0(z, j) = z_j$ on $\{0,1\}^{p_0} \times [p_0]$, with uniform laws. Then*

$$d_0 := \log p_0 + 1, \qquad \mathrm{D}(R_0) = d_0, \qquad \mathrm{rank}(R_0) = p_0 = 2^{d_0 - 1}, \qquad \mathcal{P}(R_0; 2^{-L}, d_0).$$

For $L = \lceil \log n \rceil$ and $n \geq 4$, the seed has polynomial size in n and can be printed in polynomial time.

The proof is in Section A.

Proposition 3.5. *Iterate Theorem 3.3 from R_0 using the same admissible graph and schedule. Put $B_i = d_0 + ih$, $H_i = B_i + i$, and $r_i = 2^{B_i - 1}$. A YES graph gives $\mathrm{D}(R_i) \le B_i$. A NO graph gives $\mathrm{D}(R_i) = H_i$ and $\mathcal{P}(R_i; 2^{-L}, H_i)$. On either promise, $\mathrm{rank}(R_i) \le r_i$ and $\mathrm{D}(R_i) \le H_i$.*

Proof. The seed supplies the case $i = 0$. At each round the two cost bounds increase by h and $h+1$, respectively, while rank grows by at most 2^h. The NO profile is restored at every round, so the induction continues. □

4 Final sampling and growing error

The final sampled table keeps the absolute communication threshold from the full-product construction while reducing the printed size to a rank-sensitive sample.

4.1 Counting columns served by short protocols

We first identify the objects that the sampling union bound must count. For a target threshold H, making the final table hard requires every depth-$(H-1)$ protocol to miss a sampled column, so the count must be uniform over retained row sets. The right objects to count are therefore the column sets that one protocol with Bob's maps unspecified can serve on one retained row set.

Definition 4.1 (Compatible-column set)**.** A *partial protocol tree* fixes Alice's behaviour and the leaf outputs while leaving Bob's maps to be chosen separately for each column. More precisely, a depth-t protocol tree P specifies a complete binary tree of depth t, the speaker at each internal node, all Alice maps, and all leaf outputs; Bob's maps are unspecified. For a Boolean matrix $F : X \times Y \to \{0,1\}$ and $U \subseteq X$, let $G_P(U)$ be the set of columns for which some assignment of bits to Bob's nodes makes the tree correct on every row in U. The assignment may depend on the column, but not on Alice's row.

This definition turns the restriction question into a containment question: a short protocol on a restriction is exactly one such tree whose compatible set contains all retained columns.

Observation 4.2. *For retained sets U, V, with correctness on an empty restriction understood vacuously,*

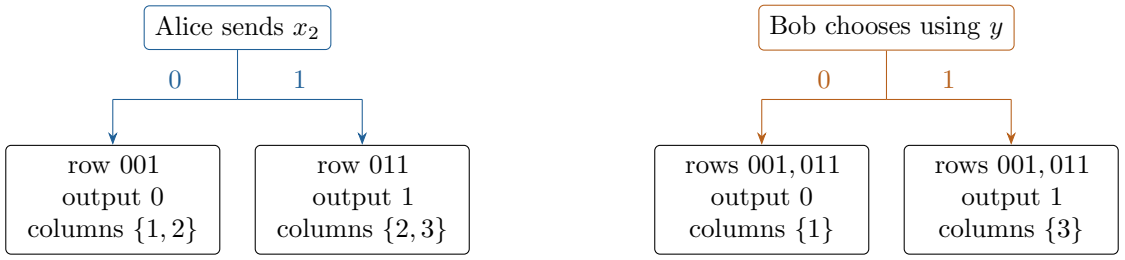
$$\mathrm{D}(F[U,V]) \le t \quad \Longleftrightarrow \quad V \subseteq G_P(U) \text{ for some depth-}t \text{ protocol tree } P \text{ of this kind.} \tag{1}$$

Proof. Pad a protocol of depth at most t with fixed dummy messages and extend Alice's maps to all rows. Its Bob maps witness the required containment. Conversely, for each retained column choose a witnessing assignment at Bob's nodes; these assignments define Bob's maps for a correct protocol on $U \times V$. Empty restrictions satisfy both assertions vacuously. □

The two one-bit trees in Figure 3 use the same small lookup table. They show why leaf requirements combine by intersection at an Alice node and by union at a Bob node: Bob's choice can depend on the column, but must work for both retained rows.

One lookup table for both trees: $F(x, y) = x_y$.

	Bob's question y		
Alice's row x	1	2	3
001	0	0	1
011	0	1	1



(a) **Both required: intersection.** $\{1, 2\} \cap \{2, 3\} = \{2\}$.

(b) **Either works: union.** $\{1\} \cup \{3\} = \{1, 3\}$.

Figure 3: Alice's branch depends on the row, so a compatible column must satisfy both leaf conditions. Bob's branch depends on the column, so either leaf may work, but his chosen leaf must answer both rows correctly.

The count below uses the row-sum argument of Oum, Sæther and Vatshelle [18, Theorem 4.2]. Counting sums, rather than the rows that produce them, is what keeps the final sample small.

Theorem 4.3 (Compatible-column count)**.** *Let r, t be nonnegative integers. If a Boolean matrix F has real rank at most r, then, over all depth-at-most-t partial protocol trees and all retained row subsets, there are at most*

$$2^{(r+1)\lceil \log(r+2) \rceil\, 2^{t+1}} \tag{2}$$

distinct compatible-column sets. Repeated row and column labels are allowed.

Proof. Stack the rows of F and $\mathbf{1} - F$ into one Boolean matrix A. Its row space lies in the span of the rows of F and the all-one vector, so its rank is at most $r + 1$. Fix a set of basis columns of A. For any subset of its rows, choose a basis of their span and sum those basis rows. Nonnegativity makes the zero coordinates of this sum exactly the columns vanishing on the chosen row subset. The sum is determined by its basis-column coordinates, each an integer between 0 and $r + 1$. Hence there are at most $(r + 2)^{r+1}$ common-zero sets. The zero sum includes empty row subsets.

Both leaf outputs are covered: output 0 asks for a common-zero set of rows of F, and output 1 asks for one of their complements. Write $a = (r + 1)\lceil \log(r + 2) \rceil \geq 1$, so there are at most 2^a leaf conditions. At an Alice node the child conditions intersect; at a Bob node they unite. These operations use the rows consistent with Alice's messages on each path, and therefore include every retained row set and every choice of Alice's maps. After padding to depth t, the count is at most

$$2^{2^t - 1}(2^a)^{2^t} \leq 2^{a 2^{t+1}}.$$

The argument also covers rank zero and repeated row or column labels. □

4.2 Sampling columns and then rows

We sample the same number of rows and columns and round that number up to a power of two. This gives the required square table directly. The proof exposes the columns first: their guarantee must hold for every sufficiently heavy row set before the independent row sample is drawn.

Theorem 4.4 (Two-sided compression)**.** *Let $F : X \times Y \to \{0,1\}$ have real rank at most r, where $r \geq 0$ is an integer, and let μ_X, μ_Y be input laws on nonempty finite indexed sets. Suppose an integer $H \geq 1$ satisfies*

$$\mu_X(U) \geq 3/5, \quad \mu_Y(V) \geq 3/5 \quad \Longrightarrow \quad \mathrm{D}(F[U,V]) \geq H.$$

Put

$$m = \mathrm{pow}_2\big(16(r+1)\,\lceil \log(r+2) \rceil\; 2^H\big),$$

where $\mathrm{pow}_2(u)$ is the least power of two at least u. Draw m rows and m columns independently, retaining separate indices for repeated draws. The resulting $m \times m$ matrix has cost at least H except with probability less than $1/100$. Every protocol for F works on every sample outcome.

Proof. Repeated labels can always be decoded locally; conversely, a protocol on repeated labels can be restricted to one occurrence of each label. We use this observation throughout.

For every U with $\mu_X(U) \geq 3/5$ and every depth-$(H-1)$ partial protocol P, the compatible set $G_P(U)$ has mass less than $3/5$: otherwise Observation 4.2 contradicts the restriction hypothesis. The *whole family* of these sets, as both U and P vary, has size at most $2^{m/16}$ by Theorem 4.3. Thus the probability that all sampled columns lie in any one of them is at most $2^{m/16}(3/5)^m$. Outside this event, the sampled-column matrix $F_Y(x,j) = F(x,y_j)$ has cost at least H on every original row set of mass at least $3/5$.

Condition on such a column outcome. The transpose F_Y^{T} still has rank at most r. Every compatible set of its columns, using all its rows, has μ_X-mass less than $3/5$, or transposing the resulting protocol would contradict the preceding guarantee. There are again at most $2^{m/16}$ such sets. The row draws remain independent with law μ_X, so the probability that they all lie in one of these sets is at most $2^{m/16}(3/5)^m$. Any short protocol on the final matrix would give precisely such a containment. Since $m \geq 32$, the total failure probability is at most

$$2 \cdot 2^{m/16}(3/5)^m \leq 2^{1-7m/16} \leq 2^{-13} < 1/100.$$

The universal upper bound follows by decoding sampled indices and running a protocol for F. □

Compression is used only after the full-product amplification is complete. It preserves the final threshold; no further amplification of the sample is needed. The probability bound does not give a test for whether a particular sample succeeds.

4.3 Printing the sample and comparing errors

The full-product table is too large to print. We instead generate complete labels for the sampled inputs, then evaluate only their intersections.

Take a source graph on n vertices that passes the computable admissibility test in Lemma F.1. Every sufficiently large source does so. Use its schedule at every round and write

$$L = \lceil \log n \rceil, \qquad B_k = d_0 + kh, \qquad H_k = B_k + k, \qquad r_k = 2^{B_k - 1}$$

as in Proposition 3.5. The known bound r_k suffices; the reduction never computes the rank of the implicit table.

Lemma 4.5. *For the level-k iteration from the lookup seed in Proposition 3.5, use r_k, H_k in Theorem 4.4. A joint sampler and entry evaluator use $n^{O(k+1)}$ time and random bits on every tape, with an absolute implied constant independent of k. Every output label is an actual input of R_k, and the sample counts and tape length are fixed. The joint law is within total variation less than $1/100$ of the ideal independent sample.*

Here total variation is the maximum difference in probabilities assigned to an event. The recursive construction and its fixed-bit implementation are proved in Section E. Sampling is approximate; every printed entry is exact.

Theorem 4.6 (Uniform randomised reduction)**.** *One randomised algorithm, given an admissible source graph on n vertices and an integer $k \geq 1$, outputs a total Boolean matrix M and an integer B. The table is square with power-of-two side length. On every tape the algorithm uses $n^{O(k+1)}$ time and random bits, with an absolute implied constant independent of k. For a YES source,* $\mathrm{D}(M) \leq B$ *on every tape. For a NO source,*

$$\Pr[\mathrm{D}(M) = B + k] \geq 49/50.$$

Here $B = B_k$ and the tape-independent entry count is

$$m = \mathrm{pow}_2\big(16(r_k + 1)\,\lceil \log(r_k + 2) \rceil\; 2^{H_k}\big), \qquad T = m^2.$$

Algorithm 1 Printing the sampled table and its threshold

Require: An admissible source graph G and an integer $k \geq 1$.

1: Compute its seed and schedule using Lemma F.1.
2: Compute B_k, H_k, r_k, m and reserve the fixed random tape.
3: Generate m columns and m rows using Lemma 4.5.
4: Evaluate all stored row–column pairs, keeping repeated labels as distinct indices.
5: **return** (M, B_k).

Proof. The resource and dimension guarantees follow from Lemma 4.5. On a YES source, local decoding transfers the bound $\mathrm{D}(R_k) \leq B_k$ to every sample, since all generated labels are legal.

On a NO source, Proposition 3.5 gives

$$\mathrm{D}(R_k) = H_k, \qquad \mathcal{P}(R_k; 2^{-L}, H_k), \qquad \mathrm{rank}(R_k) \leq r_k.$$

Because $2^{-L} \leq 1/2$, the $j = 0$ profile bound applies whenever both retained masses are at least $3/5$. Thus Theorem 4.4 gives ideal sampling failure less than $1/100$. Replacing its law by the implemented sampler adds less than $1/100$, by Lemma 4.5. Except with probability less than $1/50$, the output therefore has cost at least H_k; local decoding gives cost at most H_k on every tape. Hence its cost is exactly $B_k + k$. □

Proof of Theorem 1.1. Suppose a bounded-error randomised polynomial-time estimator achieves additive error $a(T) = c \log T / \log\log T$ for some fixed $0 \leq c < 1/8$ on every sufficiently large square table of power-of-two side length. We obtain a bounded-error $2^{o(N)}$ algorithm for sparse 3-SAT.

For a formula with N variables, Theorem 3.2 produces a source graph with n vertices, where

$$2N/\Delta \leq n \leq N(\log(N + 2))^p$$

for a fixed p. The lower bound follows from $N \leq |E| \leq \Delta n/2$. Thus the graph is admissible for every sufficiently large N, and

$$L = \lceil \log n \rceil = \log N + O(\log\log N).$$

Choose $k = \left\lfloor N / \lceil \log N \rceil^2 \right\rfloor$. By Lemmas 3.4 and F.1, $h = L + O(\log L)$ and $d_0 = O(\log L)$, so

$$B_k = d_0 + kh = (1 + o(1))k \log N.$$

Substituting $r_k = 2^{B_k-1}$ and $H_k = B_k + k$ into Theorem 4.6 gives $\log m = 2B_k + k + O(\log(B_k+1))$. Hence the printed size satisfies

$$\log T = 2\log m = 4B_k + 2k + O(\log(B_k+1)) = (4+o(1))k\log N, \tag{3}$$

$$\log\log T = (1+o(1))\log N, \qquad k = \left(\frac{1}{4}+o(1)\right)\frac{\log T}{\log\log T}. \tag{4}$$

Consequently $k > 2a(T)$ for all sufficiently large N. The table already has the square power-of-two form required by the assumed estimator.

For each estimator run, test whether its output is below the rational midpoint $B_k + k/2$. An accurate estimate answers this test correctly on every YES output and every successful NO output. Repeat independently a constant number of times and take the majority answer, reducing the decision error on each such table below $1/100$. This uses only the rational-threshold comparison required by the estimator model. The reduction fails with probability at most $1/50$, so the decision error is less than $3/100 < 1/3$.

Both construction time and random-bit usage are $n^{O(k+1)} = 2^{o(N)}$ on every tape. The estimator also takes $2^{o(N)}$ time because $\log T = O(N/\log N)$. For all sufficiently large N, the chosen k is positive, the graph is admissible, and T meets the estimator's size threshold. The remaining finitely many input sizes can be decided exhaustively within a fixed bound. This contradicts randomised sparse ETH. In particular, an estimator working on all explicit total tables is also impossible. □

5 Proof of robust amplification

5.1 Encoding the graph

We combine the graph's activity pattern with lookup in a base matrix. The widths s, q_2, w, Q remain symbolic here; Lemma F.1 fixes their public dyadic values and admissibility tests.

In a YES instance, the players will reduce the new lookup to one lookup in R using three messages: Alice names a bin, Bob reveals its activity coordinate, and Alice selects the relevant base question. A proper colouring ensures that an active item is unique within its bin. In a NO instance, some bin must contain a collision, and the rest of the proof shows that the collision cannot be avoided cheaply. The template rows below force the first messages; the source rows carry the graph constraint.

Definition 5.1 (Source construction)**.** Let $R : X \times Y \to \{0,1\}$ have positive rational input laws, and let $s \geq 4$ be dyadic. Integer coordinates below are uniform; products use the specified factor laws.

Activity. An item is a vertex or auxiliary label in one of $4s$ groups: $\mathcal{I} = [4s] \times (V \sqcup [s-3])$. In each group, each edge occurrence has a coordinate activating its two endpoints and all $s-3$ auxiliary items. All other items are inactive. Pad these $4s|E|$ coordinates with inactive coordinates to length $q_0 = \mathrm{pow}_2(4s|E|)$, then repeat each coordinate q_2/q_0 times, where q_2 is a dyadic multiple of q_0. Write $v_i(\alpha)$ for item i's activity at coordinate α, and π_α for the order-preserving injection of active items into $[s]$. Activity does not depend on any assignment of items to bins.

Template lookup. Put $q_1 = s-2$, leaving two codes for active and inactive source items. Define

$$\Gamma = Y^{2s-1}, \qquad J = [q_1] \times X, \qquad S = J^{q_2}, \qquad J_2 = [q_2] \times \Gamma.$$

The first q_1 positions of a word $\gamma \in \Gamma$ answer template questions; its last $s+1$ positions will answer source questions. The two lookup matrices are

$$A(\gamma, (r,u)) = R(u, \gamma_r), \qquad R_{\mathrm{temp}}(z, (\alpha, \gamma)) = A(\gamma, z_\alpha).$$

Raw entries. The raw matrix has template rows (p, z) and source rows (i, u):

$$\mathcal{K} = ([s] \times S) \sqcup (\mathcal{I} \times X).$$

Set $\lambda = 255/256$ and $\mu = 1/256$. Template rows have total mass λ; vertex rows and auxiliary rows each have mass $\mu/2 = 1/512$. Within a type, bin or item labels are uniform and the other labels follow their specified laws. For a column $Z = (\xi, j_1, \dots, j_s)$, with $\xi \in Y^{s+1}$ and $j_p = (\alpha_p, \gamma_p) \in J_2$, set

$$B_{\mathrm{raw}}((p, z), Z) = R_{\mathrm{temp}}(z, j_p),$$

$$B_{\mathrm{raw}}((i, u), Z) = \begin{cases} R(u, \xi_{\pi_\alpha(i)}), & \alpha_1 = \cdots = \alpha_s = \alpha,\ v_i(\alpha) = 1, \\ R(u, \xi_{s+1}), & \text{otherwise.} \end{cases}$$

Thus a source row uses its active position when all selectors agree, and the common inactive position otherwise.

Two column families. Ordinary columns $\Omega = Y^{s+1} \times J_2^s$ have their product law. Diagonal columns are

$$\mathcal{D} = \{(\mathrm{tail}(\gamma), j, \dots, j) : j = (\alpha, \gamma) \in J_2\}, \qquad \mathrm{tail}(\gamma) = (\gamma_{q_1+1}, \dots, \gamma_{2s-1}),$$

with the pushforward of the J_2 law. Give these families the masses

Family	Mass	Role
Ω	λ	Reference inputs and the initial selector prefix
$\mathcal{D}$	μ	Collision witnesses and protected inputs.

In particular, the ordinary-column mass equals the template-row mass λ. Keep the two family tags distinct even when their raw columns coincide. At every bin p, the label j_p has the J_2 law in either family.

The two interlaces. Transpose the raw matrix and name its reference restriction:

$$N = B_{\mathrm{raw}}^{\mathsf{T}}, \qquad N_{\mathrm{ref}} = N[\Omega, \mathcal{K}],$$

where the Ω law is normalised. Finally set

$$U = \mathsf{I}_w(N), \qquad W = U^{\mathsf{T}}, \qquad R^+ = (\mathsf{I}_Q(W))^{\mathsf{T}}.$$

The ordinary inputs force the protocol to assign items to bins; the diagonal inputs then expose a collision through distinct positions of one base word. We call the diagonal inputs *protected inputs*. The internal interlace tracks their destinations under arbitrary speaking orders. On a NO instance, the argument gives its transpose W the required full cost, with a restriction profile still one bit below that cost. The final interlace restores the profile at the new cost, so the next round can begin.

The following $s = 4$ example compares two bin assignments with the same activity. The endpoints u, v and auxiliary item a are active; item h, from another group, is inactive. Take u, v to have active ranks $1, 2$. In the diagonal family they request γ_3, γ_4, while h requests $\xi_{s+1} = \gamma_7$.

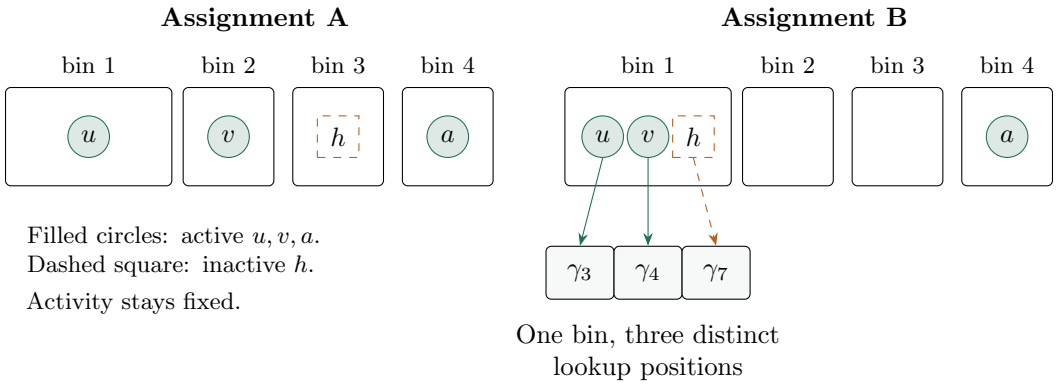


Figure 4: The same active and inactive items can have different bin assignments. In assignment B, one bin contains the active endpoints u, v and the inactive item h. Their diagonal inputs request three distinct positions. These are two illustrative assignments, not steps of a protocol.

The collision in Figure 4 is the local witness after a protocol has fixed an assignment of items to bins. The lemma below makes that witness uniform over the allowed deletions, which is the graph-specific input needed by the amplification proof.

Lemma 5.2. *A YES graph admits an assignment of $\mathcal{I}$ to $[s]$ with at most one active item in each bin at every coordinate. For a NO graph, delete fewer than $(s/2)\delta_v n$ vertex items and fewer than $s/2$ auxiliary items, where*

$$\delta_v = \frac{\gamma|E|}{2\Delta n}, \qquad \zeta = \frac{\gamma}{24}.$$

Every assignment of the remaining items to the s bins then has at least ζq_2 coordinates at which one bin contains two active items and an inactive item.

The same bin assignment gives a short protocol on YES instances.

Proposition 5.3. *If $\mathrm{D}(R) \le y$, then $\mathrm{D}(N) \le y + 2\log s + \log q_2 + 1$. For a YES graph, the final $+1$ can be omitted. Both bounds remain valid after restricting either input set.*

Proof. For a YES graph, fix a public bin assignment χ from Lemma 5.2. Alice holds a template row (p, z) or a source row (i, u); Bob holds $Z = (\xi, j_1, \ldots, j_s)$ with $j_p = (\alpha_p, \gamma_p)$. They exchange:

Alice	bin p (use $p = \chi(i)$ for a source row)	$\log s$ bits,
Bob	α_p	$\log q_2$ bits,
Alice	one of q_1 template codes, active, inactive	$\log s$ bits.

The third message has $q_1 + 2 = s$ possibilities. For a template row, write $z_{\alpha_p} = (r, u)$ and send code r; the answer is $R(u, (\gamma_p)_r)$. For a source row send active precisely when $v_i(\alpha_p) = 1$. If all α_j agree and the code is active, Bob identifies the unique active item in bin p and uses $\xi_{\pi_{\alpha_p}(i)}$. Otherwise he uses ξ_{s+1}. Bob can test agreement from his own input. The entry rule therefore reduces both row types to one execution of a cost-y protocol for R.

For an arbitrary graph, use any public assignment and replace the active code by the s possible values of $\pi_{\alpha_p}(i)$. There are now $q_1 + s + 1 = 2s - 1$ codes, so the third message costs one additional bit. The same rules work for every column tag. Exchanging the players gives the protocol for $N = B^{\mathsf{T}}_{\mathrm{raw}}$, and restriction preserves its cost. □

This protocol sets the YES baseline. We next prove that collisions force the extra bit and that the two interlaces preserve it.

5.2 A collision survives the conversation

The graph promise must apply to one assignment fixed by the protocol before an activity coordinate is selected. We use the items and matrices of Definition 5.1, with the widths fixed by Lemma F.1. The combinatorial count in Lemma 5.2 applies to every assignment of the retained items.

Proof of Lemma 5.2. For a YES graph, let $c : V \to [3]$ be a proper three-colouring supplied by Theorem 3.2. Assign each vertex v to bin $c(v)$, and assign the $s-3$ auxiliary labels to the remaining bins. Every real coordinate activates an edge's two endpoints and those auxiliaries in one group, so all its active items occupy distinct bins. Padded coordinates activate no items.

Now suppose that G is NO. By the deletion hypotheses in Lemma 5.2, fewer than $s/2$ groups lose an auxiliary item and fewer than $s/2$ groups lose more than $\delta_v n$ vertices. Hence more than $3s$ groups retain every auxiliary item and all but at most $\delta_v n$ vertices.

Fix such a group. If two auxiliaries share a bin, they collide at every edge coordinate. Otherwise the auxiliaries occupy $s-3$ distinct bins, leaving three free bins. Colour the retained vertices in those free bins by their bins, and extend the colouring arbitrarily to all other vertices. At least $\gamma|E|$ edge occurrences are monochromatic by the NO promise, and at most $\Delta\delta_v n = \gamma|E|/2$ meet a deleted vertex. Every remaining monochromatic edge gives an active collision: an endpoint in an auxiliary bin collides with its auxiliary; if neither endpoint is in an auxiliary bin, the two endpoints share a free bin. Thus this group has at least $\gamma|E|/2$ edge coordinates with a collision.

For each counted collision, choose one bin containing the two active items. Discard a group from the count if one of its chosen bins contains retained items from that group alone. Every discarded group has an exclusive chosen bin, and two groups cannot have the same exclusive bin; hence at most s groups are discarded. In each of the more than $2s$ remaining groups, every chosen collision bin also contains a retained item from another group. That item is inactive at the chosen coordinate. We have therefore found more than $\gamma s|E|$ witnesses with two active items and one inactive item. Since $q_0 < 8s|E|$, their fraction after padding exceeds $\gamma/8$, and hence the stated $\zeta = \gamma/24$. Uniform repetition to width q_2 preserves that fraction. □

Retained coordinate blocks give lower bounds on the conversation that remains after a prefix. The first estimate below handles a constant retained column mass; the second permits exponentially smaller mass when more blocks remain. Their proofs, and the verification of the source profiles and forced prefix, are in Section A.

Lemma 5.4. *Suppose $\mathcal{P}(M; x, H)$ holds for a rationally weighted matrix M, where $H \geq 2$. Let $u \geq 2$ be dyadic with $ux \leq 1$. Every full-product interlace restriction retaining at least $u/2+1$ blocks, each of base row mass at least ux, and a column set of mass at least $9/25$, has cost at least $H + \log u$. In particular, two blocks of base row mass at least $2x$ and column mass at least $9/25$ suffice for cost $H+1$.*

Lemma 5.5. *Suppose $\mathcal{P}(M; x, H)$ holds for a rationally weighted matrix M, where $H \geq 3$. Let $u \geq 16$ be dyadic with $ux \leq 1$. Every full-product interlace restriction retaining at least $9u/16$ blocks, each of base row mass at least ux, and a column set of mass at least $2^{-u/1024}$, has cost at least $H + \log u$.*

These estimates give the template profiles and the extra bit forced by three distinct tail positions.

Lemma 5.6. *Use the full-product schedule of Lemma F.1, and suppose $\mathrm{D}(R) = d \geq 3$ and $\mathcal{P}(R; \rho, d)$, where $\rho = 2^{-L}$. Write $x = 2^{-10L}$ for the scheduled block threshold, and set*

$$H_1 := d + \log s, \qquad H_2 := H_1 + \log q_2, \qquad C := H_2 + \log s + 1, \qquad x_{\mathrm{cap}} := 2s\rho.$$

Then

$$\mathrm{D}(A) = H_1, \quad \mathcal{P}(A; x, H_1), \qquad \mathrm{D}(R_{\mathrm{temp}}) = H_2, \quad \mathcal{P}(R_{\mathrm{temp}}; x, H_2), \qquad \mathcal{P}(N_{\mathrm{ref}}; x, C-1).$$

In the interlace of R on Γ, the first q_1 positions together with any three distinct tail positions require at least $H_1 + 1$ bits whenever every position retains base row mass at least x_{cap} and the word set has mass greater than $1/2$.

The next lemma follows collisions until Alice must speak. Its proof uses one conserved count: an intact collision counts twice, and after it splits each of the two resulting parts counts once. Some continuation must then carry more requests than its remaining communication budget permits.

Lemma 5.7 (Following collision requests)**.** *Let $q \geq 2$ be dyadic, let $0 \leq b < q$ be an integer, and let $x > 0$ satisfy $2qx \leq 1/8$. Suppose a protocol with at most $H + \log q$ bits left has $q - b$ usable coordinate fibres, each containing word mass at least $3/4$. Here a fibre consists of the word inputs having one fixed coordinate label. Mark $w_0 > b$ fibres as collision witnesses. Assume the following lower bounds, with all masses measured in the original word and selector laws.*

(i) Until Alice speaks, a fibre of word mass at least x, against the unchanged selector set, costs at least $H > 0$.

(ii) At the first Alice query, one child retains a selector set on which, for every dyadic $u \geq 2$ with $ux \leq 1$, any $u/2 + 1$ distinct fibres of word mass at least ux cost at least $H + \log u$.

(iii) A marked fibre retaining word mass greater than $1/2$ costs at least $H + 1$ before Alice speaks.

Then no such protocol exists.

Proof. Follow the fibres through Bob queries and stop at the first Alice query. At depth r, maintain word mass at least $2qx2^{-r}$ in every tracked fibre, called its support. An unmarked support follows a child retaining at least half its mass. A marked support splits into two unmarked supports if both children meet the next threshold; otherwise it follows the qualifying child and keeps its mark. Thus every mark splits at most once. Before a split, the discarded word mass is less than

$$\sum_{r \geq 0} 2qx2^{-r-1} = 2qx \leq 1/8,$$

so an unsplit mark always retains word mass greater than $1/2$.

By (i), a support cannot end at a leaf, nor reach Bob depth $\log q + 1$: it would still have mass at least x but only $H - 1$ bits left. The stopping nodes are prefix-free. A node at depth r_v has *prefix capacity* $\kappa_v = q2^{-r_v} \geq 1$, the remaining width budget; Kraft's inequality gives $\sum_v \kappa_v \leq q$. Write h_v for its support count and $z_v \leq h_v$ for its unsplit marks. Counting an ordinary support once and an unsplit mark twice is conserved: a split replaces two by one plus one. Consequently

$$\sum_v (h_v + z_v) = q - b + w_0 > q \geq \sum_v \kappa_v.$$

Some node therefore has $h + z > \kappa$. Its supports have distinct coordinates, because descendants of a split cannot rejoin, and each has mass at least $2\kappa x$.

If $\kappa \geq 2$, the inequalities $h + z \leq 2h$ and $h + z > \kappa$ give $h \geq \kappa/2 + 1$, since κ is even. Thus (ii), with $u = \kappa$, requires $H + \log \kappa$ bits after Alice's query, but only $H + \log \kappa - 1$ remain. For $\kappa = 1$ and $h \geq 2$, use (ii) with $u = 2$ and the available masses $2x$. In the remaining case $h = 1$, the inequalities force $z = 1$. By (iii), the unsplit collision needs $H + 1$ bits at this stopping node, where only H remain. □

Theorem 5.8 (Source hardness after deletion)**.** *Use either the full-product source of Definition 5.1 or the indexed source of Definition 7.3, with a NO graph and* $\mathrm{D}(R) = d \geq 3$*. Set* $H_1 = d + \log s$*,* $H_2 = H_1 + \log q_2$*,* $C = H_2 + \log s + 1$*, and* $x_{\mathrm{cap}} = 2s\rho$*. Suppose the three profiles and the three-tail bound stated in Lemma 5.6 hold with these parameters. Assume the scalar margins* (9)*, that the selector law on* S *is exactly eight-wise independent with marginal* J*, and that the ordinary family's bin tuple has an exactly eight-wise independent law with marginal* J_2*. Then* $\mathrm{D}(N) = C$*, and every restriction retaining at least* $1 - \eta$ *of each input law has cost at least* C*. In the full-product case these hypotheses are supplied by Lemmas 5.6 and F.1.*

Proof. We first recover one assignment from the template prefix. Its graph collisions then satisfy the three lower bounds in Lemma 5.7, which rules out the proposed communication budget.

Suppose a retained restriction has a protocol of depth at most $C - 1$, and transpose it to the raw orientation. Each template bin loses at most $\eta s/\lambda < 1/1000$ of its S law. The ordinary family Ω retains normalised mass at least $1 - \eta/\lambda > 99/100$. Project its retained inputs onto the whole bin tuple (onto its whole index label for an indexed source), forgetting the tail. This image has mass at least $99/100$. Choose one retained completion per image label, independently of the template row. Since template entries ignore the tail, these representatives realise an interlace of R_{temp}.

Apply Theorem A.3 to the inherited, unpruned protocol with $u = s$ and $\xi = 999/1000$. Its hypotheses follow from the profile of R_{temp} and the scalar margins $999/1000 > 2sx$ and $999/1000 - 2(s-1)x > 998/1000$. Every binary history of length below $\log s$ is reached by template inputs, and every such node is Alice's. They are actual nodes of the original protocol, so the same complete Alice prefix governs all its retained inputs, including item rows and diagonal columns. Its final parts have dominant template sets $A_1, \ldots, A_s \subseteq S$, each of mass greater than $998/1000$. The representatives were used only to establish this prefix; henceforth we use all originally retained columns. The remaining budget is H_2, since $C = H_2 + \log s + 1$.

We next fix the item assignment before choosing an activity coordinate. Call an item deficient if less than half of its base row mass remains in the retained restriction. By the row masses in Definition 5.1, each item type has total mass $1/512$; hence fewer than $4096\eta sn$ vertex items and fewer than $4096\eta s(s-3)$ auxiliary items are deficient. These bounds are within the deletion allowances of Lemma 5.2 by (9). Every nondeficient item can be assigned to one prefix part receiving at least $1/(2s)$ of its base mass, by averaging over these s prefix parts. The schedule gives $4s^2\rho \leq 1$, so

$$\frac{1}{2s} \geq 2s\rho = x_{\mathrm{cap}}.$$

Thus the assigned part retains at least x_{cap} of the item's base row law. This assignment is fixed before the activity coordinate is selected. By Lemma 5.2, at least ζq_2 coordinates contain two active items and an inactive item in one bin. Assign each such coordinate to one prefix part witnessing its collision, and write $U_j \subseteq [q_2]$ for the resulting disjoint sets.

We remove the coordinates on which the assigned rows or diagonal words are too depleted. For a coordinate α, call it a template-bad coordinate for A_j if some one of the q_1 blocks of its J-projection retains less than x_{cap} of the base law. Since $x_{\mathrm{cap}} \leq 1/8$ by (9), such a block misses more than $7/8$ of its $1/q_1$ mass, hence the J-projection misses at least $7/(8q_1)$. If there were $8q_1$ such coordinates, choose one missed event at each and let Z count these events on a random selector. Pairwise independence of the coordinates gives $\mathbb{E}Z \geq 7$ and $\operatorname{Var} Z \leq \mathbb{E}Z$. Every selector in A_j has $Z = 0$, so

$$\Pr(A_j) \leq \Pr(Z = 0) \leq \frac{\operatorname{Var} Z}{(\mathbb{E}Z)^2} \leq \frac{1}{7},$$

contrary to $\Pr(A_j) > 998/1000$. Only pairwise independence was used. Thus A_j has fewer than $8q_1$ template-bad coordinates. Likewise, fewer than $4\eta q_2/\mu$ coordinates retain less than $3/4$ of their

diagonal word law. The diagonal activity label is uniform, so each such fibre deletes more than $\mu/(4q_2)$ of the total column law, whose total deletion is at most η. Let B_j be the union of these two bad-coordinate sets and put $b_j := |B_j|$. The schedule in (9) gives

$$\sum_{j=1}^{s} b_j < 8sq_1 + 4s\eta q_2/\mu < \zeta q_2/4.$$

Since $\sum_j |U_j| \geq \zeta q_2$, some j satisfies $|U_j| > 4b_j$. Fix this j, write $b := b_j$, and define $w_0 := |U_j \setminus B_j|$. Then $w_0 \geq |U_j| - b > 3b$.

Fix a usable collision coordinate $\alpha \in U_j \setminus B_j$. Let i_1, i_2 be its two active items and let the inactive item be the third item in their bin, as supplied by Lemma 5.2. Choose retained template rows for the first q_1 projected blocks and retained rows of i_1, i_2 and the inactive item. By the entry rule in Definition 5.1, against a diagonal word (α, γ) these rows read

$$R(u, \gamma_\ell) \quad (\ell \in [q_1]), \qquad R(u, \gamma_{q_1+\pi_\alpha(i_1)}), \qquad R(u, \gamma_{q_1+\pi_\alpha(i_2)}), \qquad R(u, \gamma_{q_1+s+1}).$$

The three tail positions are distinct by the collision definition and the tail convention in Definition 5.1. Each selected row group has base mass at least x_{cap}, because $\alpha \notin B_j$ and the item assignment above was fixed with that threshold. The representatives depend on the row data and not on γ. The assumed three-tail bound therefore requires $H_1 + 1$ bits whenever the retained diagonal-word mass is greater than $1/2$.

Apply Lemma 5.7 with $q = q_2$ and $H = H_1$. Its usable fibres and marked fibres are exactly the $q_2 - b$ coordinates and $w_0 > 3b$ witnesses just obtained; the schedule gives $2q_2x \leq 1/8$. We check its three lower bounds explicitly.

Until Alice speaks, the selector set A_j is unchanged. Its projection onto each J-coordinate has mass at least $\Pr(A_j) > 998/1000 > 3/5$. Choosing a surviving selector for every projected label therefore gives an actual restriction of A, so its profile supplies (i). At the first Alice query, a denser child retains selector mass greater than $499/1000 > 9/25$. After transposition, its entries on supported coordinates are $A(\gamma, z_\alpha)$: they form the interlace of A with those word sets as row blocks. The exactly eight-wise selector law and Lemma 5.10 supply (ii). All representatives are chosen from the retained selector set, independently of the word input. The three-tail bound proved above supplies (iii), before that Alice query. Thus the tracking lemma excludes depth $C - 1$. The upper protocol in Proposition 5.3 has depth C, proving equality. □

5.3 What remains after restricting tuple inputs

A protocol can create dependencies by restricting its tuple inputs. For example, after Bob announces that two independent coordinates agree, either coordinate determines the other. We therefore use independence only in the original tuple law. The next lemma says that conditioning on an event of specified mass changes the small marginals substantially at only a bounded number of coordinates.

For an event-mass lower bound $0 < v \leq 1$ and an error cap $0 < \tau \leq 1$, set

$$\varepsilon(v, \tau) := \min\{\tau, 2^{-30} v^{1/256}\},$$
$$e(v, \tau) := 8 \left\lceil 2^{20} \varepsilon(v, \tau)^{-2} v^{-1/512} \right\rceil, \qquad e(v) := e(v, 1).$$

The constant independence order $8192 = 8 \cdot 1024$ lets us test 1024 disjoint events, each involving at most eight coordinates.

Lemma 5.9 (Conditioning outside exceptional coordinates)**.** *Let ν be the law of an indexed exactly* 8192*-wise independent tuple list on Z^q with marginal μ, and let $0 < v, \tau \le 1$. For every label event Y with $\nu(Y) \ge v$, there is a set $E \subseteq [q]$, with $|E| \le e(v,\tau)$, such that*

$$d_{\mathrm{TV}}\big((\nu \mid Y)_J, \mu^J\big) \le \varepsilon(v,\tau) \qquad (J \subseteq [q] \setminus E,\ |J| \le 8).$$

Here total variation is the maximum absolute difference between the two laws' probabilities of an event. Coordinates outside this E are called good for Y*. The event and exceptional set are proof objects; neither must be computed by the reduction.*

Write $\pi_J K$ for the decoded projection of a label set K onto coordinates J. If $K \subseteq Y$ and J consists of at most eight good coordinates, then the inverse image of $\pi_J K$ contains K, so

$$\mu^J(\pi_J K) \ge (\nu \mid Y)_J(\pi_J K) - \varepsilon(v,\tau) \ge (\nu \mid Y)(K) - \varepsilon(v,\tau).$$

Choose one actual retained label for each projected tuple. On the row blocks indexed by J, these representatives realise the projected matrix. Its column mass is measured in the original product law μ^J; no product assumption on $\nu \mid Y$ is needed. Before conditioning, exact marginals give the same inequality without the error term.

The next two results turn surviving blocks into communication lower bounds. They apply to full products as well as to the explicit lists constructed later. Their proofs, including the small-coordinate cases and the moment estimates behind Lemma 5.9, are in Section C.

Lemma 5.10 (Surviving blocks in an exact tuple list)**.** *Suppose $d \ge 2$ and $\mathcal{P}(M; x, d)$. Let the tuple source for an interlace be exactly eight-wise independent with the base-column marginal. If an integer $a \ge 1$ satisfies $2^a x \le 1$, and a restriction retains at least $2^{a-1}+1$ blocks, each of base row mass at least $2^a x$, together with a tuple-label set of source mass at least $9/25$, then its complexity is at least $d + a$.*

Theorem 5.11 (Surviving blocks after conditioning)**.** *Suppose $d \ge 2$, $\mathcal{P}(M; x, d)$, and ν is an exactly* 8192*-wise independent tuple source with the base-column law. Let $\nu(Y) \ge v > 0$, choose the exceptional set from Lemma 5.9 with any cap $0 < \tau \le 1$, and let an integer $a \ge 1$ satisfy $2^a x \le 1$. If a restriction retains at least $2^{a-1}+1$ good blocks, each of base row mass at least $2^a x$, and a column set $K \subseteq Y$ with $(\nu \mid Y)(K) \ge 1/2$, then its complexity is at least $d + a$.*

5.4 Where displaced inputs can go

We now prove the internal-interlace bound once for both constructions. Protected inputs are the diagonal inputs in $\mathcal{D}$. In $U = \mathsf{I}_w(N)$, call a copy an input's *origin*. Measure transferred mass in units of one complete copy of N, without division by w; reference losses within Ω are normalised. Recall that Ω has mass $\lambda = 255/256$ and $\mathcal{D}$ has mass $\mu = 1/256$.

Here are the inputs to the common argument. The source profiles are $\mathcal{P}(A; x, H_1)$, $\mathcal{P}(R_{\mathrm{temp}}; x, H_2)$ and $\mathcal{P}(N_{\mathrm{ref}}; x, C-1)$, with $H_2 = H_1 + \log q_2$, $C = H_2 + \log s + 1$, and $H_1, H_2 \ge 2$. The source has exact cost C, including on every restriction retaining $1-\eta$ of both laws. The selector law S is exactly 8192-wise independent on J coordinates; the ordinary bin tuple has that same independence order and marginal J_2. The internal tuple law ν has the raw $\mathcal{K}$ marginal and the following representation by three independent complete streams: metadata selects a bin or an item, base labels have the X law, and selectors have the S law. Each stream is exactly 8192-wise independent across origins. The decoder uses (p, z_i) at a template position and (a, u_i) at an item position. These hypotheses imply the exact 8192-wise $\mathcal{K}$ marginals. They also imply exact 8192-wise independence of the flattened

selector coordinates (i, α): group at most 8192 requests by origin, apply outer independence, then the independence within each S selector. Fixing the other complete streams leaves this ambient selector law unchanged.

Both schedules satisfy (9) and the following bounds, where $C_{128} = (2^{14}/(\lambda - 1/2)^2)^{128}$ is the flag-moment constant used in the proof of Lemma 5.13:

$$\begin{gathered}\delta_U = \eta/(16w), \qquad e_0 \geq e(\theta, \eta/2), \qquad E_{\mathrm{exc}} = 2e_0, \qquad K_* = 512\mu(\log w + 2)s, \\ E_{\mathrm{exc}} \leq w/4, \qquad K_* E_{\mathrm{exc}} \leq \eta w/16, \\ e(\theta) \leq w/128, \qquad e(\theta/4) \leq sq_2, \qquad C_{128} s^{-128} \leq \theta/4.\end{gathered} \tag{5}$$

The numerical consequences of the scalar margins include $2wx + \eta/(16\lambda) < 1$, $wx < 1/256$ and $1 - \eta s/(2\lambda) > 2sx$; all row thresholds below are at most one.

For full products the stream representation is an auxiliary probability space, not a change to the column alphabet $\mathcal{K}^w$. Independently sample complete metadata, base-label and selector streams, even the values that the decoder will ignore. Metadata has exactly the type/bin/item law of $\mathcal{K}$. The decoder therefore pushes this probability space forward to the original product law on $\mathcal{K}^w$. Pulling back a retained event preserves its mass, and pulling back a protocol preserves its entries and depth. There is no enumeration or size requirement on this auxiliary space. In the explicit construction these complete streams are the actual indexed input labels.

This distinction matters only in the receipt proof: first choose the actual retained tuple event (or its denser Bob child), pull it back, and then fix metadata and base streams by averaging. Only afterwards condition on the surviving selector section. The selector law before that last step is unchanged; no independence after conditioning is asserted. Selected positions have template metadata, so their projected selector values have surviving actual-column representatives. Distinct selected values decode to distinct actual columns. Unused stream values need not be represented in the original full-product matrix.

The full-product construction meets these inputs by Lemmas 5.6 and F.1 and Theorem 5.8 and product independence. The explicit construction will verify the same inputs in Section 7.

Lemma 5.12. *Under the common inputs above, suppose a restriction of U retains row mass at least $1 - \delta_U$ and a tuple event Y' of ambient mass at least θ, with a protocol of depth at most $C + \log w - 1$. Apply Lemma 5.9 to this one event with cap $\eta/2$, removing at most e_0 origins. At an Alice node of depth r and capacity $\kappa = w2^{-r}$, track each remaining reference support of normalised mass at least $2\kappa x$ into every child retaining mass at least κx. An unsplit support has weight one; each descendant after its first split has weight one-half.*

Stop at the first Bob query or at depth $\log w$, whichever comes first; nodes at that depth are called final, regardless of the next speaker. Complete paths without supports by dummy Alice queries. A first-Bob node of capacity $\kappa \geq 2$ has weight at most $\kappa/2$, and a final node has at most one support. At least $w - E_{\mathrm{exc}}$ final nodes carry unsplit supports of distinct origins. Call these nodes their origins' owners*. All other destinations together have capacity at most E_{exc}.*

Proof. Initial deletion removes at most $w\delta_U = \eta/16$ complete copies. Every origin starts with normalised reference mass at least $1 - \eta/(16\lambda) > 2wx$. A support always has a child retaining at least half its mass. Its first split preserves total weight, and further splits can only increase it. Descendants of an origin never rejoin. Thus the total tracked weight is at least $w - e_0$.

Until Bob speaks, the tuple set is the fixed Y'. For every good origin, its projection has ambient $\mathcal{K}$ mass at least $1 - \eta/2 > 3/5$: the conditional law gives this image mass one, and the marginal guarantee loses at most $\eta/2$. One surviving tuple per projected value realizes a restriction of N_{ref}. A

support therefore cannot reach a leaf, by its profile and row mass at least $2x$. Padding support-free paths gives a complete frontier whose capacities sum to w.

At a first-Bob node of capacity $\kappa \geq 2$, suppose there were at least $\kappa/2 + 1$ supports. Their origins are distinct and good for the same event Y'. The denser Bob child has conditional mass at least one-half. Apply Theorem 5.11 to N_{ref} with $a = \log \kappa$: each support has mass $2\kappa x \geq \kappa x$. The child requires $C - 1 + \log \kappa$ bits, but only $C + \log \kappa - 2$ remain. Consequently there are at most $\kappa/2$ supports and at most that weight. This also covers $\kappa = 2$.

At a final node, two supports of mass $2x$ would, by the same theorem with $a = 1$ and all of Y', require C bits against a budget of $C - 1$. Thus a final node has at most one support. It is an owner if that support is unsplit; otherwise its weight is at most one-half. If c is the number of owners, every other node has at most half its capacity in weight, whence

$$w - e_0 \leq c + \tfrac{1}{2}(w - c), \qquad w - c \leq 2e_0 = E_{\mathrm{exc}}.$$

Each owner has capacity one. An unsplit support follows only one path, so owners have distinct origins and all other destinations have the claimed total capacity. □

Two receipt bounds complete the argument. Their proofs in Section B use the common stream representation above.

Lemma 5.13. *In the protocol and frontier of Lemma 5.12, the protected mass from all origins reaching a first-Bob node of capacity κ is less than $K_*\kappa$. At a final node it is less than K_*. Mass is measured in units of one complete copy of N.*

Lemma 5.14. *In the same protocol, let ε_i be the normalised reference mass missing at owner i. Every other origin that has an owner sends at most $\mu\delta_i$ protected mass to owner i, where*

$$\delta_i = 2q_2 x + 4\varepsilon_i.$$

Lemma 5.15 (Conservation of protected mass)**.** *Measure mass in units of one complete source copy. Suppose $w \geq 2$ origins send surviving rows to a complete Alice-only frontier, and at least $3w/4$ origins have distinct owners. Each copy has reference mass $\lambda = 255/256$ and diagonal mass $\mu = 1/256$. The following properties cannot all hold:*

(i) Each owner misses more than η of its own copy.

(ii) The missing normalised reference masses satisfy $\sum_i \varepsilon_i \leq \eta/(8\lambda)$, and initial deletion removes at most $\eta/16$ protected mass from these origins in total.

(iii) Destinations other than owners receive at most $\eta w/16$ protected mass from these origins.

(iv) Every other owner-origin sends at most $\mu(2qx + 4\varepsilon_i)$ protected mass to owner i, where $4\mu w q x \leq \eta/32$.

Proof. Let P_i be the surviving protected mass from origin i sent outside its owner, and a_i its initially deleted protected mass. Missing reference rows, initially deleted protected rows, and protected rows elsewhere partition the missing part of its copy. Thus $P_i + a_i + \lambda\varepsilon_i > \eta$, and

$$\sum_i P_i > \frac{3\eta w}{4} - \frac{\eta}{16} - \frac{\eta}{8} > \frac{\eta w}{2}.$$

But other owners receive at most

$$\mu w \sum_i (2qx + 4\varepsilon_i) \le w\left(2\mu w q x + \frac{\mu\eta}{2\lambda}\right) \le \eta w\left(\frac{1}{64} + \frac{1}{510}\right) < \frac{\eta w}{32}.$$

Together with (iii), this is less than $3\eta w/32$. Every surviving row has one destination because only Alice has spoken, contradicting the outgoing mass. □

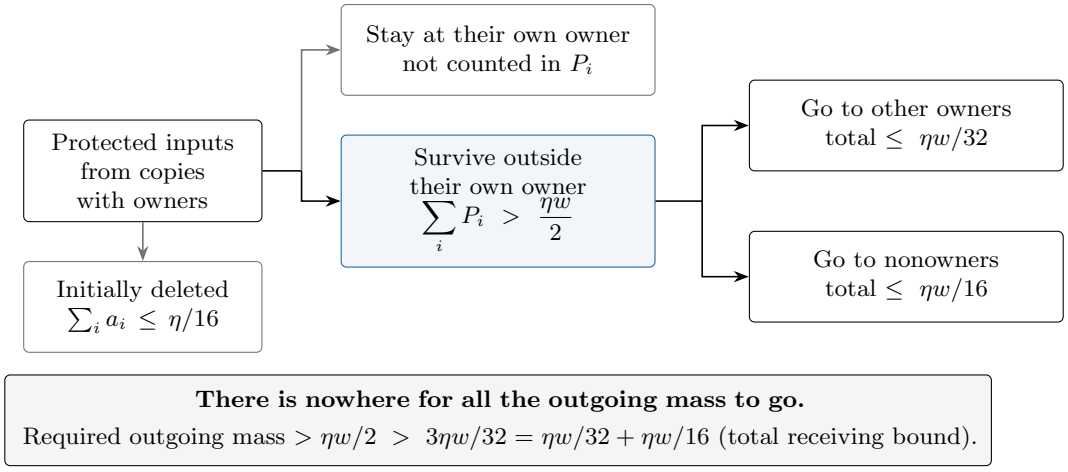


Figure 5: The conservation contradiction. Every surviving protected input outside its own owner must reach one of the two destinations on the right, whose combined receipt bound is too small. Nonowners are first-Bob or final nonowner nodes. Deletion and retention at the own owner are accounted for separately. All masses are in complete-copy units; arrow widths carry no scale.

Theorem 5.16 (Lower bound for the internal interlace)**.** *Under the common inputs of Section 5.4,*

$$\mathrm{D}(W) = C + \log w, \qquad \mathcal{P}(W; \theta, C + \log w - 1).$$

Every restriction of W retaining row mass at least θ and column mass at least $1 - \delta_U$ has cost at least $C + \log w$.

Proof. Suppose a restriction violates the near-full bound, and transpose to U. Its retained tuple event Y' has mass at least θ. By Lemma 5.12, at least $w - E_{\mathrm{exc}} \ge 3w/4$ origins have owners, each with only $C - 1$ bits left. Conditioning on Y' with cap $\eta/2$ gives every owner-origin a column projection of ambient mass at least $1 - \eta/2$. Surviving representatives realize a restriction of N. Since its near-full cost is C, the owner must miss more than η of its own row law.

An unsplit reference support discards less than $\sum_{r=0}^{\log w - 1} wx2^{-r} = 2(w-1)x$ on its path. Initial deletion costs at most $\eta/16$ complete copies, so

$$\sum_i \varepsilon_i \le \frac{\eta}{16\lambda} + 2w(w-1)x \le \frac{\eta}{8\lambda}.$$

This also bounds initially deleted protected mass by $\eta/16$. Exceptional destinations have total capacity at most E_{exc}; Lemma 5.13 bounds their receipt by $K_* E_{\mathrm{exc}} \le \eta w/16$. Finally Lemma 5.14 and $4\mu w q_2 x \le \eta/32$ supply the last hypothesis of Lemma 5.15, with $q = q_2$. Its contradiction proves the near-full bound.

For the profile, fix $j \in \{0,1\}$ and transpose a restriction of W with row mass at least θ and column mass at least $\beta_j = (3/5)2^{-j}$. In U its tuple event Y has mass at least θ. Discard the diagonal rows. The average normalised reference retention is at least $\beta'_j = (\beta_j - (1-\lambda))/\lambda$. Put $\kappa = w2^{-j}$ and apply Lemma 5.9 to Y with cap one. If m good origins retain reference mass at least κx, block averaging gives $w\beta'_j \le m + e(\theta) + w\kappa x$ and hence

$$\frac{m}{\kappa} \ge 2^j\beta'_j - wx - \frac{4e(\theta)}{w} \ge \frac{(3/5) - 4(1-\lambda)}{\lambda} - \frac{1}{256} - \frac{1}{32} = \frac{44}{75} - \frac{1}{256} - \frac{1}{32} > \frac{1}{2}.$$

Thus at least $\kappa/2 + 1$ good blocks remain. Apply Theorem 5.11 to N_{ref} with $a = \log\kappa$, using all of Y, of conditional mass one. The schedules give $\kappa \ge 2$ and $\kappa x \le 1$. The lower bound is $C - 1 + \log\kappa = C + \log w - 1 - j$, proving the profile after transposition. The near-full bound applies to the whole matrix. Conversely the selector protocol sends the copy index and then runs the cost-C protocol for N, giving equality $\mathrm{D}(W) = C + \log w$. □

5.5 Restoring the bound for the next round

Repeated amplification needs a restriction profile at the exact cost, whereas Theorem 5.16 leaves that profile one bit lower. Its stronger bound when almost all columns remain is the additional hypothesis used by Theorem 5.17 below. The starting matrix must meet the full profile as well; the lookup seed and the restoration theorem are proved in Section A.

Theorem 5.17 (Restoration of the restriction profile)**.** *Let W have $m \ge 1$ rows and $n \ge 1$ columns, and suppose its exact cost satisfies $\mathrm{D}(W) \ge 3$. Let $\theta > 0$ and $0 < \delta \le 1/64$. Assume that every restriction retaining row mass at least θ and column mass at least $1 - \delta$ has cost at least $\mathrm{D}(W)$, and that $\mathcal{P}(W; \theta, \mathrm{D}(W) - 1)$ holds. For an integer $b \ge 1$, choose any power of two $Q = 2^r$ satisfying $Q\delta \ge 80b$, and assume $Q\theta < 1/20$. Then $R^+ = (\mathsf{I}_Q(W))^{\mathsf{T}}$ satisfies*

$$\mathrm{D}(R^+) = \mathrm{D}(W) + r, \qquad \mathcal{P}(R^+; 2^{-b}, \mathrm{D}(W) + r).$$

Independently of the lower-bound hypotheses, any upper bound on $\mathrm{D}(W)$ gives an upper bound larger by r on $\mathrm{D}(R^+)$. The matrix R^+ has exactly n^Q rows and Qm columns.

Lemma 5.18. *In the source construction, let $h = 2\log s + \log q_2 + \log w + \log Q$ and suppose $|\mathcal{I}| \le 2sq_2$. Then $\mathrm{rank}(R^+) \le 2^h \mathrm{rank}(R)$. The same bound holds if any word alphabet is replaced by an indexed list of legal words, including products of such lists.*

Proof. A width-t interlace stacks t blocks obtained by selecting or repeating columns of its base matrix, and hence multiplies rank by at most t. The template part therefore has rank at most $sq_1q_2\,\mathrm{rank}(R)$. For each item i, its block has entries $R(u, \psi_i(Z))$, where the selected column $\psi_i(Z)$ is independent of u; its rank is at most $\mathrm{rank}(R)$. After the outer interlaces, rank subadditivity gives

$$\mathrm{rank}(R^+) \le wQ(sq_1q_2 + |\mathcal{I}|)\,\mathrm{rank}(R) \le wQsq_2(q_1+2)\,\mathrm{rank}(R) = 2^h\,\mathrm{rank}(R),$$

using $q_1 + 2 = s$. Restricting word lists, duplicating indexed labels, and transposing preserve this argument. □

Proof of Theorem 3.3. We first apply the restoration theorem to the NO case, then derive the YES upper bound and the rank bound, which does not use the graph promise.

For a NO graph, Theorem 5.8 gives $\mathrm{D}(N) = C$, and Theorem 5.16 gives

$$\mathrm{D}(W) = C + \log w, \qquad \mathcal{P}(W; \theta, C + \log w - 1),$$

together with the near-full restriction bound required by Theorem 5.17. By Lemma F.1, $0 < \delta_U \le 1/64$, $Q = w^2$ satisfies $Q\delta_U \ge 80L$, and $Q\theta = 1/512 < 1/20$. Applying Theorem 5.17 with $b = L$, $\delta = \delta_U$, and W as above yields

$$\mathrm{D}(R^+) = C + \log w + \log Q, \qquad \mathcal{P}(R^+; 2^{-L}, C + \log w + \log Q).$$

The definitions in Lemma 5.6 give $C = d + 2\log s + \log q_2 + 1$. Thus, with

$$h := 2\log s + \log q_2 + \log w + \log Q,$$

these bounds become

$$\mathrm{D}(R^+) = d + h + 1, \qquad \mathcal{P}(R^+; 2^{-L}, d + h + 1).$$

On either source promise, Proposition 5.3 and the two outer selector protocols from Definition 5.1 give the upper bound $\mathrm{D}(R^+) \le \mathrm{D}(R) + h + 1$; on a YES source the source protocol omits the final bit by Proposition 5.3, giving $\mathrm{D}(R^+) \le \mathrm{D}(R) + h$. The overhead estimate $h = L + O_{\Delta,\gamma}(\log L)$ and the effective admissibility cutoff are supplied by Lemma F.1. Recursive input generation and entry evaluation are supplied by Lemma 4.5.

The source-coordinate bound $q_2 \ge 4s|E|$ and $|E| \ge n/2$ imply $|\mathcal{I}| = 4s(n + s - 3) \le 2sq_2$. Thus Lemma 5.18 supplies the rank bound, independently of the graph promise. □

6 Interlacing on small input lists

6.1 Constructing exact marginals

The deterministic reduction replaces each full product by an explicit tuple list, as defined in Definition 2.9. The list agrees with the product on every small set of coordinates; Section 5.3 explains why this remains useful after a protocol restricts its inputs.

The construction appends one coordinate at a time and removes words through linear dependences that preserve every small marginal. This is the sequential construction idea of Koller and Megiddo [14]; its implementation and bit bound are given in Section D. We store the rational weights directly and measure restrictions in these laws.

Lemma 6.1 (Exact rational supports)**.** *Fix an integer $\mathsf{D} \ge 1$. Given a rational law μ on an alphabet of size m and an integer $q \ge 1$, a deterministic algorithm constructs an exactly D-wise independent source $\mathsf{Code}_{\mathsf{D}}(q, \mu)$ with support size at most $(1 + qm)^{\mathsf{D}}$. For fixed D, the construction time, total output length and rational-weight bit lengths are polynomial in q, m and the binary encoding length of μ.*

Every construction call below uses Code_{8192}.

6.2 Restoring the full restriction profile

Restoration uses the base matrix's full cost on near-full column sets when Bob has not yet spoken. Once he has spoken, the conditioned block bound supplies the extra bit from the weaker profile.

Theorem 6.2. *Let $0 < \phi \le 1$, $0 < \delta \le 1/64$ and $\theta > 0$. Suppose W, with input laws μ_X, μ_Y, satisfies $\mathrm{D}(W) = C \ge 3$, $\mathcal{P}(W; \theta, C - 1)$, and*

$$\mathrm{D}(W[A, B]) \ge C \quad \textit{whenever } \mu_X(A) \ge \theta \textit{ and } \mu_Y(B) \ge 1 - \delta.$$

Choose a power of two Q satisfying

$$Q \geq 128\big(e(\phi, \delta/2) + 1\big), \qquad Q\theta \leq 1/20.$$

Interlace W Q times on any exactly 8192*-wise independent tuple source with the base-column law, and transpose the result to obtain R^+. Then*

$$\mathrm{D}(R^+) = C + \log Q, \qquad \mathcal{P}(R^+; \phi, C + \log Q).$$

Independently of the lower-bound hypotheses, a protocol of depth U for W yields one of depth $U + \log Q$ for R^+.

Proof. Fix $j \in \{0, 1\}$ and a restriction of R^+ retaining row mass at least ϕ and column mass at least $(3/5)2^{-j}$. Undo the transpose. The columns now form an event Y of source mass at least ϕ, and the rows have average block mass at least $(3/5)2^{-j}$. Apply Lemma 5.9 to Y with cap $\delta/2$, and put

$$e = e(\phi, \delta/2), \qquad k = Q2^{-j}, \qquad a = \log k.$$

The width assumptions make a a positive integer. Call a block qualifying when its coordinate is good and its retained base-row mass is at least $k\theta$. There are more than $k/2$ qualifying blocks: otherwise the block-averaging observation gives $(3/5)Q2^{-j} \leq k/2 + e + Qk\theta$, and hence

$$\frac{3}{5} \leq \frac{1}{2} + \frac{2^j e}{Q} + Q\theta \leq \frac{1}{2} + \frac{1}{64} + \frac{1}{20} < \frac{3}{5}.$$

Retain the qualifying blocks and all of Y.

Suppose this restriction has a protocol of depth at most $C + a - 1$. Follow the mass-halving choices from Theorem 5.11 for a messages, starting with conditional column mass one. Pad an early leaf with dummy Alice messages if necessary. If α messages are Alice's and β are Bob's, then $\alpha + \beta = a$ and the surviving restriction has

$$\text{more than } 2^{\beta-1} \text{ good blocks}, \qquad \text{row mass per block at least } 2^\beta\theta,$$
$$(\nu \mid Y)(K') \geq 2^{-\beta}, \qquad \text{at most } C - 1 \text{ bits remaining.}$$

If $\beta = 0$, the columns are still all of Y. At least one good block remains, with row mass at least θ. Its projected column set has base mass at least $1 - \varepsilon(\phi, \delta/2) \geq 1 - \delta$, by Lemma 5.9. Choosing actual retained representatives gives a restriction of W that requires C bits by the near-full hypothesis.

If $\beta \geq 1$, apply the short-continuation bound Lemma C.3 with $d = C - 1$, $b = \beta - 1$, and $x = \theta$. The integer block count is at least $2^b + 1$, and the row and column masses meet its thresholds. Also $d \geq 2$ and $2^{b+1}x = 2^\beta\theta \leq k\theta \leq 1$. Again C bits are required. Both cases contradict the remaining budget.

Thus each restriction requires at least $C + a = C + \log Q - j$ bits. Transposing back proves the profile and the lower bound on the whole matrix. For the upper bound, send the block index in $\log Q$ bits, decode that coordinate, and run the depth-U protocol for W. Exchanging the players gives the same cost for R^+. Taking $U = C$ proves equality. □

7 Deterministic gap tables

For every requested integer $k \geq 1$, the explicit-list construction below preserves a deterministic gap of k bits after k rounds, with the YES and NO costs and the complete encoding bound given in Theorem 7.5.

We use the subscript g for the explicit-list construction. For the source graph G, choose an admissible dyadic $\mathsf{N} \geq |V(G)|$ as in Lemma F.2, and put $L_{\mathrm{g}} = \log \mathsf{N}$ and $\rho_{\mathrm{g}} = 2^{-L_{\mathrm{g}}}$. That lemma supplies all subscripted widths and thresholds below. They depend on N and the fixed graph promise, and are kept unchanged as the base matrix R passes through successive rounds.

The one-round statement records what the construction must preserve. Its proof below verifies the source profiles and the common internal-interlace hypotheses, then restores the profile for another round.

Theorem 7.1. *Under the admissible schedule of Lemma F.2, the construction of Definition 7.3 is deterministic and polynomial in* N *and the explicit rational encoding length of* R*. Set*

$$h_{\mathrm{g}} = 2\log s_{\mathrm{g}} + \log q_{2,\mathrm{g}} + \log w_{\mathrm{g}} + \log Q_{\mathrm{g}} = 12L_{\mathrm{g}} + 2\lceil L_{\mathrm{g}}/4 \rceil .$$

Then every upper bound $\mathrm{D}(R) \leq y$ *gives*

$$\mathrm{D}(R_{\mathrm{g}}^{+}) \leq y + h_{\mathrm{g}} + 1,$$

and, on a YES graph, $\mathrm{D}(R_{\mathrm{g}}^{+}) \leq y + h_{\mathrm{g}}$*. If the graph is NO,* $\mathrm{D}(R) = d \geq 3$*, and* $\mathcal{P}(R; \rho_{\mathrm{g}}, d)$*, then*

$$\mathrm{D}(R_{\mathrm{g}}^{+}) = d + h_{\mathrm{g}} + 1, \qquad \mathcal{P}(R_{\mathrm{g}}^{+}; \rho_{\mathrm{g}}, d + h_{\mathrm{g}} + 1).$$

On either promise, $\mathrm{rank}(R_{\mathrm{g}}^{+}) \leq 2^{h_{\mathrm{g}}}\,\mathrm{rank}(R)$*. Every generated input decodes to an input of the corresponding full-product construction with the same entries.*

7.1 Replacing products by explicit lists

In Definition 7.3, we replace each product list by an explicit rational support while preserving the selector laws needed for another round. We retain labels even when decoded inputs repeat, and keep the lists for different roles independent before exposing any restriction.

Lemma 7.2. *Let an indexed list on* $Z^{q_{\mathrm{in}}}$ *be exactly* 8192*-wise independent with marginal* μ_Z*. Choose its indices at* q_{out} *positions using an exactly* 8192*-wise independent list whose marginal is the inner index law. Then the decoded coordinates, indexed by outer–inner pairs, are exactly* 8192*-wise independent with marginal* μ_Z*. If this complete outer list is independent of finitely many other complete lists, fixing any positive-probability joint value of those lists leaves its decoded ambient law unchanged.*

Proof. Write ν for the inner index law and $z_r(\ell)$ for the inner decoder. For at most 8192 prescribed outer–inner pairs, group the requests by outer position: let P be the positions used and T_p the inner coordinates requested at position p. If I_p is the chosen inner index, then

$$\Pr[z_r(I_p) = a_{p,r} \text{ for all requested pairs}] = \prod_{p \in P} \Pr_{\ell \sim \nu}[z_r(\ell) = a_{p,r}\ (r \in T_p)] = \prod_{p \in P} \prod_{r \in T_p} \mu_Z(a_{p,r}).$$

The first equality uses outer independence and the second inner independence; each involves at most 8192 coordinates. Independence of complete lists also means that conditioning on values of other lists does not change this law. □

The construction uses independent complete lists of metadata, base labels and selector indices. In the receipt argument, fix the metadata and base-label streams first; the ambient selector law is unchanged by Lemma 7.2. Then condition on the surviving selector section, using the exceptional-coordinate bounds.

Definition 7.3. Let $R : X \times Y \to \{0,1\}$ be a rationally weighted base matrix and use the admissible schedule of Lemma F.2 and the item-activity construction of Definition 5.1. Replace the word alphabets by the indexed lists

$$\Gamma_{\mathrm{g}} = \mathsf{Code}_{8192}(2s_{\mathrm{g}} - 1, \mu_Y), \qquad J_{\mathrm{g}} = [q_{1,\mathrm{g}}] \times X, \qquad S_{\mathrm{g}} = \mathsf{Code}_{8192}(q_{2,\mathrm{g}}, \mu_{J_{\mathrm{g}}}),$$

$$J_{2,\mathrm{g}} = [q_{2,\mathrm{g}}] \times \Gamma_{\mathrm{g}}, \qquad \mathcal{J}_{\mathrm{g}} = \mathsf{Code}_{8192}(s_{\mathrm{g}}, \mu_{J_{2,\mathrm{g}}}).$$

Integer factors are uniform, and all other factors carry their base or indexed laws. On decoded labels use

$$A_{\mathrm{g}}(\gamma, (r,u)) = R(u, \gamma_r), \qquad R_{\mathrm{temp,g}}(z, (\alpha,\gamma)) = A_{\mathrm{g}}(\gamma, z_\alpha).$$

Keep the ordinary and diagonal column families separately tagged. The ordinary family Ω_{g} has mass $\lambda = 255/256$ and pairs an independent $\mathcal{J}_{\mathrm{g}}$ index with the tail pushforward of the Γ_{g} law. The diagonal family $\mathcal{D}_{\mathrm{g}}$ has mass $\mu = 1/256$ and is the pushforward of $J_{2,\mathrm{g}}$ under

$$(\alpha,\gamma) \longmapsto (\mathrm{tail}(\gamma), (\alpha,\gamma), \dots, (\alpha,\gamma)).$$

The raw-row alphabet is

$$\mathcal{K}_{\mathrm{g}} = ([s_{\mathrm{g}}] \times S_{\mathrm{g}}) \sqcup (\mathcal{I}_{\mathrm{g}} \times X),$$

with type masses $\lambda = 255/256$, $1/512$, and $1/512$, and the same conditional item laws as in Definition 5.1. These decoded rules define $B_{\mathrm{raw,g}}$, $N_{\mathrm{g}} = B_{\mathrm{raw,g}}^{\mathsf{T}}$, and the normalised restriction to the Ω_{g} inputs, denoted $N_{\mathrm{ref,g}}$. The internal interlace uses three independent complete streams of length w_{g}, each exactly 8192-wise independent: metadata has the type/bin/item pushforward of the raw-row law, base labels have law μ_X, and selector indices have the S_{g} law. At each position, the metadata selects either its bin and selector or its item and base label. Their coordinatewise combination gives an exactly 8192-wise independent $\mathcal{K}_{\mathrm{g}}$-tuple list with the raw-row marginal. This list defines U_{g} from N_{g}; set $W_{\mathrm{g}} = U_{\mathrm{g}}^{\mathsf{T}}$. Interlace W_{g} at width Q_{g} using Code_{8192} with its column law, then transpose to obtain R_{g}^{+}. Retain the complete-list products, rational weights, and index labels explicitly, including repeated decoded values.

Fixing the metadata and base-label streams leaves the ambient selector stream unchanged by Lemma 7.2. Selecting a surviving section is a further conditioning event, governed by Lemma 5.9. For projected restrictions, surviving representatives certify the entries; their masses are measured in the original projected law.

7.2 Verifying the source profiles

We verify the source profiles separately because their retained events can be much rarer than those in the common internal proof. Once these profiles are available, the collision and destination arguments already proved apply to the indexed construction without change.

Lemma 7.4. *Suppose* $\mathrm{D}(R) = d \geq 3$ *and* $\mathcal{P}(R; \rho_{\mathrm{g}}, d)$. *Set*

$$H_{1,\mathrm{g}} = d + \log s_{\mathrm{g}}, \qquad H_{2,\mathrm{g}} = H_{1,\mathrm{g}} + \log q_{2,\mathrm{g}}, \qquad C_{\mathrm{g}} = H_{2,\mathrm{g}} + \log s_{\mathrm{g}} + 1, \qquad x_{\mathrm{cap,g}} = 2s_{\mathrm{g}}\rho_{\mathrm{g}}.$$

Then

$$\mathrm{D}(A_{\mathrm{g}}) = H_{1,\mathrm{g}}, \qquad \mathcal{P}(A_{\mathrm{g}}; x_{\mathrm{g}}, H_{1,\mathrm{g}}),$$

$$\mathrm{D}(R_{\mathrm{temp,g}}) = H_{2,\mathrm{g}}, \qquad \mathcal{P}(R_{\mathrm{temp,g}}; x_{\mathrm{g}}, H_{2,\mathrm{g}}), \qquad \mathcal{P}(N_{\mathrm{ref,g}}; x_{\mathrm{g}}, C_{\mathrm{g}} - 1).$$

In the interlace of R *decoded by* Γ_{g}, *retaining the first* $q_{1,\mathrm{g}}$ *positions and any three distinct tail positions, each with base row mass at least* $x_{\mathrm{cap,g}}$ *against word mass greater than* $1/2$, *requires* $H_{1,\mathrm{g}} + 1$ *bits.*

Proof. Suppress the subscript g and put $\lambda = 255/256$. The block-averaging observation proves all three profiles. For $j \in \{0,1\}$ put $\beta_j = (3/5)2^{-j}$. Transpose each profile restriction to view it as an interlace. The following table lists its base, the target width u, and a lower bound for the number m of blocks whose retained base-row mass is at least u times the base profile threshold.

profile to prove	base and threshold	u	m/u is at least
A	R, ρ	$s2^{-j}$	$\frac{3}{5}(1-2/s) - q_1\rho$
R_{temp}	A, x	$q_2 2^{-j}$	$\frac{3}{5} - q_2 x$
N_{ref}	R_{temp}, x	$s2^{-j}$	$\frac{2^j(\beta_j-(1-\lambda))}{\lambda} - sx$

For the first row there are $q_1 = s-2$ actual blocks. For the last row we first discard the non-template rows, of total mass $1-\lambda$. Each displayed bound follows from $q\beta \le m + qt$, at threshold $t = u\rho$ or ux. The schedule makes all three bounds greater than $9/16$.

In each row, the retained tuple-label event has mass at least x. For N_{ref}, take its image on the $\mathcal{J}$ label, forgetting the independent tail; this image has mass at least x. Entries of template rows ignore the tail, so one surviving reference label per image label realises precisely this projected restriction. By Lemma 5.9, at most

$$e(x) \le s/128 \le u/32$$

coordinates are exceptional. More than $17u/32 > u/2$ qualifying blocks remain, and their integer count is at least $u/2+1$. Apply Theorem 5.11 to all of the conditioned event, whose conditional mass is one. Successively using the preceding row's profile gives lower bounds $H_1 - j$, $H_2 - j$, and $C - 1 - j$. This proves all three profiles; the schedule ensures $u \ge s/4 \ge 16$ and the required row thresholds are at most one.

The $j = 0$ lower bounds, together with the protocols sending the q_1 or q_2 selector and then solving the base problem, give $\mathrm{D}(A) = d + \lceil \log q_1 \rceil = H_1$ and $\mathrm{D}(R_{\mathrm{temp}}) = H_2$. Finally, a collision uses $q_1 + 3 = s + 1 = (2s)/2 + 1$ distinct positions of Γ. Its ambient law is at least eight-wise independent, each retained row mass is $2s\rho$, and its word mass exceeds $1/2 > 9/25$. Applying Lemma 5.10 with $a = \log(2s)$ gives $d + \log(2s) = H_1 + 1$ bits. □

7.3 Iteration, encoding size and deterministic ETH

Each round must restore the profile as well as enlarge the gap. The one-round guarantee Theorem 7.1 does this; to iterate it as a uniform reduction, we also bound the complete rational encoding of every intermediate round.

Proof of Theorem 7.1. We bound the construction cost, apply restoration to the NO lower bound, and then verify the upper protocols, rank, and decoding. By Lemma F.2, every width in Definition 7.3 is polynomial in N. By Lemma 6.1, each exact-support call of fixed order 8192 takes polynomial time in its width and input-law encoding length and produces a polynomial-length rational encoding. The construction in Definition 7.3 uses a fixed number of such nested calls and products of complete lists. Composing their polynomial bounds therefore bounds the entire round, including its indexed labels and rational weights, by a fixed polynomial in N and the rational encoding length of R.

For a NO graph, Lemma 7.4 supplies all source profiles and the three-tail bound required by Theorem 5.8. The laws of S_{g} and $\mathcal{J}_{\mathrm{g}}$ are exactly 8192-wise independent, and the scalar margins hold by Lemma F.2. Thus the common source proof gives $\mathrm{D}(N_{\mathrm{g}}) = C_{\mathrm{g}}$, even after deleting at most η_{g} of each law.

We verify the remaining inputs of Section 5.4. The internal tuple list in Definition 7.3 is formed from independent complete metadata, base-label and selector streams with the correct raw-row

marginal. The flattened selector coordinates have exact 8192-wise independence by Lemma 7.2; fixing the other streams leaves their ambient law unchanged. The ordinary bin law is $\mathcal{J}_{\mathrm{g}}$, and the diagonal image in every bin is exactly $J_{2,\mathrm{g}}$. Finally Lemma F.2 supplies (5) with $e_{0,\mathrm{g}} \geq e(\theta_{\mathrm{g}}, \eta_{\mathrm{g}}/2)$ and $E_{\mathrm{exc,g}} = 2e_{0,\mathrm{g}}$. Consequently Theorem 5.16 gives, for $D_0 = C_{\mathrm{g}} + \log w_{\mathrm{g}}$,

$$\mathrm{D}(W_{\mathrm{g}}) = D_0, \qquad \mathcal{P}(W_{\mathrm{g}}; \theta_{\mathrm{g}}, D_0 - 1),$$

and cost at least D_0 when row mass is at least θ_{g} and column mass at least $1 - \delta_{U,\mathrm{g}}$. Since $D_0 \geq 3$, these are precisely the matrix hypotheses of Theorem 6.2. Apply that theorem with $\phi = \rho_{\mathrm{g}}$, $\theta = \theta_{\mathrm{g}}$, $\delta = \delta_{U,\mathrm{g}}$, and width Q_{g}. The numerical hypotheses follow from Lemma F.2:

$$Q_{\mathrm{g}} \geq 128\big(e(\rho_{\mathrm{g}}, \delta_{U,\mathrm{g}}/2) + 1\big), \qquad Q_{\mathrm{g}}\theta_{\mathrm{g}} = 1/512 < 1/20, \qquad 0 < \delta_{U,\mathrm{g}} \leq 1/64.$$

The outer source has the required exact marginals by Definition 7.3 and Lemma 6.1. Thus Theorem 6.2 gives exact cost and profile value

$$D_0 + \log Q_{\mathrm{g}} = d + 2\log s_{\mathrm{g}} + \log q_{2,\mathrm{g}} + \log w_{\mathrm{g}} + \log Q_{\mathrm{g}} + 1 = d + h_{\mathrm{g}} + 1,$$

with profile threshold ρ_{g}, as required.

Every indexed input decodes locally to a legal input of the corresponding full-product construction, with the same entry. Thus the source protocol of Proposition 5.3, followed by the two outer selector protocols, gives cost $y + h_{\mathrm{g}} + 1$, or $y + h_{\mathrm{g}}$ on YES graphs. The source protocol has s_{g} local selector choices in the YES case and $2s_{\mathrm{g}} - 1$ in general; no feature of the ambient product law is used.

The source-coordinate bound implies $|\mathcal{I}_{\mathrm{g}}| \leq 2s_{\mathrm{g}}q_{2,\mathrm{g}}$, so Lemma 5.18 also applies to these indexed lists. It gives $\operatorname{rank}(R_{\mathrm{g}}^{+}) \leq 2^{h_{\mathrm{g}}} \operatorname{rank}(R)$. If decoded labels repeat, retaining them separately or identifying them preserves communication cost by Observation 2.5; weights of identified labels are added. The construction retains the labels and their rational weights explicitly for use in the next round. □

Iteration uses one public schedule at every round because restoration returns the same profile hypothesis. The following theorem makes the round loop and its complete encoding bound explicit.

Theorem 7.5. *There are effective absolute constants $\mathsf{A} \geq 2$ and $C_0 > 0$ and a deterministic algorithm with the following guarantee. Given a source graph G and an integer $k \geq 1$, it produces an explicit total Boolean matrix $F_{G,k}$ and a public integer B_k^{g} such that*

$$\textit{YES: } \mathrm{D}(F_{G,k}) \leq B_k^{\mathrm{g}}, \qquad \textit{NO: } \mathrm{D}(F_{G,k}) = B_k^{\mathrm{g}} + k.$$

For every admissible dyadic $\mathsf{N} \geq |V(G)|$, its running time and full binary output length are at most $\mathsf{N}^{C_0\mathsf{A}^k}$. The constants depend only on the fixed source reduction and construction. The output may be square with power-of-two side length at least N. Its threshold is

$$B_k^{\mathrm{g}} = d_{0,\mathrm{g}} + kh_{\mathrm{g}},$$

where

$$d_{0,\mathrm{g}} = \log \operatorname{pow}_2(64(L_{\mathrm{g}} + 1)) + 1, \qquad L_{\mathrm{g}} = \log \mathsf{N}.$$

Proof. Choose the schedule of Lemma F.2 and the lookup seed of Lemma 3.4 with $L_{\mathrm{g}} = \log \mathsf{N}$. Apply Definition 7.3 k times, storing each indexed table and its rational input laws. The same schedule is used throughout. Finally, forget the weights and duplicate inputs to a square dyadic side at least N.

The seed has exact cost $d_{0,\mathrm{g}} \geq 3$ and profile threshold $\rho_{\mathrm{g}} = 2^{-L_{\mathrm{g}}}$. At each round, Theorem 7.1 increases the YES upper bound by h_{g} and the NO exact cost by $h_{\mathrm{g}} + 1$, restoring the same profile threshold. Induction therefore gives

$$\mathrm{D}(R_i) \leq d_{0,\mathrm{g}} + ih_{\mathrm{g}} \quad \text{on YES}, \qquad \mathrm{D}(R_i) = d_{0,\mathrm{g}} + i(h_{\mathrm{g}} + 1) \quad \text{on NO}.$$

Forgetting weights and the local duplication of Corollary 2.6 change neither cost. This proves the gap with $B_k^{\mathrm{g}} = d_{0,\mathrm{g}} + kh_{\mathrm{g}}$.

For the uniform time bound, let S_i include all indexed labels, table entries, and rational laws after round i. The fixed polynomial construction in Theorem 7.1 gives constants a, c independent of i and k such that

$$S_{i+1}, \text{ time for round } i+1 \leq c(\mathsf{N} + S_i + 2)^a.$$

Choose a fixed power of two $\mathsf{A} \geq 2$ large enough to absorb the coefficient and the additional term $\mathsf{N} + 2$. With $Z_i = \mathsf{N} + S_i + 2$, we then have $Z_{i+1} \leq Z_i^{\mathsf{A}}$. The polynomial seed encoding gives $Z_0 \leq \mathsf{N}^{C_1}$ for a fixed constant C_1, and hence

$$Z_i \leq \mathsf{N}^{C_1 \mathsf{A}^i}, \qquad \text{total round time} \leq k\mathsf{N}^{C_1 \mathsf{A}^k} \leq \mathsf{N}^{(C_1+1)\mathsf{A}^k}.$$

The final square side is less than $2\max\{\mathsf{N}, m_k, n_k\} \leq 2Z_k$, where m_k, n_k are the table dimensions. Printing its entries costs a fixed polynomial in Z_k. Enlarging a single effective constant C_0 therefore covers the schedule, seed, all rounds, threshold computation, and final output within $\mathsf{N}^{C_0 \mathsf{A}^k}$. This bound uses one fixed polynomial per round, not a polynomial whose degree can depend on k. □

Proof of Theorem 1.3. For fixed k, apply the graph reduction of Theorem 3.2 and then Theorem 7.5. The first admissible dyadic size obeys $\mathsf{N} \leq 2\max\{|V(G)|, \mathsf{N}_0\}$ for an effective constant N_0, by Lemma F.2. Since k is fixed, $\mathsf{N}^{C_0 \mathsf{A}^k}$ is polynomial in the formula length. The resulting public threshold separates costs at most B_k^{g} from cost exactly $B_k^{\mathrm{g}} + k$, on a square table with power-of-two side. □

Proof of Theorem 1.2. We choose enough rounds to separate the estimator's error intervals while keeping both construction and estimation subexponential in the sparse SAT size. Suppose the deterministic estimator excluded by Theorem 1.2 exists for the coefficient c_* chosen below. Given a sparse formula on N variables, apply Theorem 3.2 and write $n = |V(G)|$. Its size bound and Lemma F.2 give an admissible dyadic $\mathsf{N} \geq \max\{N, n\}$ with $\mathsf{N} = N^{O(1)}$: start the schedule search at the least dyadic size at least $\max\{N, n\}$, and use its effective cutoff. Enlarge the effective constant A of Theorem 7.5 to a power of two if necessary. For sufficiently large N, choose

$$k = \left\lfloor \frac{\lfloor \log N \rfloor}{2 \log \mathsf{A}} \right\rfloor,$$

Then $k \geq 1$, $k = \Theta(\log N)$, and $k \log \mathsf{A} \leq (\log N)/2$, so $\mathsf{A}^k \leq \sqrt{N}$. Apply Theorem 7.5 and use its square output, with T entries and threshold B_k^{g}. Its encoding bound and minimum side give

$$N^2 \leq \mathsf{N}^2 \leq T \leq \mathsf{N}^{C_0 \mathsf{A}^k}, \qquad \log T \leq C_0 \mathsf{A}^k \log \mathsf{N} = O(\mathsf{A}^k \log N).$$

Taking another logarithm yields

$$\log \log T \leq k \log \mathsf{A} + O(\log \log N).$$

Choose an effective positive c_* with $2c_* \log \mathsf{A} < 1$. The preceding bound and $k = \Theta(\log N)$ imply

$$\frac{2c_* \log\log T}{k} \le 2c_* \log \mathsf{A} + O\left(\frac{\log\log N}{\log N}\right) < 1$$

for all sufficiently large N. Write $a_T = c_* \log\log T$ and let $\widehat{D}$ be the estimator's output. The promises in Theorem 7.5 and the estimator guarantee then give

$$\begin{aligned} &\text{YES:} \quad \widehat{D} \le B_k^{\mathrm{g}} + a_T < B_k^{\mathrm{g}} + k/2, \\ &\text{NO:} \quad \widehat{D} \ge B_k^{\mathrm{g}} + k - a_T > B_k^{\mathrm{g}} + k/2. \end{aligned}$$

Thus comparison with the public binary rational $B_k^{\mathrm{g}} + k/2$, permitted by Definition 2.1, decides satisfiability using Theorem 3.2.

By Theorem 7.5, construction takes at most

$$\mathsf{N}^{C_0 \mathsf{A}^k} = 2^{O(\sqrt{N} \log N)} = 2^{o(N)}$$

time. Any fixed polynomial in T, including the estimator's running time and the comparison, has the same subexponential bound by the displayed upper bound on T. The source reduction is polynomial-time by Theorem 3.2. Finally, $T \ge N^2$ ensures that the estimator's guarantee on every sufficiently large table applies for all sufficiently large N. Exhaustive search handles the fixed finite remainder, including the values for which $k < 1$. We have obtained a deterministic $2^{o(N)}$-time algorithm for sparse SAT, contradicting Definition 2.3. □

8 Strictly alternating protocols

We now apply the sparsification principle to a lookup construction designed for strict alternation. The unrestricted construction's short protocols use multi-bit messages. Simulating them by inserting the missing turns can double the cost and consume the additive gap, so we need a protocol and lower bound adapted to the alternating model.

Start conceptually with a table in which Bob may hold any Boolean array on the lookup positions. A balanced hypergraph colouring gives a short alternating protocol. In a NO instance, the absence of a suitable colouring forces many lookup requests to share Alice's early response strategy, leaving too few remaining response patterns. Uniform arrays give the pattern-probability bound needed for this contradiction, but give an exponential column family. Replacing that family by a small explicit list preserves the pattern bound used in the proof. This yields a polynomial-size table for every fixed gap, using the fixed turn order in place of the repeated interlacing amplifier. The polynomial's degree may grow with the requested gap.

8.1 Starter, output and lookup inputs

We first compare the choices of starter and output; these constant losses are used when transferring the gap to other conventions.

Lemma 8.1. *On nonempty input sets, each cost $C \in \{D_A^{\mathrm{alt}}, D_B^{\mathrm{alt}}, D_{\mathrm{free}}^{\mathrm{alt}}, Y_{\mathrm{free}}\}$ satisfies*

$$|C(M) - D_A^{\mathrm{alt}}(M)| \le 2.$$

Proof. Prepending one fixed bit changes the starter, so $|D_A^{\mathrm{alt}} - D_B^{\mathrm{alt}}| \le 1$ and

$$D_A^{\mathrm{alt}} - 1 \le D_{\mathrm{free}}^{\mathrm{alt}} \le D_A^{\mathrm{alt}}.$$

Public output also gives receiver output. Conversely, at a nonzero-depth terminal transcript the receiver is the next scheduled speaker and can announce the answer in one bit. Public stopping tells both players when to make this announcement. At depth zero, choose the designated output player as the starter and let that player announce the answer. Thus

$$Y_{\mathrm{free}} \le D_{\mathrm{free}}^{\mathrm{alt}} \le Y_{\mathrm{free}} + 1,$$

and the claimed two-bit bound follows. These transformations preserve public stopping and use an input-independent starter. □

Next we identify the source promise used by the lookup construction. It supplies a small colouring for the intended protocol and forbids the larger colouring that a short competing protocol would induce. We use the balanced hypergraph conversion of Guruswami and Lee [7], including distinct vertices within each edge, recalled in Section G.

Proposition 8.2 (Balanced hypergraph source)**.** *Fix an integer $t \ge 4$, and set*

$$s_{\mathrm{a}} = 2^t, \qquad K_{\mathrm{a}} = s_{\mathrm{a}}^2, \qquad S_{\mathrm{a}} = 2^{s_{\mathrm{a}}}.$$

There is a deterministic polynomial-time reduction from 3-SAT to K_{a}-uniform hypergraphs with nonempty vertex sets and distinct vertices in every edge such that:

1. *a satisfiable formula has an s_{a}-colouring with at least $s_{\mathrm{a}} - 1$ vertices of every colour in every edge;*
2. *an unsatisfiable formula has no proper S_{a}-colouring.*

The polynomial may depend on t.

The table turns this source promise into a lookup problem. Alice's vertex and one stored address select a bit in Bob's array. Bob specifies which address to use and an ordered edge. The address supplies the first array coordinate, and the vertex's position in the edge supplies the second. If the vertex is outside the selected edge, or the edge index is unused, the answer is zero.

Definition 8.3. Let $\mathcal{H} = (V, E)$ be a source hypergraph with an ordering of the vertices in each edge. Set

$$q_{\mathrm{a}} = 2^{t-1} = s_{\mathrm{a}}/2, \qquad H_{\mathrm{a}} = 2^b,$$

where H_{a} is the least power of two at least $\max\{1, |E|\}$. Index the edges by distinct $\alpha \in [H_{\mathrm{a}}]$, and let $\iota_\alpha(v)$ be the position of v in that edge, or zero if v is absent or the index is unused. Use every address tuple once in

$$\mathcal{C} = [H_{\mathrm{a}}]^{q_{\mathrm{a}}}.$$

For any labelled Boolean array list $\mathcal{W} = (w^\ell)$ on

$$\Omega_{\mathrm{a}} = [H_{\mathrm{a}}] \times [K_{\mathrm{a}}],$$

define

$$M_{\mathrm{a}}((v, c), (g, \alpha, \ell)) := \begin{cases} w^\ell(c_g, \iota_\alpha(v)), & \iota_\alpha(v) > 0, \\ 0, & \iota_\alpha(v) = 0. \end{cases}$$

Rows are indexed by a vertex and an address tuple. Column labels lie in $[q_{\mathrm{a}}] \times [H_{\mathrm{a}}] \times [|\mathcal{W}|]$. Equal arrays retain their separate labels.

The two selections in Figure 6 determine the row and column of Bob's array separately: g selects an address from Alice's tuple, and her vertex determines its position in Bob's edge. Their intersection is the requested bit. A vertex outside the edge needs no array lookup.

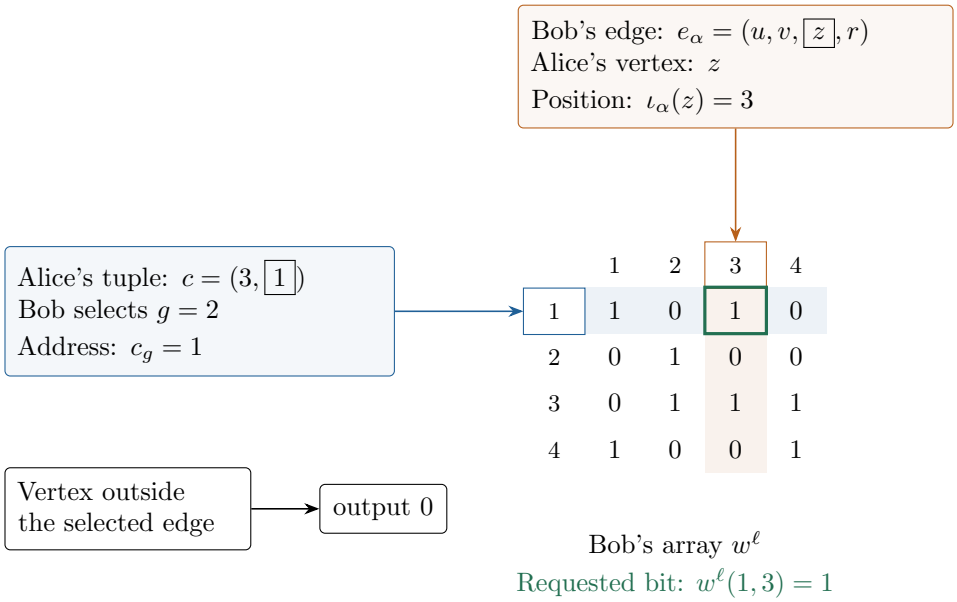


Figure 6: The address selects a row and the vertex's edge position selects a column. Their intersection gives $w^\ell(1,3) = 1$; a vertex outside the edge bypasses the array and gives zero. The small indices illustrate the entry rule independently of the source parameters; the arrows describe the lookup, not a protocol's messages.

8.2 The protocol supplied by a colouring

We next use a balanced colouring to build the short protocol. It shortens the position message after Alice has learnt the edge. The intended schedule exchanges colour and coordinate, then edge and address, then the position within the colour, and its public cost is

$$B_{\mathrm{a}} := (2t-1) + 2b + 2(t+1) + 1 = 2b + 4t + 2.$$

Proposition 8.4. *For the table of Definition 8.3, an s_{a}-colouring with at least $s_{\mathrm{a}} - 1$ vertices of every colour in every edge gives*

$$D_A^{\mathrm{alt}}(M_{\mathrm{a}}) \le B_{\mathrm{a}}.$$

Proof. Fix a row (v, c) and a column (g, α, ℓ). With a balanced colouring χ, an edge contains at least $s_{\mathrm{a}} - 1$ vertices of each of the other $s_{\mathrm{a}} - 1$ colours. Consequently the vertices whose colour is $\chi(v)$ occupy at most

$$K_{\mathrm{a}} - (s_{\mathrm{a}} - 1)^2 = 2s_{\mathrm{a}} - 1$$

positions. Add one message code for "outside the edge." There are then at most $2s_{\mathrm{a}} = 2^{t+1}$ slots, encoded in $t+1$ bits. Fix their ordering publicly for every colour and edge index; an unused edge index has only the outside slot. Alice determines her slot from v, α and $\chi(v)$.

First Alice sends the t bits of $\chi(v)$, while Bob sends the $t-1$ bits of g between consecutive Alice bits. This uses $2t-1$ bits and leaves Alice knowing g. Next Bob sends the b bits of α, while Alice sends the b bits of c_g; this uses $2b$ bits, and Alice knows α before she sends the slot. Finally Bob

sends $t+1$ dummy bits and Alice sends the $t+1$ slot bits, after which Bob sends one answer bit. He answers zero on the outside slot and otherwise reads the specified array entry. The cost is

$$(2t-1)+2b+2(t+1)+1 = B_{\mathrm{a}}.$$

□

8.3 Few response patterns cannot cover many requests

Bob's early messages may depend on the whole array, so one realised transcript does not describe Alice's behaviour across columns. To analyse a competing protocol, we therefore track Alice's responses on hypothetical Bob histories. For a fixed row, a complete early response strategy specifies Alice's rth bit after every possible history of $r-1$ Bob bits, including hypothetical histories; her own preceding bits are then determined. Shorter branches are completed with fixed alternating bits. For example, her first two turns require three response bits: the initial bit and one response to each possible first Bob bit. In general, her first t turns require $\sum_{r=1}^{t} 2^{r-1} = 2^t - 1$ response bits, giving at most 2^{2^t-1} complete early strategies.

A uniformly random Boolean array matches a pattern on k distinct positions with probability 2^{-k}. We need only the weaker bound $2^{-k}+\varepsilon$, which permits a small explicit list. More precisely, for a labelled list of Boolean arrays $\mathcal{W} = (w^\ell)$ on a finite position set Ω, we ask that its probability law ν satisfy

$$\Pr_{\ell\sim\nu}[w^\ell|_I = z] \le 2^{-|I|} + \varepsilon \qquad (I \subseteq \Omega,\ z \in \{0,1\}^I). \tag{6}$$

A union bound controls any family of patterns. All probabilities below use this original law, including after we restrict the columns.

The lookup proof keeps a surplus of requests through a prefix of the protocol. The remaining response tables then describe too few patterns to cover all retained arrays.

Observation 8.5. *Two remaining Bob–Alice pairs permit at most eight output patterns. On eight distinct array positions, these patterns have original-law mass at most* $8(2^{-8}+\varepsilon) \le 1/16$ *when* $\varepsilon \le 2^{-8}$.

The next lemma uses m for the total number of Alice turns, and t for the number whose complete response strategies agree across the rows.

Lemma 8.6 (Alternating lookup)**.** *Let $m > t \ge 1$ be integers with $m - t \ge \lceil \log m \rceil$. Suppose the array law ν satisfies Equation* (6) *with $\varepsilon \le 2^{-4m}$. If $J \subseteq \Omega$ has at least 2^{m-t+1} positions, then the lookup table*

$$L_J(j,\ell) := w^\ell(j)$$

has no Alice-first alternating protocol of depth at most $2m-1$ whose rows share one complete response strategy for their first t turns.

Proof. Suppose such a protocol exists. Pad every shorter branch to depth $2m-1$ with fixed alternating bits. At the final Alice node, her row and the transcript determine the old output leaf, so replace her last bit by that output. This uses no extra bit and, since $m > t$, preserves her first t responses.

Because Alice's first t responses are common to all rows after every Bob history, the first $2t-1$ bits partition only the columns, into at most 2^{t-1} classes. Retain a class of mass at least $2^{-(t-1)}$, keeping every row.

Put

$$h = \lceil \log m \rceil, \qquad Q = 2^h, \qquad i = m - t - h \geq 0.$$

For the next i Bob–Alice pairs, retain a Bob child with at least half the current column mass and an Alice child with at least half the current rows. The resulting rectangle $X \times Y$ satisfies

$$|X| \geq 2^{m-t+1-i} = 2Q, \qquad \nu(Y) \geq 2^{-(m-h-1)}.$$

Exactly h Bob–Alice pairs remain. Bob's complete suffix response table has one entry for every hypothetical history of earlier Alice bits, hence $1 + 2 + \cdots + 2^{h-1} = Q - 1$ entries. Fixing that table and Alice's maps determines her final output on every retained row. On any $2Q$ of those rows, there are therefore at most 2^{Q-1} output patterns.

Since $m \leq Q < 2m$, we have $\varepsilon \leq 2^{-2Q}$. Correctness, Equation (6), and a union bound now give

$$\nu(Y) \leq 2^{Q-1}(2^{-2Q} + \varepsilon) \leq 2^{-Q}.$$

This contradicts the retained-mass bound because $Q \geq m > m - h - 1$. □

The pattern bound holds for every set of positions, so the same array list works for every proposed protocol. The law conditioned on a transcript need not satisfy that bound. The finite-field construction in Section H supplies the list we need.

Corollary 8.7. *For every fixed $t \geq 4$, a nonempty labelled array list on Ω_{a} whose uniform law satisfies Equation* (6) *with*

$$\epsilon_{\mathrm{a}} := 2^{-4(b+3t)}$$

can be generated in deterministic time polynomial in H_{a}.

8.4 From a hypergraph to an explicit gap

We colour each vertex by a common early strategy used by many of its address tuples. A monochromatic source edge then supplies enough distinct requests to apply the lookup lower bound.

Proposition 8.8. *Use the complete address list and the explicit array list of Corollary 8.7 in M_{a}. A balanced source colouring gives*

$$D_A^{\mathrm{alt}}(M_{\mathrm{a}}) \leq B_{\mathrm{a}},$$

whereas the absence of a proper S_{a}-colouring gives

$$D_A^{\mathrm{alt}}(M_{\mathrm{a}}) \geq B_{\mathrm{a}} + 2t - 8.$$

For each fixed t, the full table can be printed in deterministic time polynomial in $|V| + H_{\mathrm{a}}$.

Proof. The upper bound is Proposition 8.4. For the lower bound, put $m = b + 3t - 3$, so $2m - 1 = B_{\mathrm{a}} + 2t - 9$, and suppose that an Alice-first protocol has depth at most $2m - 1$. Pad it to that depth. For each vertex v, partition its address tuples by Alice's complete response strategy on her first t turns. There are at most $2^{2^t - 1} \leq S_{\mathrm{a}}$ strategies by the complete response-table count above. A largest class C_v therefore has

$$|C_v| \geq H_{\mathrm{a}}^{q_{\mathrm{a}}} / S_{\mathrm{a}}.$$

Colour v by this class's strategy. The source promise gives an edge e_α whose vertices all receive the same strategy.

For $v \in e_\alpha$ and $g \in [q_\mathrm{a}]$, let $J_{v,g} = \{c_g : c \in C_v\}$ be the set of addresses occurring in coordinate g. Since C_v is contained in the product of these projections,

$$\prod_{g=1}^{q_\mathrm{a}} |J_{v,g}| \geq |C_v| \geq H_\mathrm{a}^{q_\mathrm{a}}/S_\mathrm{a}.$$

AM–GM, summed over the vertices of e_α, gives a coordinate g with

$$\begin{aligned} N := \sum_{v \in e_\alpha} |J_{v,g}| &\geq K_\mathrm{a} H_\mathrm{a} S_\mathrm{a}^{-1/q_\mathrm{a}} \\ &= \frac{K_\mathrm{a} H_\mathrm{a}}{4} = 2^{m-t+1}, \end{aligned}$$

where $S_\mathrm{a} = 4^{q_\mathrm{a}}$. For each $v \in e_\alpha$ and $u \in J_{v,g}$, choose a tuple $c(v,u) \in C_v$ with $c(v,u)_g = u$. Retain the corresponding rows and all columns (g, α, ℓ). Their entries are

$$M_\mathrm{a}((v, c(v,u)), (g, \alpha, \ell)) = w^\ell(u, \iota_\alpha(v)).$$

These are N distinct lookup positions: different vertices occupy different positions in the edge, and different addresses give different first coordinates. All retained rows share the chosen early strategy. To apply Lemma 8.6, note that $m - t \geq t + 1 \geq 5$ and $m \leq 2(m-t) - 1 \leq 2^{m-t}$, so $m - t \geq \lceil \log m \rceil$; also $\epsilon_\mathrm{a} \leq 2^{-4m}$. The resulting lookup restriction contradicts that lemma. Integer depths give $D_A^{\mathrm{alt}}(M_\mathrm{a}) \geq 2m = B_\mathrm{a} + 2t - 8$.

There are $|V| H_\mathrm{a}^{q_\mathrm{a}}$ rows and $q_\mathrm{a} H_\mathrm{a} |\mathcal{W}|$ columns. For fixed t, the number q_a is constant and Corollary 8.7 constructs the arrays in polynomial time. Enumerating all tuples and evaluating the displayed entry rule therefore prints the entire table in polynomial time. □

Proof of Theorem 1.4 and Corollary 1.6. Given a fixed requested gap $k \geq 1$, choose $t = \lceil k/2 \rceil + 4$. Then $2t - 8 = 2\lceil k/2 \rceil \geq k$. Compose the source reduction with Proposition 8.8. It gives, in deterministic polynomial time, a table with YES cost at most B_a and NO cost at least $B_\mathrm{a} + k$, proving Theorem 1.4.

An additive-u estimate of any cost in Lemma 8.1 is, by the triangle inequality, an additive-$(u+2)$ estimate of D_A^{alt} with the same success probability. Choose any fixed integer $k > 2(u+2)$ and apply Observation 1.5 to the reduction just proved. This gives $\mathsf{P} = \mathsf{NP}$ for a deterministic estimator and $\mathsf{NP} \subseteq \mathsf{BPP}$ for a bounded-error randomised estimator, proving Corollary 1.6. □

The address length $q_\mathrm{a} = 2^{t-1}$ and the source reduction's degree may both grow with t. Thus the construction is polynomial for each fixed gap; this argument supplies no growing-error theorem for strict alternation.

9 Remaining approximation questions

The program has moved from controlling communication in interlaced problems to encoding exact hardness, amplifying a gap, and preserving it in small explicit tables. This yields fixed-error NP-hardness in both models and quantitative growing-error hardness for unrestricted protocols. The remaining limitation is how the explicit table size grows with the requested gap. Improving that dependence would give stronger approximation lower bounds.

Our deterministic construction has the upper bound $N^{2^{O(k)}}$ on its size. Can smaller lists preserve the restriction bounds needed for the next round, or can another construction improve the $\log \log T$ error scale?

For strict alternation, the construction is polynomial for each fixed gap, but its polynomial may depend on that gap. A construction with better control of this dependence could give growing-error hardness in Yao's model as well.

A Interlacing estimates and the lookup seed

We prove the restriction profiles and forced Alice prefix needed in Lemma 5.6 and Theorem 5.8 using the estimates below.

A.1 Projection and probability mass

Fix a Boolean matrix $M : X \times Y \to \{0,1\}$ with positive input laws μ and ν. Every mass below is measured under these original laws or their products. For a nonempty $K \subseteq Y^p$, define its logarithmic mass loss by $\Delta_p(K) = \log(1/\nu^p(K))$. The empty-coordinate projection has loss zero. For $p \geq 1$ and $0 < x, z \leq 1$, let $\mathcal{B}(M; p, x, z)$ contain the restrictions of $\mathsf{I}_p(M)$ retaining base row mass at least x in each block and product column mass at least z, and put

$$\mathcal{D}_M(p; x, z) := \min_{G \in \mathcal{B}(M;p,x,z)} \mathrm{D}(G).$$

Lemma A.1. *Let $K \subseteq Y^p$ be nonempty, and write $\pi_J K$ for its projection onto the coordinates in $J \subseteq [p]$. Then*

$$\Delta_{|J|}(\pi_J K) \leq \Delta_p(K),$$
$$\Delta_{|J_0|}(\pi_{J_0} K) + \Delta_{|J_1|}(\pi_{J_1} K) \leq \Delta_p(K) \qquad (J_0 \sqcup J_1 = [p]).$$

For every integer $1 \leq l \leq p$, some l-element set J satisfies

$$\Delta_l(\pi_J K) \leq \frac{l}{p} \Delta_p(K).$$

Each projected matrix, with arbitrary retained rows in its selected blocks, is isomorphic as a Boolean matrix to a restriction of the original matrix. Consequently, for admissible $p' \leq p$, $x' \leq x$, and $z' \leq z$,

$$\mathcal{D}_M(p'; x', z') \leq \mathcal{D}_M(p; x, z).$$

Proof. The first two inequalities follow from $K \subseteq \pi_J^{-1}(\pi_J K)$ and $K \subseteq \pi_{J_0} K \times \pi_{J_1} K$, using the product law.

For the averaged bound, let Z have law $Q = \nu^p(\cdot \mid K)$. For probability laws P, V on a finite set, their relative entropy is $D_{\mathrm{KL}}(P\|V) = \sum_z P(z) \log(P(z)/V(z))$, and entropy is $\mathsf{H}(P) = -\sum_z P(z) \log P(z)$, with zero terms omitted. Conditioning on K gives

$$D_{\mathrm{KL}}(Q\|\nu^p) = \Delta_p(K).$$

Since Q_J is supported on $\pi_J K$, nonnegativity of relative entropy gives

$$\Delta_{|J|}(\pi_J K) \leq D_{\mathrm{KL}}(Q_J\|\nu^J).$$

For a uniformly chosen l-subset J, the entropy chain rule gives

$$\mathbb{E}_J \mathsf{H}(Z_J) \geq \mathbb{E}_J \sum_{i \in J} \mathsf{H}(Z_i \mid Z_1, \ldots, Z_{i-1}) = \frac{l}{p} \mathsf{H}(Z).$$

Also each coordinate contributes to $-\mathbb{E}\log\nu^J(Z_J)$ with probability l/p. Subtracting the entropy inequality therefore yields

$$\mathbb{E}_J\Delta_l(\pi_J K) \le \mathbb{E}_J D_{\mathrm{KL}}(Q_J\|\nu^J) \le \frac{l}{p} D_{\mathrm{KL}}(Q\|\nu^p) = \frac{l}{p}\Delta_p(K).$$

Some J satisfies the desired bound. The conditioned law Q need not be a product; all projection masses above use the original product law.

For each tuple in $\pi_J K$, choose one tuple in K above it. The entries on the selected row blocks depend only on the selected coordinates, so these representatives realise the projected Boolean matrix (Observation 2.5). Their original weights need not be preserved: the projection estimate uses ν^J, while representatives certify the communication lower bound. Selecting $p' \le p$ blocks leaves projected mass at least z and base row mass at least x, proving the final monotonicity. □

A.2 Two or three surviving blocks

The first-message arguments needed below use only the two thresholds in the restriction profile. We state their three small cases together.

Lemma A.2 (Small block bounds)**.** *Suppose $\mathcal{P}(M; x, H)$ with $H \ge 2$. The following restrictions of a full-product interlace have the indicated lower bounds, whenever the displayed row threshold is at most one. Row mass is measured separately in each block.*

blocks	*row mass per block*	*product column mass*	*cost*
2	$2x$	$9/50$	H
2	$2x$	$9/25$	$H+1$
3	$4x$	$9/50$	$H+1$

Proof. For the first row, Lemma A.1 gives a singleton column projection of mass at least $\sqrt{9/50} > 3/10$. Its profile cost is at least $H - 1 \ge 1$, so the restriction is nonconstant. If Alice speaks first, follow this block into a child retaining base row mass at least x; the suffix costs at least $H - 1$. If Bob speaks first, his denser child has column mass at least $9/100$, and hence a singleton projection of mass at least $3/10$. Its suffix again costs at least $H - 1$. Including the first bit gives H.

For the second row, the first bound already gives nonconstancy. If Alice speaks, a singleton projection has column mass at least $3/5$; a child retaining half this block's rows costs at least H. If Bob speaks, his denser child has column mass at least $9/50$, so the first bound gives the same suffix cost. Including the first bit gives $H + 1$.

For the third row, projecting onto any two blocks and applying the first bound excludes a constant restriction. If Alice speaks, assign each block to a child retaining half its rows. One child receives two blocks of mass at least $2x$, and the first bound gives suffix cost H. If Bob speaks, his denser child retains column mass at least $9/100$. The averaged projection inequality gives two of its three coordinates with product mass at least

$$(9/100)^{2/3} > 9/50.$$

Their row masses are still at least $4x \ge 2x$, so the first bound again gives suffix cost H. Including the first bit proves the claim. All projected restrictions are realised by surviving representatives as in Lemma A.1; their masses use the original product laws. □

Proof of Lemma 5.4. A full product is exactly eight-wise independent. Apply Lemma 5.10 with $a = \log u$: the hypotheses are precisely $u/2 + 1$ retained blocks, base row mass at least ux, and tuple mass at least $9/25$. This includes $u = 2$ and gives $H + \log u$ bits. □

At the exact base-cost budget, an early Bob query leaves too many surviving copies for the remaining depth. The next theorem formalises this forced prefix and identifies the dominant block in each resulting rectangle. It applies both to full products and to exact tuple lists. Its only additional input is the exact block bound Lemma 5.10, whose proof uses Lemma A.2 and the pair-projection estimate, independently of this theorem.

Theorem A.3. *Suppose $\mathcal{P}(M;\rho,d)$ with $d \geq 2$. Let $u \geq 2$ be dyadic and $1 \geq \xi > 2u\rho$. Interlace M at width u using an exactly eight-wise independent tuple list with the base-column marginal, as in Definition 2.9. Every restriction retaining base row mass at least ξ in every block and tuple-label mass at least $18/25$, with a protocol of depth at most $d + \log u$, has a complete prefix of $\log u$ Alice queries. Its u final parts are in bijection with the blocks: each part retains more than $\xi - 2(u-1)\rho$ of its assigned block and less than 2ρ of every other block. Full products are a special case, and fixed queries count as queries.*

Proof. At depth $r < \log u$, maintain $\kappa = u2^{-r}$ assigned blocks at each node, each retaining mass at least $\xi 2^{-r} > 2\kappa\rho$, with the original retained column set unchanged. The root has this property. A singleton projection has base-column mass at least $18/25 > 3/5$, by the exact marginal. One retained label per projected value realises that base restriction, so the profile excludes a leaf. The inequality $\xi > 2u\rho$ also ensures that every row threshold used below is at most one.

If Bob speaks, one child retains column mass at least $9/25$ and all κ assigned blocks. By Lemma 5.10 with $a = \log\kappa$, it needs $d + \log\kappa$ bits, but has only $d + \log u - r - 1 = d + \log\kappa - 1$ left. Thus Alice speaks. Assign each block to a child retaining at least half its mass. If a child received more than $\kappa/2$ blocks, it would contain at least $\kappa/2 + 1$ blocks of mass $\xi 2^{-r-1} > \kappa\rho$, and the same lemma would again exceed its budget. Both children therefore receive exactly $\kappa/2$ blocks and remain live, proving the invariant and the complete prefix. In particular, a fixed Alice query is impossible, since it would assign every block to one child.

At depth $\log u$, each part has one assigned block of mass greater than 2ρ. Two blocks of mass at least 2ρ would require $d+1$ bits by Lemma 5.10 with $a = 1$, whereas only d remain. Every other block thus has mass less than 2ρ. Each original block is assigned to exactly one part; its mass elsewhere totals less than $2(u-1)\rho$, proving the final bound. All column masses in this argument are measured in the original tuple law; no independence after restriction is asserted. □

A.3 Absorbing column loss

For a restriction G of $\mathsf{I}_p(M)$ with surviving columns K, $\Delta_p(K) = \log(1/\nu^p(K))$ records the loss created by several Bob messages. The following induction charges this loss to surviving copies and uses only lower bounds on the base matrix; exact costs enter later through a matching selector protocol.

Lemma A.4. *Let $C \geq 3$, $0 < \theta \leq 1/4$, and $0 < \delta \leq 1/64$. Suppose*

$$\mathcal{D}_M(1;\theta,1-\delta) \geq C, \qquad \mathcal{P}(M;\theta,C-1).$$

Let $e \in \{0,1\}$, let $a \geq 2e$ be an integer with $2^a\theta \leq 1$, and let $p > 2^{a-1}$ be an integer. If a restriction G of $\mathsf{I}_p(M)$ retains base row mass at least $2^a\theta$ in every block and its nonempty column set K satisfies

$$\Delta_p(K) \leq (p - 2^{a-1})\delta + \frac{4e}{3},$$

then $\mathrm{D}(G) \geq C + a - e$. Here C is only a lower bound; equality $\mathrm{D}(M) = C$ is not assumed.

Proof. We prove the lower bound by simultaneous induction on

$$\mathfrak{m}(a, e) := 2a + 1 - e.$$

The induction tracks the number of surviving blocks and the logarithmic column-density loss that they can absorb. Alice's bit halves row thresholds; Bob's bit increases the defect by at most one, after which Lemma A.1 supplies a suitable projection.

The profile $\mathcal{P}(M; \theta, C-1)$ and $C - 1 \geq 2$ allow Lemma 5.10 on full products. Its cases $a = 1, 2$ give

$$\mathcal{D}_M(2; 2\theta, 9/25) \geq C, \qquad \mathcal{D}_M(3; 4\theta, 9/25) \geq C + 1.$$

The numerical margin used to invoke these bounds is

$$3\delta + \frac{4}{3} \leq \frac{265}{192} < \frac{7}{5} < -\log(9/25).$$

The last inequality follows from $2^7 \cdot 9^5 < 25^5$.

The initial induction states are as follows. For $(a, e) = (0, 0)$, the averaged projection bound in Lemma A.1 gives a singleton $J \subseteq [p]$ with

$$\Delta_1(\pi_J K) \leq \frac{p - 1/2}{p}\delta < \delta.$$

Since $2^{-v} \geq 1 - v$ for $0 \leq v \leq 1$, the projected column density is greater than $1 - \delta$, and the near-full hypothesis gives cost at least C.

For $(a, e) = (2, 1)$, the defect allowance and Lemma A.1 give a three-coordinate projection with defect at most

$$\frac{3}{p}\left((p-2)\delta + \frac{4}{3}\right) \leq 3\delta + \frac{4}{3} < -\log(9/25).$$

Its column mass exceeds $9/25$, and each block retains mass at least 4θ. The three-copy bound above gives cost at least $C + 1 = C + 2 - 1$. There is one further base case, $(a, e, p) = (3, 1, 5)$. Its defect is at most $\delta + 4/3 < -\log(9/25)$, and its rows have mass at least 8θ. Apply Lemma 5.10 with $a = 3$ to these five blocks, giving $C + 2$ bits.

For $(a, e) = (1, 0)$, a singleton projection has defect at most $(p-1)\delta/p < \delta$, so its column density is greater than $1 - \delta$; the near-full hypothesis makes this restriction nonconstant. After a Bob bit, the denser child has a two-coordinate projection of defect at most

$$\frac{2}{p}\left((p-1)\delta + 1\right) \leq 2\delta + 1 < 3\delta + \frac{4}{3},$$

so the two-copy bound gives cost at least C after that bit. Every other valid state has $a \geq 2$, $p \geq 3$, and base row mass at least 4θ; its three-coordinate projection has defect at most $3\delta + 4/3$, so the three-copy bound makes the restriction nonconstant before the first-bit analysis.

Consider such a non-base state and a protocol node for the restriction. If Alice speaks, assign every block to a child retaining at least half its row mass. For child $i \in \{0, 1\}$, let p_i be the number of assigned blocks and let L_i be the defect after projecting the retained column set onto those blocks. Then $p_0 + p_1 = p$, and the partition inequality in Lemma A.1 gives $L_0 + L_1 \leq \Delta_p(K)$. Put

$$s_i = \max\{p_i - 2^{a-2}, 0\}.$$

Since $s_0 + s_1 \geq p - 2^{a-1} > 0$, at least one s_i is positive. The allowances of the children with $s_i > 0$ satisfy

$$\sum_{i:s_i>0} \left(s_i\delta + \frac{4e}{3} \right) \geq (p - 2^{a-1})\delta + \frac{4e}{3} \geq \sum_{i:s_i>0} L_i.$$

At least one eligible child contributes the constant $4e/3$; the final inequality uses the assumed defect bound on K and $L_0 + L_1 \leq \Delta_p(K)$. Consequently some child with $s_i > 0$ meets its allowance. Each assigned block retains mass at least $2^{a-1}\theta$, and there are $p_i > 2^{a-2}$ such blocks. This is state $(a-1, e)$; the exceptional state $(a, e) = (2, 1)$ was already proved. The induction hypothesis gives cost at least $C + a - 1 - e$ in that child, and Alice's bit gives $C + a - e$. The same argument includes the transition $(1, 0) \to (0, 0)$.

If Bob speaks, one child has at least half the current column mass, so its defect is at most $\Delta_p(K) + 1$. When $e = 0$ and $a \geq 2$, this child satisfies state $(a, 1)$, because

$$\Delta_p(K) + 1 \leq (p - 2^{a-1})\delta + \frac{4}{3}.$$

Its induction cost is at least $C + a - 1$, and adding Bob's bit proves the claim.

It remains to handle $e = 1$ and $a \geq 3$. The five-block base case already handles $p = 5$, so $p \geq 6$. Put $p' = \lceil p/2 \rceil$. Then

$$\frac{1}{2} \leq \frac{p'}{p} \leq \frac{4}{7}, \qquad \frac{p'}{p}(p - 2^{a-1}) \leq p' - 2^{a-2}, \qquad \frac{4}{7}\left(\frac{4}{3} + 1\right) = \frac{4}{3}.$$

By the averaged projection bound in Lemma A.1, some p' coordinates of the denser child therefore have defect at most

$$\frac{p'}{p}\left((p - 2^{a-1})\delta + \frac{4}{3} + 1\right) \leq (p' - 2^{a-2})\delta + \frac{4}{3}.$$

The projected child retains base row mass at least $2^{a-1}\theta$ in each of its $p' > 2^{a-2}$ blocks, so it is state $(a-1, 1)$. Its induction cost is at least $C + a - 2$; adding Bob's bit gives $C + a - 1$, as required. □

Proof of Lemma 5.5. Let G be the given restriction. Project to $p = 9u/16$ retained blocks and apply Lemma A.4 with

$$p = \frac{9u}{16}, \qquad a = \log u, \qquad \delta = \frac{1}{64}, \qquad \theta = x, \qquad C = H, \qquad e = 0.$$

Since $u \geq 16$ and $ux \leq 1$, we have $x \leq 1/16 \leq 1/4$, $2^a\theta = ux \leq 1$, and $p = 9u/16 > u/2 = 2^{a-1}$. The profile $\mathcal{P}(M; x, H)$ implies $\mathcal{P}(M; x, H-1)$, and its $j = 0$ case gives $\mathcal{D}_M(1; x, 1-\delta) \geq H$ because $1 - \delta = 63/64 \geq 3/5$. Thus all hypotheses of Lemma A.4 are available.

The retained column density $2^{-u/1024}$ gives defect at most $u/1024$. For the chosen parameters,

$$(p - 2^{a-1})\delta = \left(\frac{9u}{16} - \frac{u}{2}\right)\frac{1}{64} = \frac{u}{1024}.$$

If more than $9u/16$ blocks are retained, select exactly p of them. By Lemma A.1, projecting the retained column set to these blocks does not increase its defect, and surviving representatives realise an actual restriction. Applying Lemma A.4 to this projected restriction gives

$$\mathrm{D}(G) \geq C + a = H + \log u.$$

The original restriction contains this projected restriction, so its cost is at least the same by the monotonicity conclusion of Lemma A.1. □

Proof of Theorem 5.17. Put $C = \mathrm{D}(W)$ and $f = \lceil b/\delta \rceil$. The assumptions imply

$$f\delta \geq b, \qquad (f-1)/Q < 1/80, \qquad f < Q/40, \qquad Q \geq 5120.$$

Indeed, $b/\delta \geq 64$, $Q\delta \geq 80b$, and $f \leq b/\delta + 1 < 2b/\delta$. Fix $j \in \{0,1\}$ and put

$$a_j = r - j, \qquad p_j = 2^{a_j - 1} + f, \qquad t_j = 2^{a_j}\theta.$$

These choices give $a_j \geq 0$, $2^{a_j-1} < p_j < Q$, and $t_j < 1$.

Transpose a profile restriction of R^+ to $\mathsf{I}_Q(W)$. It retains tuple mass at least 2^{-b} and total block-row mass at least $(3/5)2^{-j}$. If fewer than p_j blocks had base row mass at least t_j, total row mass would be less than

$$\frac{p_j - 1}{Q} + t_j < 2^{-j-1} + \frac{1}{80} + \frac{1}{20 \cdot 2^j} \leq \frac{3}{5}\,2^{-j},$$

a contradiction; the last inequality uses $j \leq 1$. Select p_j qualifying blocks J. For the retained tuple set K, Lemma A.1 gives

$$\Delta_{p_j}(\pi_J K) \leq \Delta_Q(K) \leq b \leq f\delta = (p_j - 2^{a_j-1})\delta.$$

The projected restriction therefore meets Lemma A.4 with $e = 0$: the near-full and weaker profile hypotheses are those for W, $\theta < 1/4$ follows from $Q\theta < 1/20$, and the row threshold is t_j. It requires $C + a_j = C + r - j$ bits. Representatives realise it as a Boolean restriction of the original matrix, so transposing back proves $\mathcal{P}(R^+; 2^{-b}, C + r)$.

The selector protocol costs $r + \mathrm{D}(W)$ bits, giving equality on the whole matrix and the claimed transfer of any upper bound. By definition the transpose has n^Q rows and Qm columns. □

A.4 The initial lookup matrix and template layers

The lookup seed starts from $R_0(z, j) = z_j$ on $\{0,1\}^{p_0} \times [p_0]$, and rank after deleting query coordinates supplies the restriction profile needed for amplification. Its proof forces leaves of each output value and then uses the elementary query-and-answer protocol for equality.

Proof of Lemma 3.4. We first lower-bound the rank of every retained lookup submatrix, then count leaves for the two output values, and finally record the exact size and upper bounds of the seed.

Write $a = \log p_0$. Fix $j \in \{0,1\}$, a retained row set $V \subseteq \{0,1\}^{p_0}$ with $|V| \geq 2^{p_0 - L}$, and a retained query set $J \subseteq [p_0]$ with $|J| \geq \lceil (3/5)2^{-j}p_0 \rceil$. Let v be the dimension of the real span of V. A v-dimensional subspace has v coordinate functionals whose restrictions form a basis, so projection to those v coordinates is injective on the subspace. Since the Boolean cube has at most 2^v points in those coordinates, V has at most 2^v points. The lower bound on $|V|$ therefore implies $v \geq p_0 - L$.

Deleting the $p_0 - |J|$ coordinates outside J can lower the row rank by at most $p_0 - |J|$. Thus the restricted lookup matrix has rank r satisfying

$$r \geq |J| - L \geq \frac{3}{5}\,2^{-j}p_0 - L > 2^{a-j-1}.$$

The strict inequality follows from

$$\frac{3}{5}\,2^{-j}p_0 - 2^{a-j-1} = \frac{p_0}{10 \cdot 2^j} \geq \frac{p_0}{40} > L,$$

where $p_0/40 - L > 0$ is part of the choice in the lemma statement.

Let F denote the restricted lookup matrix. Consider a protocol for F. The indicator of each 1-leaf rectangle has rank at most one, and the sum of these indicators is the restricted matrix, so there are at least r one-leaves. Let $\mathbf{1}$ be the all-ones matrix; it has rank one. The complement of the restricted matrix therefore has rank at least $r-1$, by $\operatorname{rank}(\mathbf{1}-F) \geq \operatorname{rank}(F) - \operatorname{rank}(\mathbf{1})$, so there are at least $r-1$ zero-leaves. Hence

$$\#\text{leaves} \geq r + (r-1) = 2r - 1 > 2^{a-j}.$$

A depth-t binary protocol has at most 2^t leaves, so the restriction has depth at least $a+1-j$. At $j=0$, this gives $\mathrm{D}(R_0) \geq a+1$. Conversely, Bob sends the query coordinate in a bits and Alice sends the queried bit, so $\mathrm{D}(R_0) \leq a+1$; hence $\mathrm{D}(R_0) = a+1 = d_0$. The unit-vector rows form the identity submatrix, so $\operatorname{rank}(R_0) = p_0 = 2^{d_0-1}$.

By minimality of the dyadic choice, $p_0 < 128(L+1)$. If $L = \lceil \log n \rceil$ and $n \geq 4$, then $L+1 \leq \log n + 2 \leq 2\log n$, so $p_0 < 256 \log n$ and $2^{p_0} < n^{256}$. Enumerating the 2^{p_0} words and the p_0 query coordinates therefore prints the table deterministically in polynomial time. □

Proof of Lemma 5.6. Apply the block-averaging observation to the three interlaces in order. Fix $j \in \{0,1\}$, put $\beta_j = (3/5)2^{-j}$, and write $\lambda = 255/256$ for the template mass. After transposition, the relevant data are

target	base	q	u	average block retention
A	R	$q_1 = s-2$	$s2^{-j}$	β_j
R_{temp}	A	q_2	$q_2 2^{-j}$	β_j
N_{ref}	R_{temp}	s	$s2^{-j}$	$(\beta_j - (1-\lambda))/\lambda$

The retained tuple mass is at least x in each case. For N_{ref}, discard the non-template rows after transposition and project each reference label onto its J_2^s selector tuple, forgetting its tail. The displayed average retention follows by subtracting non-template mass; projection keeps mass at least x and representatives preserve the template entries.

Count blocks at thresholds $u\rho$, ux, and ux, respectively. The bound $q\beta \leq h + qt$ gives the following lower bounds for their counts divided by u:

$$\frac{3}{5}(1-2/s) - q_1\rho, \qquad \frac{3}{5} - q_2 x, \qquad \frac{2^j(\beta_j - (1-\lambda))}{\lambda} - sx.$$

All three exceed $9/16$ by Lemma F.1. The same schedule gives $u \geq 16$, row thresholds at most one, and $x = 2^{-10L} \geq 2^{-u/1024}$. Applying Lemma 5.5 successively therefore gives costs at least

$$d + \log s - j = H_1 - j, \qquad H_1 + \log q_2 - j = H_2 - j, \qquad H_2 + \log s - j = C - 1 - j.$$

These are the three asserted profiles. The $j=0$ cases, together with the selector protocols of costs $d + \log s$ and $H_1 + \log q_2$, give $\mathrm{D}(A) = H_1$ and $\mathrm{D}(R_{\text{temp}}) = H_2$.

Finally, the first q_1 positions and three distinct tails give $q_1 + 3 = s + 1 = (2s)/2 + 1$ blocks. Each retains base row mass at least $2s\rho$, and the retained word mass exceeds $1/2 > 9/25$. Apply Lemma 5.4 with parameter $2s$ and the profile of R to get $d + \log(2s) = H_1 + 1$ bits. Projected representatives realise the required Boolean restriction by Lemma A.1. □

B Bounds on transferred inputs

We prove the two receipt bounds in Section 5.4 under its common hypotheses. All parameters are written without a construction subscript. Protected inputs belong to the diagonal family, of mass $\mu = 1/256$ per source copy. The template and ordinary families each have mass $\lambda = 255/256$.

B.1 Two selectors at one destination

Two independently chosen selectors force an extra bit even when their inner indices coincide. Only eight-wise independence in their original laws is needed.

Lemma B.1 (Two tagged selectors)**.** *Let $A : \Gamma \times J \to \{0,1\}$ satisfy $\mathcal{P}(A; x, H_1)$ with $H_1 \geq 2$, and let q_2 be dyadic with $2q_2x \leq 1/8$. Take two independent labelled sources S_o, S_f of J^{q_2}, each exactly eight-wise independent with the base-column marginal. Put $H_2 = H_1 + \log q_2$ and give $J_2 = [q_2] \times \Gamma$ its uniform-index times base-row law. Rows are selector pairs $(z_o, z_f) \in S_o \times S_f$. Ordinary and other-copy columns are separately tagged copies of J_2, with entries*

$$A(\gamma, (z_o)_\alpha) \quad \textit{and} \quad A(\gamma, (z_f)_\alpha),$$

respectively. The tags remain distinct even when the inner indices coincide. Retain a pair event of original product mass at least $49/64$, ordinary columns of normalised mass at least $1 - \varepsilon$, and other-copy columns of normalised mass at least δ, where

$$\varepsilon \geq 0, \qquad 0 < \delta \leq 1, \qquad \delta > 2q_2x + \varepsilon.$$

Then the retained matrix requires at least $H_2 + 1$ bits.

Proof. Set $\lambda_0 = 2q_2x$. Let b count ordinary Γ-fibres of retained mass below λ_0, and let f count other-copy fibres of mass at least λ_0. Averaging their masses gives

$$1 - \varepsilon \leq 1 - \frac{b}{q_2}(1 - \lambda_0), \qquad \delta \leq \lambda_0 + \frac{f}{q_2}(1 - \lambda_0).$$

Consequently

$$b \leq \frac{\varepsilon q_2}{1 - \lambda_0}, \qquad f \geq \frac{q_2(\delta - \lambda_0)}{1 - \lambda_0} > b.$$

Thus more than q_2 tagged fibres have mass at least λ_0.

Suppose a protocol has depth at most H_2. Follow each such fibre through column-player queries, choosing a child retaining at least half its current mass. At depth r its mass is at least $\lambda_0 2^{-r}$, while the pair set of rows is unchanged. Every tagged selector coordinate has projection mass at least $49/64 > 3/5$. The profile of A therefore excludes a leaf and also excludes depth $\log q_2 + 1$: there the fibre mass is at least x, so H_1 bits are needed but only $H_1 - 1$ remain. Stop each fibre at its first row-player query.

At a stopping node of depth r put $\kappa = q_2 2^{-r}$. These capacities are positive integers, and Kraft's inequality gives total capacity at most q_2. Some node therefore holds at least $\kappa + 1$ tagged fibres, each of mass at least $2\kappa x$. Its denser row child retains pair mass at least $49/128 > 9/25$.

The two sources together are exactly eight-wise independent on tagged coordinates: a request for at most eight coordinates factors between the independent sources, and within either source uses at most eight coordinates. After transposition, the fibres are distinct blocks of an interlace of A. Keep the surviving pair labels as tuple labels, so their mass is unchanged. Apply Lemma 5.10 with width 2κ: there are at least $(2\kappa)/2 + 1$ blocks of base row mass at least $2\kappa x$, and $2\kappa x \leq 2q_2x \leq 1/8$. The lower bound is

$$H_1 + \log(2\kappa) = H_1 + \log \kappa + 1,$$

whereas the child has only $H_2 - r - 1 = H_1 + \log \kappa - 1$ bits left. This contradiction includes $\kappa = 1$. All masses used here are measured in the original laws; no independence after conditioning is asserted. □

Claim B.2. *Use the common source laws and profiles of Section 5.4, and put* $\lambda_0 = 2q_2x$, $\tau_0 = 2sx$, $C = H_2 + \log s + 1$, *and*

$$a_{\min} = \frac{99/100 + \lambda^2 - 1}{\lambda^2}.$$

Retain ordinary columns of normalised Ω*-mass at least* $1 - \varepsilon$, *and separately tagged other-copy columns whose measure-preserving map to* J_2 *has retained image mass at least* δ. *Renumber bins so that the distinguished bin is one. It consists of a set* $P_* \subseteq S^2$ *of product mass at least* $a_{\min}$; *every other bin retains* S*-mass at least* τ_0. *Write ordinary coordinates as* $Z = (\xi, j_1, \dots, j_s)$, *with* $j_p = (\alpha_p, \gamma_p)$. *Require entries*

	$Z \in \Omega$	*other-copy image* (α, γ)
$(z_o, z_f) \in P_*$	$A(\gamma_1, (z_o)_{\alpha_1})$	$A(\gamma, (z_f)_\alpha)$
$(p, z),\ p \neq 1$	$A(\gamma_p, z_{\alpha_p})$	*arbitrary.*

If $\varepsilon \geq 0$, $0 < \delta \leq 1$, *and* $\delta > \lambda_0 + 4\varepsilon$, *then this matrix requires at least* C *bits.*

Proof. Suppose the restriction has a protocol of depth at most $C - 1$. Track one selector support per bin until the first column-player query. An ordinary support follows a row child retaining at least half its current mass. The distinguished support carries the pair set P_*. At depth r, split it into two ordinary supports if both children have first-selector projection mass at least $\tau_0 2^{-r-1}$; otherwise keep the pair in the qualifying child. A discarded pair set has mass no larger than its first-selector projection, so all unsplit steps together discard less than

$$\sum_{r \geq 0} \tau_0 2^{-r-1} = \tau_0.$$

The common numerical bounds give $\tau_0 \leq 10^{-3}$ and $a_{\min} - \tau_0 > 49/64$. An intact pair therefore retains mass greater than $49/64$, and every support at depth r has selector mass at least $\tau_0 2^{-r}$.

The hypothesis on δ implies $\varepsilon < 1/4$, so the ordinary columns retain normalised mass greater than $3/4$. Their projection onto any bin has at least that mass. The profile $\mathcal{P}(R_{\text{temp}}; x, H_2)$ excludes a support-bearing leaf and depth $\log s + 1$: at that depth a support has mass at least x, but only $H_2 - 1$ bits remain. Hence every support reaches a first column query at depth at most $\log s$.

Give an ordinary support demand one and an intact pair demand two. Following one child preserves demand, and splitting preserves it because $2 = 1 + 1$. If a stopping node v has h_v supports, of which z_v are intact pairs, then

$$\sum_v (h_v + z_v) = s + 1, \qquad \sum_v \kappa_v \leq s, \qquad \kappa_v = s 2^{-r_v}.$$

The second inequality is Kraft's inequality. Some node therefore has $h + z > \kappa$. Its supports have distinct bin coordinates, since descendants in different branches never rejoin, and each has selector mass at least $\tau_0 2^{-r} = 2\kappa x$. Since $z \leq h$, for dyadic $\kappa \geq 2$ this gives $h \geq \kappa/2 + 1$; for $\kappa = 1$, either $h \geq 2$ or $h = z = 1$.

Handle $\kappa = 1, h = z = 1$ before restricting columns or projecting the pair. Both tagged column families are still unchanged. Project ordinary columns onto j_1 and other-copy columns onto their J_2 images, choosing surviving representatives. These maps preserve the ambient laws, so the projected masses are at least $1 - \varepsilon$ and δ. The intact row pair has mass greater than $49/64$. Thus Lemma B.1 requires $H_2 + 1$ bits, while the stopping node has only $C - 1 - r = H_2$ bits left. The original law on the two selector labels is the product of their exact selector laws, as required by that lemma.

In all other cases choose the denser column child, retaining ordinary mass greater than $3/8 > 9/25$. Replace any intact pair by its first-selector projection, choosing one surviving row for each projected label. The entries now form an interlace of R_{temp} on the supported bins. Its tuple source has exactly 8192-wise independent bin coordinates in either construction. Forget the unused tail while keeping whole selector-tuple labels in the product case and whole selector-list indices in the explicit case. Their image event has at least the retained mass in its original law; one surviving completion per image label realises its entries. Thus Lemma 5.10 applies in both constructions. For $\kappa \geq 2$, use width κ: the $\kappa/2 + 1$ blocks have row mass at least $2\kappa x \geq \kappa x$, so the required cost is $H_2 + \log \kappa$ against a child budget of $H_2 + \log \kappa - 1$. For $\kappa = 1$ there are at least two blocks of mass at least $2x$; width two gives the stronger bound $H_2 + 1$. The schedules ensure all these row thresholds are at most one. This completes the contradiction without using other-copy entries outside the distinguished bin. □

B.2 Total receipt at a destination

Proof of Lemma 5.13. Put $\kappa = 1$ at a final node. Suppose a destination receives protected mass at least $K_*\kappa$. If $d_i \in [0, 1]$ is the fraction of origin i's diagonal family reaching it, then $K_* = 512\mu L_w s$, with $L_w = \log w + 2$, gives

$$D_0 := \sum_i d_i \geq 512 L_w s\kappa.$$

Set $d_i = 0$ outside the origins being counted. Discard terms below $D_0/(2w)$; their sum is less than $D_0/2$. The remaining terms occupy at most L_w dyadic classes. One class V, of size m and upper endpoint $\delta \leq 1$, therefore satisfies

$$\delta/2 < d_i \leq \delta \quad (i \in V), \qquad m\delta \geq \frac{D_0}{2L_w}, \qquad \delta \geq \frac{D_0}{2w}, \qquad m \geq 256 s\kappa.$$

All incoming row sets are fixed by the Alice-only transcript. At a first-Bob node choose a child whose column event Y^* has original mass at least $\theta/2$; at a final node set $Y^* = Y'$.

For each $i \in V$, let T_i indicate a template position in the metadata stream. The centred variables $T_i - \lambda$ are 256-wise independent in the original law. By Lemma C.1,

$$\mathbb{E}\left|\sum_{i \in V}(T_i - \lambda)\right|^{256} \leq (2^{14}m)^{128}.$$

If fewer than $m/2$ positions are template positions, the absolute sum is greater than $(\lambda - 1/2)m$. Markov's inequality and the scheduled bound give

$$\Pr\left\{\sum_{i \in V} T_i < m/2\right\} \leq C_{128} m^{-128} \leq \theta/4, \qquad C_{128} = \left(\frac{2^{14}}{(\lambda - 1/2)^2}\right)^{128}.$$

Thus Y^* intersects the event of at least $m/2$ template positions in mass at least $\theta/4$.

Average this intersection over the complete metadata and base-label streams. Some fixed configuration leaves at least $m/2$ template positions in V, with fixed bins p_i, and a selector-label section Z of original mass at least $\theta/4$. Fix only those two other streams. The remaining selector stream, flattened to coordinates (i, α), still has its original exactly 8192-wise independent J marginals. Now apply Lemma 5.9 to Z with cap one. It removes at most $e(\theta/4) \leq sq_2$ coordinates.

At every fixed template position i, the diagonal input (α, γ) reads $A(\gamma, z_{i,\alpha})$, regardless of the chosen bin p_i. Its retained J_2-mass is $d_i > \delta/2$. More than $\delta q_2/4$ of its Γ-fibres have mass at least

$\delta/4$: otherwise those fibres contribute at most $\delta/4$ and the remaining fibres less than $\delta/4$ to the total. Consequently there are more than $m\delta q_2/8$ such blocks (i,α). Set $K_0 = 2s\kappa q_2$. The class bounds and $w^2 q_2 x \le 1$ give

$$\frac{m\delta q_2}{8} \ge 32 s\kappa q_2, \qquad K_0 x \le \frac{2s\kappa}{w^2} \le \frac{\delta}{4}.$$

After removing the at most sq_2 exceptional coordinates, more than $K_0/2$ blocks remain. Retain $K_0/2+1$ of them and all selector labels in Z. Choose an actual incoming row for every retained diagonal image. Distinct (i,α,γ) images give distinct rows, and their entries on Z are exactly $A(\gamma, z_{i,\alpha})$. The row masses use the original Γ law; the column event has mass one in the fixed conditional law on Z. The chosen completions certify these entries, not an identity of their old weights with the projected weights.

Apply Theorem 5.11 with base profile $\mathcal{P}(A; x, H_1)$ and parameter $a = \log K_0$. Its row threshold is at most $\delta/4 \le 1$, and it requires

$$H_1 + \log K_0 = C + \log \kappa$$

bits. The first-Bob child has only $C + \log\kappa - 2$ bits left, and a final node only $C-1$. This contradiction proves the receipt bound for every collection of origins, including those excluded from the owner argument. □

B.3 Transfer to another origin's owner

Proof of Lemma 5.14. Let ε_i be the owner's missing normalised ordinary mass and put $\delta_i = 2q_2 x + 4\varepsilon_i$. If $\delta_i \ge 1$, the conclusion is automatic: an origin has protected mass only μ. Otherwise, a transfer exceeding $\mu\delta_i$ supplies a diagonal set from origin $j \ne i$ of actual normalised mass $\delta > \delta_i$. We keep this strict inequality throughout the argument.

Both origins are good coordinates for Y', and only Alice has spoken before the owner. By Lemma 5.9, the projection of Y' onto these two raw-row coordinates has original product mass greater than $99/100$. Restrict to the event that both are templates. Its mass is λ^2, so the retained relative mass is at least

$$a_{\min} = \frac{99/100 + \lambda^2 - 1}{\lambda^2}.$$

Average over the two bin labels and fix a pair of bins attaining this relative mass. Their selector section is a set $P_* \subseteq S^2$ of original product mass at least $a_{\min}$. Distinguish the chosen bin at i. The good singleton projection at i loses at most $\eta/2$ of its raw-row law; hence every other bin retains selector mass at least

$$1 - \frac{\eta s}{2\lambda} > 2sx.$$

Choose one surviving completion in Y' for every selector pair in P_* and for every retained selector in the other bins. The bin at i distinguishes the latter families from the pairs; the projected labels distinguish the representatives within each family. These choices realise entries of an actual restriction and do not assert independence of the law conditioned on Y'.

After transposition, the ordinary columns from origin i have normalised mass $1-\varepsilon_i$. The diagonal columns from j have J_2-image mass $\delta > \delta_i$, since that image map preserves their original law. A pair (z_o, z_f) reads $A(\gamma_{p_i}, (z_o)_{\alpha_{p_i}})$ on ordinary columns and $A(\gamma, (z_f)_\alpha)$ on the diagonal columns. Other bins have the ordinary template entries, and their diagonal entries are immaterial. This is the restriction in Claim B.2, after renaming the distinguished bin. The claim requires C bits because $\delta > 2q_2 x + 4\varepsilon_i$, whereas the owner has only $C-1$ bits left. This contradiction proves the non-strict transfer bound, including the case $\varepsilon_i = 0$. □

C Conditioning and surviving blocks

The higher moments below use the original law, where independence is exact. The conditional fourth moment uses only the approximate marginals outside the exceptional coordinates.

C.1 A moment count and the exceptional coordinates

Lemma C.1. *Let $r \geq 1$ be an integer. If $X_1, \dots, X_m$ have mean zero, $|X_i| \leq 1$, and every subfamily of at most $2r$ variables is independent, then*

$$\mathbb{E}\left|\sum_{i=1}^{m} X_i\right|^{2r} \leq (r^2 m)^r.$$

In particular, for centred indicators with 1024-wise independence,

$$\mathbb{E}\left|\sum_i X_i\right|^{1024} \leq (2^{18}m)^{512}, \qquad \mathbb{E}\left|\sum_i X_i\right|^{256} \leq (2^{14}m)^{128}.$$

Proof. The case $m = 0$ is immediate. Expand the $2r$th power. A term with an index occurring exactly once has expectation zero, by centring and independence. Every remaining term uses at most r distinct indices. There are at most $m^r r^{2r}$ such terms: encode one by its distinct indices in increasing order, pad that list to length r by repeating its first index, and record one of the r positions for each of its $2r$ factors. This encoding is injective if repeated padding positions are never used. Each expectation has absolute value at most one. Taking $r = 512$ and $r = 128$ proves the displayed special cases. □

Proof of Lemma 5.9. Put $\varepsilon = \varepsilon(v, \tau)$. Choose a maximal disjoint family $J_1, \dots, J_m$ of sets of at most eight coordinates whose conditional marginals differ from their product marginals by more than ε. If $m = 0$, nothing must be deleted. Otherwise, total variation supplies an indicator f_i on each J_i with $\mathbb{E}_{\nu|Y} f_i - \mathbb{E}_\nu f_i > \varepsilon$; take its complement if necessary. Under the original law ν, any subfamily of at most 1024 such indicators is independent, because its disjoint sets use at most 8192 coordinates.

For $S = \sum_i (f_i - \mathbb{E}_\nu f_i)$, conditional Jensen's inequality and Lemma C.1 give

$$v(m\varepsilon)^{1024} < \nu(Y)\mathbb{E}_{\nu|Y}|S|^{1024} \leq \mathbb{E}_\nu |S|^{1024} \leq (2^{18}m)^{512}.$$

Thus $m < 2^{18}\varepsilon^{-2}v^{-1/512}$. Delete $E = \bigcup_i J_i$, so $|E| \leq 8m \leq e(v, \tau)$. Every remaining set of at most eight coordinates has the desired marginal: otherwise it could be added to the disjoint family, contradicting maximality. □

C.2 Some pair has a large projection

Lemma C.2. *Fix disjoint coordinate pairs $J_1, \dots, J_m$ in a tuple list with marginal μ and law ν. Suppose every projection of a label set K has product mass $\mu^2(\pi_{J_i} K) < 9/25$. For an exactly eight-wise independent list,*

$$\nu(K) < \frac{2}{m^2} < \frac{9}{50m} \qquad (m \geq 16).$$

For an exactly 8192-wise independent list, let $K \subseteq Y$ with $\nu(Y) \geq v > 0$, and choose the pairs outside the exceptional set from Lemma 5.9, with any cap $0 < \tau \leq 1$. Then

$$(\nu \mid Y)(K) < \frac{1}{4m} \qquad (m \geq 16).$$

Thus a retained set of conditional mass at least $1/(4m)$ forces one of these pairs to have product projection mass at least $9/25$.

Proof. Write $P_i = \pi_{J_i} K$ and put

$$p_i = \mu^2(P_i) < 9/25, \qquad X_i = \mathbf{1}[z_{J_i} \in P_i] - p_i, \qquad S = \sum_{i=1}^{m} X_i.$$

On K every indicator equals one, so $S > (16/25)m$. Under the product law, the centred indicators are independent, with $\mathbb{E}X_i^2 \le 1/4$ and $\mathbb{E}X_i^4 \le 1/4$. Only terms with no singleton index survive the fourth-power expansion. Hence

$$\mathbb{E}_{\mathrm{prod}} S^4 = \sum_i \mathbb{E}X_i^4 + 6\sum_{i<j} \mathbb{E}X_i^2 \mathbb{E}X_j^2 \le \frac{m}{4} + \frac{6}{16}\binom{m}{2} \le \frac{m^2}{4}.$$

Exact eight-wise independence gives the same fourth moment under ν, since each term uses at most four pairs. Markov's inequality gives $\nu(K) < 2/m^2$, because $1/[4(16/25)^4] < 2$. For $m \ge 16$, $2/m^2 \le 1/(8m) < 9/(50m)$.

Now suppose the pairs are good for the fixed event Y. Every term in S^4 lies in $[-1, 1]$ and depends on at most eight good coordinates. Total variation changes its expectation by at most 2ε, where $\varepsilon = \varepsilon(v, \tau)$. Summing the m^4 terms and applying Markov therefore gives

$$p := (\nu \mid Y)(K) \le \frac{2}{m^2} + \frac{2\varepsilon}{(16/25)^4} \le \frac{2}{m^2} + 16\varepsilon.$$

For the other estimate, return to the original law. Any subfamily of at most 1024 pair indicators is independent under ν, and $K \subseteq Y$ with $\nu(Y) \ge v$. Thus Lemma C.1 gives

$$p = \frac{\nu(K)}{\nu(Y)} \le \frac{(2^{18}m)^{512}}{v((16/25)m)^{1024}} \le \frac{(2^{20}/m)^{512}}{v}.$$

Combining the bounds and using $\min\{u, w\} \le u^{511/512} w^{1/512}$ yields

$$\begin{aligned} mp &\le \frac{2}{m} + \min\{16\varepsilon m, 2^{20\cdot 512} v^{-1} m^{-511}\} \\ &\le \frac{2}{m} + (16\varepsilon)^{511/512} 2^{20} v^{-1/512}. \end{aligned}$$

The last term is at most

$$2^{20 - 26\cdot 511/512} v^{255/131072} < \frac{1}{32},$$

by $\varepsilon \le 2^{-30} v^{1/256}$ and $0 < v \le 1$. For $m \ge 16$, this gives $mp < 1/8 + 1/32 < 1/4$, as required. □

C.3 Small projections and short continuations

The two bounds from Lemma A.2 that we need require $d + 1$ bits:

number of blocks	row mass per block	product column mass
2	$2x$	$9/25$
3	$4x$	$9/50$.

Here $\mathcal{P}(M; x, d)$, $d \ge 2$, and the indicated row threshold is at most one. For selected coordinates J, choose one surviving label for each tuple in $\pi_J K$. By Observation 2.5, these representatives transfer the projected full-product lower bound to the labelled matrix. The projected mass is measured under μ^J, even when the labels have a conditional law.

Proof of Lemma 5.10. Suppose a protocol has depth at most $d + a - 1$. At each Alice message, assign every current block to a child retaining at least half its row mass, and keep a child assigned at least half the blocks. At each Bob message, keep a child retaining at least half the current column mass under the original law ν. Follow these choices for $a - 1$ messages, padding an early leaf with dummy Alice messages if needed. If A messages are Alice's and B are Bob's, then $A + B = a - 1$ and the remaining restriction has

$$\text{at least } 2^B + 1 \text{ blocks}, \qquad \text{row mass per block at least } 2^{B+1}x,$$
$$\nu(K) \geq (9/25)2^{-B}, \qquad \text{at most } d \text{ bits remaining}.$$

The block count follows by rounding up $(2^{a-1} + 1)/2^A > 2^B$. Every set J of at most eight surviving coordinates has $\mu^J(\pi_J K) \geq \nu(K)$ by exact marginals.

If $B = 0$, two blocks meet the first threshold above; if $B = 1$, three meet the second. If $B = 2$, five coordinates have joint product mass at least $9/100$, so Lemma A.1 supplies a pair of mass at least $(9/100)^{2/5} > 9/25$. If $B = 3$ or $B = 4$, eight coordinates have mass at least $9/400$, so some pair has mass at least $(9/400)^{1/4} > 9/25$. If $B \geq 5$, choose $m = 2^{B-1} \geq 16$ disjoint pairs from the $2m + 1$ surviving blocks. Were every pair projection smaller than $9/25$, Lemma C.2 would give $\nu(K) < 9/(50m) = (9/25)2^{-B}$, a contradiction. In each case the selected blocks meet one of the two row thresholds above, which are at most one because $B + 1 \leq a$. They require $d + 1$ further bits, contradicting the available d. □

The conditioned version needs a short-continuation estimate for each possible number of Bob messages.

Lemma C.3. *Suppose $d \geq 2$ and $\mathcal{P}(M; x, d)$. Let ν be an exactly 8192-wise independent tuple source with the base-column law, let $\nu(Y) \geq v > 0$, and choose the good coordinates from Lemma 5.9 with cap $0 < \tau \leq 1$. If an integer $b \geq 0$ satisfies $2^{b+1}x \leq 1$, then any restriction retaining at least $2^b + 1$ good blocks, each of base row mass at least $2^{b+1}x$, and a column set $K \subseteq Y$ with $(\nu \mid Y)(K) \geq 2^{-b-1}$ requires at least $d + 1$ bits.*

Proof. Keep Y and its good coordinates fixed. Set $\varepsilon = \varepsilon(v, \tau) \leq 2^{-30}$. For any set J of at most eight good coordinates, Lemma 5.9 gives

$$\mu^J(\pi_J K) \geq (\nu \mid Y)(K) - \varepsilon \geq 2^{-b-1} - \varepsilon.$$

For $b \leq 4$ this is at least $(9/25)2^{-b}$, since the smallest margin is $7/800 > 2^{-30}$. The same small-coordinate checks in the proof of Lemma 5.10 therefore supply the required pair or triple. For $b \geq 5$, choose $m = 2^{b-1} \geq 16$ disjoint good pairs. Since $(\nu \mid Y)(K) \geq 1/(4m)$, Lemma C.2 supplies a pair of product projection mass at least $9/25$.

Every selected pair retains row mass at least $2x$; the triple at $b = 1$ retains at least $4x$. These thresholds are at most one by $2^{b+1}x \leq 1$. Thus Lemma A.2 requires $d + 1$ bits in every case. Throughout, retained label masses use the single law $\nu \mid Y$; no independence after further restriction is asserted. □

Proof of Theorem 5.11. Suppose a protocol has depth at most $d + a - 1$. At each Alice message, assign every current block to a child retaining at least half its row mass, then keep a child assigned at least half the blocks. At each Bob message, keep a child with at least half the column mass in the fixed law $\nu \mid Y$. The good coordinates remain good throughout.

Follow these choices for $a-1$ messages, padding an early leaf with dummy Alice messages if needed. If A messages are Alice's and B are Bob's, then $A+B=a-1$ and the surviving restriction has

$$\begin{aligned}\text{number of good blocks} &\geq \lceil (2^{a-1}+1)/2^A \rceil \geq 2^B+1,\\ \text{row mass per block} &\geq 2^{a-A}x = 2^{B+1}x,\\ (\nu \mid Y)(K') &\geq 2^{-B-1}.\end{aligned}$$

Apply Lemma C.3 with $b=B$. Its row threshold is at most one because $B+1\leq a$. It requires $d+1$ further bits, while only d remain. □

D Constructing rational input lists

The construction follows the sequential compression idea of Koller and Megiddo [14]: append an independent coordinate, then replace the resulting distribution by one with fewer positive weights and the same small marginals. The compression uses only a linear dependence among zero–one vectors.

Proof of Lemma 6.1. Fix the independence order D and put $P=(1+qm)^{\mathsf{D}}$. Start with the empty word of weight one. At each step, append an independent μ-coordinate to every word currently retained. On the resulting candidate set $\mathcal{U}\subseteq Y^i$, the weights satisfy

$$\sum_{u\in\mathcal{U}} w_u \mathbf{1}[u_J=a] = \prod_{j\in J}\mu(a_j) \qquad (J\subseteq[i],\ |J|\leq \mathsf{D},\ a\in Y^J). \tag{$*$}$$

The empty set J gives $\sum_u w_u = 1$. Appending the coordinate preserves old marginals, and a marginal involving the new coordinate factors by independence.

There are at most P equalities in (*), since the number of assignments on at most D coordinates is at most $\sum_{j=0}^{\mathsf{D}}(qm)^j \leq (1+qm)^{\mathsf{D}}$. For a candidate word u, let $a(u)$ be its zero–one column in this system. Discard zero weights. Whenever the remaining columns are dependent, compute a nonzero integer relation $\sum_u c_u a(u)=0$ and set

$$t=\min_{c_u>0}\frac{w_u}{c_u}, \qquad w'_u = w_u - tc_u.$$

The constant-one row gives $\sum_u c_u = 0$, so the relation has both signs. The update preserves every equality, leaves all weights nonnegative, and makes at least one positive weight zero. Discard the new zero weights and repeat until the remaining columns are independent. There are then at most P words, so the next extension has at most mP candidates. After q stages, the equalities give the required D-wise independence.

To bound the arithmetic, choose each relation using a column basis and one further column. If the basis has size $r\leq P$, select r independent rows. These span the row space, so a relation on them holds in every row. The determinant minors of the resulting $r\times(r+1)$ zero–one matrix give the relation, with integer coefficients of absolute value at most $\max\{1, r^{r/2}\}$ by Hadamard's bound. Thus each coefficient uses $O(P\log(P+1))$ bits, and exact linear algebra finds it in polynomial time. Fixed orders for the columns, rows, and tied pivots make these choices deterministic.

Let L_μ be the input law's total binary encoding length and Q_μ the product of its denominators. Maintain a common denominator Q for the current weights. An extension multiplies Q by at most Q_μ. If $w_u = a_u/Q$ and j attains the minimum ratio, one elimination gives

$$w'_u = \frac{a_u c_j - a_j c_u}{Q c_j}.$$

It adds at most $O(P\log(P+1))$ denominator bits. There are at most qmP eliminations and q extensions, so every intermediate weight has bit length

$$O\big(qL_\mu + qmP^2\log(P+1)\big).$$

Numerators obey the same bound because the weights lie in $[0,1]$. All linear algebra, ratio comparisons, and updates therefore have polynomial cost for fixed D. Together with the at most P length-q words, the same bound proves polynomial total output length. □

E Printing a sample of the implicit table

We prove the access guarantee by counting complete input labels first, then implementing their elementary draws with fixed blocks of fair bits. Although one entry visits only one recursive entry at each level, generating its input labels must also produce all the coordinates that it does not visit.

Proof of Lemma 4.5. Fix the admissible graph, put $L = \lceil \log n \rceil$, and use the same schedule at every level. Write $R : X \times Y \to \{0,1\}$ for the current base matrix.

Labels and entries. The two interlaces $U = \mathsf{I}_w(N)$, $W = U^{\mathsf{T}}$ and $R^+ = (\mathsf{I}_Q(W))^{\mathsf{T}}$ give the successor labels

$$\mathbf{a} = ((i_t, Z_t))_{t\in[Q]}, \qquad \mathbf{b} = (j, (k_i)_{i\in[w]}),$$

where $i_t \in [w]$ and $j \in [Q]$ are uniform, Z_t are tagged raw columns, and $k_i \in \mathcal{K}$ are raw rows. All are independent in their prescribed laws. The entry rule is

$$R^+(\mathbf{a}, \mathbf{b}) = B_{\mathrm{raw}}(k_{i_j}, Z_j).$$

For a template raw row (p, z) and raw column $Z = (\xi, j_1, \ldots, j_s)$, with $j_p = (\alpha_p, \gamma_p)$ and $z_{\alpha_p} = (r, u)$, this entry is $R(u, (\gamma_p)_r)$. For an item row (a, u), it is $R(u, \xi_{\pi_\alpha(a)})$ when all activity indices equal α and $v_a(\alpha) = 1$, and $R(u, \xi_{s+1})$ otherwise. Thus, after reading the stored selectors and activity data, evaluation makes exactly one base-entry call.

Choose a template row with probability $255/256$; vertex and auxiliary types each have probability $1/512$. A template row uses a uniform bin and q_2 independent labels from $J = [q_1] \times X$; an item row uses a uniform item of its type and one base row.

For a raw column, choose Ω with probability $255/256$ and $\mathcal{D}$ otherwise. For Ω, draw $s+1$ independent base columns for ξ and s independent labels from $J_2 = [q_2] \times Y^{2s-1}$. For $\mathcal{D}$, draw one $j = (\alpha, \gamma) \in J_2$, copy it into all s positions, and put $\xi = \mathrm{tail}(\gamma)$. These are the actual full-product labels; no extra streams used in the lower-bound proof are generated.

The recursive draw counts are therefore

label	base-row draws	base-column draws
raw template row	q_2	0
raw item row	1	0
Ω column	0	$(s+1) + s(2s-1) = 2s^2+1$
$\mathcal{D}$ column	0	$2s-1$
successor row	0	$\le Q(2s^2+1)$
successor column	$\le wq_2$	0

Selectors, copying and activity-list scans have polynomial cost in $n + s + q_2 + w + Q + L + 1$, with no colouring required. Every width is $2^{O(L)}$ by Lemma F.1. Hence common computable integer

bounds J_i for label length, generation time, elementary draw count and entry-evaluation time can be chosen with

$$J_0 = O(L), \qquad J_{i+1} \le 2^{c_0 L} J_i, \qquad J_k = 2^{O((k+1)L)},$$

where c_0 is fixed and effective. Here an elementary draw is charged one operation; its bit implementation is accounted for below. At the lookup seed, a row consists of $p_0 = O(L)$ independent bits, a column is a uniform index in $[p_0]$, and an entry can be read in $O(L)$ time. This proves the initial bound and ties the resource estimate to the seed iteration.

Since $h = O(L)$ and $d_0 = O(L)$, we have $B_k = O((k+1)L)$. The direct sample size in Theorem 4.4 then gives $m = 2^{O((k+1)L)}$. Generate and store all $2m$ labels, then evaluate their m^2 intersections. Including the stored tuples and printed entries, this costs

$$O((2m + m^2)J_k) = 2^{O((k+1)L)} = n^{O(k+1)}.$$

All constants are independent of k.

Fixed-bit implementation. Let $N_{\mathrm{draw}} \ge 1$ be a computable upper bound on the elementary draws for all $2m$ labels. The preceding bounds give $N_{\mathrm{draw}} = 2^{O((k+1)L)}$. A tape-independent bound on every elementary alphabet size and mixture denominator is

$$A = \max\{512, 4s(n+s), q_2, w, Q, p_0\} = 2^{O(L)}.$$

It covers type draws, item indices, bins, activity and interlace selectors, and seed draws. All base inputs are generated recursively. Set $b = \lceil \log(200 A N_{\mathrm{draw}}) \rceil$ and reserve N_{draw} blocks of b fair bits. For a draw from $[a]$, read the next block as a uniform integer $U \in \{0, \dots, 2^b - 1\}$ and return

$$\mathrm{Draw}(a) = 1 + \left\lfloor aU/2^b \right\rfloor. \tag{7}$$

For a rational mixture, draw from its denominator and map consecutive intervals to types with their prescribed numerator lengths. Ignore unused blocks.

Every result lies in $[a]$. Each value has either $\lfloor 2^b/a \rfloor$ or $\lceil 2^b/a \rceil$ preimages, so the draw's total-variation distance from uniform is at most

$$a/2^b \le A/2^b \le 1/(200 N_{\mathrm{draw}}).$$

Mapping to a mixture type cannot increase this distance. Conditional on a common history of the ideal and implemented generators, couple their next draw with disagreement probability at most this bound. Fresh blocks permit this even when the history determines the next alphabet. Before the first disagreement the generators take the same branches; summing over its at most N_{draw} possible positions bounds the joint total variation by $1/200 < 1/100$.

All elementary choices are legal, so the generation and copying rules assemble actual inputs on every tape. There are exactly m labels of each kind and a fixed reserved tape length. Entry evaluation is exact. Finally, $b = O((k+1)L)$, and each rounded draw uses arithmetic on $O(b + \log A)$-bit integers. This polynomial overhead preserves the bound $n^{O(k+1)}$ for time and random bits on every tape, completing the proof. □

F Effective amplification widths

The widths must leave room for source coordinates, deleted inputs, and the final restoration step. We use generous margins. Their exact exponents do not enter the conclusions, except that the full-product overhead must remain $L + O(\log L)$.

Write $n = |V|$, $\lambda = 255/256$, $\mu = 1/256$, and $\zeta = \gamma/24$. The same value λ is the mass of the raw template rows and, separately, of the ordinary column family. The source promise gives

$$n/2 \le |E| \le \Delta n/2, \qquad \delta_v = \frac{\gamma|E|}{2\Delta n} \ge \frac{\gamma}{4\Delta}.$$

To make the conditioning budget easy to compute, for an integer $b \ge 0$ and rational $0 < \tau \le 1$ define

$$\overline{e}(2^{-b}, \tau) := \left\lceil 2^{90}\tau^{-2}2^{\lceil b/64 \rceil} \right\rceil, \qquad \overline{e}(v) := \overline{e}(v, 1).$$

The definition in Lemma 5.9 gives

$$e(v, \tau) \le 8 + 2^{23}\tau^{-2}v^{-1/512} + 2^{83}v^{-5/512} \le 2^{84}\tau^{-2}v^{-1/64}.$$

Thus $e(v, \tau) \le \overline{e}(v, \tau)$ for dyadic v, and $\overline{e}(v, \tau) = O(\tau^{-2}v^{-1/64})$. Both schedules use

$$\begin{gathered} \delta_U = \eta/(16w), \qquad \theta = (512Q)^{-1}, \\ e_0 = \overline{e}(\theta, \eta/2), \qquad E_{\mathrm{exc}} = 2e_0, \qquad K_* = 512\mu(\log w + 2)s = 2(\log w + 2)s, \\ C_{128} = \left(\frac{2^{14}}{(\lambda - 1/2)^2}\right)^{128}. \end{gathered} \tag{8}$$

Here E_{exc} bounds exceptional destination capacity and K_* bounds incoming protected mass per unit capacity, as in Lemmas 5.12 and 5.13.

In both schedules s, q_2, w, Q are powers of two and $q_1 = s - 2$. The following common tests pay for the source restrictions and the shared internal-interlace proof:

$$\begin{gathered} \text{Source coordinates and collision survival:} \\ s \ge 256, \qquad s\eta \le 2^{-16}\min\{1, \delta_v, \mu\zeta\}, \\ q_2 \ge \mathrm{pow}_2(4s|E|), \qquad 4s^2\rho \le 1, \qquad 8sq_1 \le \zeta q_2/16, \\ \text{Protected inputs and transfers:} \\ \max\{q_2x, sx, wx\} \le 10^{-4}, \qquad 2w^2x \le \eta/(16\lambda), \\ w^2q_2x \le 1, \qquad 4\mu w q_2 x \le \eta/32, \\ \text{Conditioning, owner loss, and incoming mass:} \\ \overline{e}(\theta) \le w/128, \qquad \overline{e}(\theta/4) \le sq_2, \qquad C_{128}s^{-128} \le \theta/4, \\ E_{\mathrm{exc}} \le w/4, \qquad K_*E_{\mathrm{exc}} \le \eta w/16. \end{gathered} \tag{9}$$

As usual, $\mathrm{pow}_2(a)$ is the least power of two at least a.

Several further bounds follow automatically. No isolated vertices gives $q_2 \ge 4s|E| \ge 2sn \ge 2(n + s - 3)$, since $sn - n - s + 3 = (s - 1)(n - 1) + 2 > 0$. Also $s\rho \le 1/(4s) \le 1/1024 < 10^{-3}$, and $2wx + \eta/(16\lambda) < 1$ follows from $wx \le 10^{-4}$ and $s\eta \le 2^{-16}$. For $j = 0, 1$ and $\beta_j = (3/5)2^{-j}$, the source population ratios obey

$$\begin{aligned} \frac{3}{5}(1 - 2/s) - q_1\rho &\ge \frac{3}{5} - \frac{6}{5 \cdot 256} - 10^{-3} > \frac{9}{16}, \\ \frac{3}{5} - q_2x &\ge \frac{3}{5} - 10^{-4} > \frac{9}{16}, \\ \frac{2^j(\beta_j - (1 - \lambda))}{\lambda} - sx &\ge \frac{44}{75} - 10^{-4} > \frac{9}{16}. \end{aligned}$$

The same conservative ratio $44/75 = ((3/5) - 4(1-\lambda))/\lambda$ applies to the reference profile. After conditioning, its surviving fraction is at least

$$\frac{44}{75} - wx - \frac{4e(\theta)}{w} \geq \frac{44}{75} - \frac{1}{256} - \frac{1}{32} > \frac{1}{2}.$$

For item deletions and discarded coordinates, the common tests also give

$$4096\eta sn \leq \delta_v n/16 < (s/2)\delta_v n, \qquad 4096\eta s(s-3) \leq (s-3)/16 < s/2,$$
$$8sq_1 + 4s\eta q_2/\mu \leq (1/16 + 2^{-14})\zeta q_2 < \zeta q_2/4, \qquad 2s\rho \leq 1/(2s) \leq 1/8.$$

Template bins retain more than $999/1000$ of their mass because $\eta s/\lambda < 10^{-3}$. Projecting the retained ordinary columns onto whole selector labels preserves mass at least $1 - \eta/\lambda > 99/100$. Consequently the forced-prefix argument has the required margins $999/1000 > 2sx$ and $999/1000 - 2(s-1)x > 998/1000$. The remaining fixed comparisons are

$$\frac{99}{200} - 2(1-\lambda) > \frac{12}{25}, \qquad \frac{99/100 + \lambda^2 - 1}{\lambda^2} - 10^{-3} > \frac{49}{64},$$
$$\frac{1}{64} + \frac{\mu}{2\lambda} = \frac{1}{64} + \frac{1}{510} < \frac{1}{32}.$$

The last inequality bounds all foreign receipts by $\eta w/32$: the transfer threshold is $2q_2x + 4\varepsilon_i$, and $\sum_i \varepsilon_i \leq \eta/(8\lambda)$. Together with the exceptional allowance $\eta w/16$, this is less than the outgoing mass $\eta w/2$.

Finally, the flag test comes directly from Lemma C.1. For $m \geq s$ template indicators of mean λ with 256-wise independence,

$$\Pr[\text{fewer than } m/2 \text{ template flags}] \leq \frac{(2^{14}m)^{128}}{((\lambda - 1/2)m)^{256}} = C_{128}m^{-128} \leq \theta/4.$$

The cap $\eta/2$ at an owner also leaves each other selector bin with mass at least $1 - \eta s/(2\lambda) > 999/1000 > 2sx$. These checks apply to both schedules.

F.1 Full products

The source-coordinate width contributes the only leading term to the logarithmic overhead. All other widths are powers of $\log n$.

Lemma F.1. *Put $L = \lceil \log n \rceil$ and $\ell = \lceil \log(L+1) \rceil$. Choose*

$$s = 2^{16+\ell}, \qquad q_1 = s - 2, \quad q_2 = 2^L s^2,$$
$$w = 2^{8\ell}, \qquad \rho = 2^{-L}, \quad x = 2^{-10L},$$
$$\eta = \frac{\min\{1, \delta_v, \mu\zeta\}}{2^{16}s}, \quad Q = w^2,$$

and use (8)*. Call the source admissible when* (9) *holds and*

$$Q\delta_U \geq 80L, \qquad L \geq 16, \qquad d_0 \leq L, \qquad h \leq 7L,$$

where d_0 is the seed cost in Lemma 3.4 and

$$h = 2\log s + \log q_2 + \log w + \log Q = L + 64 + 28\ell.$$

Every sufficiently large promised source is admissible, with an effective finite exceptional range depending only on Δ, γ. All choices and tests have polynomial bit complexity, depend on neither the base matrix nor the round count, and give $h = L + O_{\Delta,\gamma}(\log L)$.

Proof. All implicit constants below are effective and depend only on the source promise. Since $L+1 \le 2^\ell < 2(L+1)$,

$$s=\Theta(L), \quad \eta=\Theta(L^{-1}), \quad q_2=2^L\Theta(L^2), \quad w=\Theta(L^8), \quad Q=\Theta(L^{16}), \quad \theta=\Theta(L^{-16}).$$

The conditioning and receipt budgets therefore satisfy

$$\overline{e}(\theta),\overline{e}(\theta/4)=O(L^{1/4}), \qquad E_{\mathrm{exc}}=O(L^{9/4}), \qquad K_*=O(L\log L),$$
$$\frac{K_*E_{\mathrm{exc}}}{\eta w}=O(L^{-15/4}\log L), \qquad \frac{s^{-128}}{\theta}=O(L^{-112}), \qquad \frac{L}{Q\delta_U}=O(L^{-6}).$$

Each budget is eventually below its tested allowance. For the remaining smallness tests, the losses divided by their variable allowances are

quantity	upper bound	quantity	upper bound
$s^2\rho$	$O(L^2 2^{-L})$	sq_1/q_2	$O(2^{-L})$
$\max\{q_2x, sx, wx\}$	$O(L^8 2^{-9L})$	w^2q_2x	$O(L^{18}2^{-9L})$
w^2x/η	$O(L^{17}2^{-10L})$	wq_2x/η	$O(L^{11}2^{-9L})$.

The source coordinates fit because $4s|E| \le 2\Delta s 2^L \le q_2$ once $s \ge 2\Delta$. Since q_2 is dyadic, this also proves its required dyadic ceiling. The product-block width tests follow directly from $q_2 \ge s$ and $s/4096 \ge 16(L+1) > 10L$.

For restoration,

$$0<\delta_U \le 2^{-20}/(sw) < 1/64, \qquad Q\theta = 1/512 < 1/20.$$

Together with $Q\delta_U \ge 80L$, these are the width conditions in Theorem 5.17. All logarithms defining h are integers, and substitution gives its displayed exact value. The seed has $d_0=\ell+7$, so $h \le 7L$, $d_0 \le L$, and $L \ge 16$ also hold eventually.

These estimates give a uniform effective cutoff. For $\ell \ge 2$, the bounds $2^{\ell-2} \le L < 2^\ell$ turn every inverse-power or logarithmic error above into $C(\ell+1)^r 2^{-a\ell}$, and every CL^d2^{-L} into $C2^{d\ell-2^{\ell-2}}$, with effective rational C, fixed $a>0$, and fixed nonnegative integers r,d. Each decreases to zero beyond a computable point. Searching beyond those points until all tolerances hold gives a cutoff valid at every larger size. The actual tests use only rational arithmetic, integer powers and ceilings, with operands of polynomial bit length. No approximation of a fractional power is required. □

F.2 Explicit supports

For explicit supports we can afford widths polynomial in the graph size. This also leaves room for the separate restoration argument.

Lemma F.2. *Fix a rational constant*

$$0 < c_{\mathrm{del}} \le 2^{-16}\min\{1, \gamma/(4\Delta), \mu\zeta\}.$$

For a dyadic $\mathsf{N} \ge n$*, choose*

$$L_{\mathrm{g}}=\log \mathsf{N}, \qquad s_{\mathrm{g}}=2^{\lceil L_{\mathrm{g}}/4\rceil}, \qquad q_{1,\mathrm{g}}=s_{\mathrm{g}}-2, \qquad q_{2,\mathrm{g}}=\mathsf{N}^4,$$
$$w_{\mathrm{g}}=\mathsf{N}^2, \qquad Q_{\mathrm{g}}=\mathsf{N}^6, \qquad \rho_{\mathrm{g}}=\mathsf{N}^{-1}, \qquad x_{\mathrm{g}}=\mathsf{N}^{-8}, \qquad \eta_{\mathrm{g}}=c_{\mathrm{del}}/s_{\mathrm{g}}.$$

Use (8) *with these subscripted quantities. Call* N *admissible when* (9) *holds and*

$$\overline{e}(x_{\mathrm{g}}) \le s_{\mathrm{g}}/128, \qquad Q_{\mathrm{g}} \ge 128\big(\overline{e}(\rho_{\mathrm{g}}, \delta_{U,\mathrm{g}}/2)+1\big).$$

There is an effective dyadic N_0*, depending only on the graph promise and* c_{del}*, such that every dyadic* $\mathsf{N} \ge \mathsf{N}_0$ *passes uniformly for graphs with* $n \le \mathsf{N}$*. The first passing dyadic* $\mathsf{N} \ge n$ *is therefore at most* $2\max\{n, \mathsf{N}_0\}$*. The tests have polynomial bit complexity and ensure every numerical hypothesis of the explicit-support construction and restoration.*

Proof. Omit subscripts in the proof. All implicit constants are effective and depend only on the fixed graph promise and c_{del}. We have

$$\mathsf{N}^{1/4} \le s < 2\mathsf{N}^{1/4}, \qquad \eta = \Theta(\mathsf{N}^{-1/4}), \qquad \delta_U = \Theta(\mathsf{N}^{-9/4}), \qquad \theta = 2^{-9}\mathsf{N}^{-6}.$$

The conservative budgets fit with room to spare:

budget	upper bound	available width
$\overline{e}(x)$	$O(\mathsf{N}^{1/8})$	$s = \Theta(\mathsf{N}^{1/4})$
$\overline{e}(\theta)$	$O(\mathsf{N}^{1/8})$	$w = \mathsf{N}^2$
$\overline{e}(\theta/4)$	$O(\mathsf{N}^{1/8})$	$sq_2 = \Theta(\mathsf{N}^{17/4})$
$\overline{e}(\rho, \delta_U/2) + 1$	$O(\mathsf{N}^5)$	$Q = \mathsf{N}^6$
E_{exc}	$O(\mathsf{N}^{3/4})$	$w = \mathsf{N}^2$.

For the two nonconstant caps the exponents are $1/2 + 6/64 < 3/4$ and $9/2 + 1/64 < 5$. Since $K_* = O(\mathsf{N}^{1/4}\log\mathsf{N})$,

$$\frac{K_* E_{\mathrm{exc}}}{\eta w} = O(\mathsf{N}^{-3/4}\log\mathsf{N}) \longrightarrow 0.$$

Also $s^{-128}/\theta = O(\mathsf{N}^{-26})$, proving the flag test. The remaining vanishing losses are

quantity	upper bound	quantity	upper bound
$s^2\rho$	$O(\mathsf{N}^{-1/2})$	sq_1/q_2	$O(\mathsf{N}^{-7/2})$
$\max\{q_2x, sx, wx\}$	$O(\mathsf{N}^{-4})$	w^2x/η	$O(\mathsf{N}^{-15/4})$
wq_2x/η	$O(\mathsf{N}^{-7/4})$	η	$O(\mathsf{N}^{-1/4})$.

The remaining product condition is exact: $w^2q_2x = 1$, which is allowed. The choice of c_{del} supplies the deletion inequality. Source coordinates fit because $4s|E| \le 2\Delta s\mathsf{N} = O(\mathsf{N}^{5/4})$, whereas $q_2 = \mathsf{N}^4$.

To check the source-profile consumers explicitly, their block parameter is $u = s2^{-j}$ or $q_2 2^{-j}$ for $j = 0, 1$, so $u \ge s/2 \ge 128$. The population fraction exceeds $9/16$, and removing at most $e(x) \le \overline{e}(x) \le s/128 \le u/32$ coordinates leaves more than $u/2$ blocks. In the first profile $u\rho \le s\rho \le 1$; in the other two, $ux \le \max\{q_2x, sx\} \le 1$. The common checks above supply the outer profile, incoming-mass, owner, and transfer margins.

For restoration,

$$0 < \delta_U \le 2^{-20}/(sw) < 1/64, \qquad w\delta_U = \eta/16, \qquad Q\theta = 1/512 < 1/20.$$

Together with the tested lower bound on Q, these are the hypotheses of Theorem 6.2 with $\phi = \rho$ and $\delta = \delta_U$. All widths are polynomial in N, and the independence order stays fixed at 8192. By Lemma 6.1, the fixed number of support constructions in one round has a fixed polynomial cost in N and the complete base encoding, including its rational weights; its exponent is independent of the round number. The overhead is

$$h_{\mathrm{g}} = 2\log s + \log q_2 + \log w + \log Q = 12L_{\mathrm{g}} + 2\lceil L_{\mathrm{g}}/4 \rceil.$$

Every asymptotic comparison above has a positive power gap. Its error ratio is bounded by $C(1 + L_{\mathrm{g}})2^{-aL_{\mathrm{g}}}$ for effective rational C and fixed rational $a > 0$. These bounds decrease to zero

once $L_g + 1 \geq 2/a$; searching beyond this point gives a uniform effective cutoff. Rational exponents can be removed by taking a fixed integer power. The actual admissibility tests use only integer exponents, rational arithmetic and ceilings, on operands with $O(L_g)$ bits. Searching successive dyadic sizes from n therefore takes polynomial time and finds a passing size by $2\max\{n, \mathsf{N}_0\}$. None of these choices uses the base matrix, its optimum, or the number of rounds. □

G Colouring promises from SAT

G.1 Almost-linear bounded-degree graph colouring

The graph source must preserve a constant colouring gap and the almost-linear size of sparse SAT. Both properties survive the standard encoding; we give the accounting because the size bound determines the final approximation ratio.

Proof of Theorem 3.2. Apply Dinur's PCP construction, amplification, and a final bounded-degree reduction [4, Lemma 8.3, Theorem 1.5, and Lemma 1.7]. After removing isolated variable blocks, the resulting constraint system has a fixed alphabet, maximum degree d, perfect completeness, and a fixed gap $\rho > 0$: every assignment on a NO input rejects at least $\rho|F|$ constraint occurrences. Here F counts occurrences, including repetitions, and the number of variable blocks is at most $2|F|$.

Encode each alphabet value by one Boolean variable. For each block impose exactly one true value, and for each constraint occurrence forbid its rejected value pairs. Convert the clauses to width three with private auxiliary variables. A true clause has a satisfying extension; a false clause leaves at least one converted clause false. Fixed alphabet and degree give fixed constants C, k such that the resulting formula has $m \leq C|F|$ clauses, $n \leq 3m$ used variables, and at most k occurrences of each variable. In particular, $m \leq kn$. Perfect completeness is preserved.

On a NO input, decode every valid block and assign arbitrary values to invalid blocks. At least $\rho|F|$ constraints are rejected. If half have valid endpoints, their separate forbidden-pair clauses give at least $\rho|F|/2$ false clauses. Otherwise at least $\rho|F|/(2\max\{1, d\})$ distinct blocks are invalid, and each falsifies one of its own validity clauses. Thus every Boolean assignment violates at least ϵm clauses for some fixed rational $0 < \epsilon \leq 1$.

The bounded-occurrence colouring construction gives a loopless graph G_0 of fixed maximum degree, with at most $C_g n$ edges, and preserves completeness [2, Theorem 14 and its proof, pp. 98–99]. We use the proof's global decoding estimate: if $r \leq kn/8$ edges are deleted, a proper colouring of the remaining graph yields an assignment violating at most $8r$ clauses. For any colouring of a NO graph, delete its r monochromatic edge occurrences. If $r < \epsilon m/8$, then $r < kn/8$, so decoding contradicts the Boolean gap. Consequently

$$r \geq \epsilon m/8, \qquad |E(G_0)| \leq C_g n \leq 3C_g m.$$

The monochromatic-edge fraction is therefore at least $\delta := \epsilon/(24C_g) > 0$.

Remove isolated vertices and add a disjoint edge, obtaining H. On a NO instance G_0 has an edge, so the added edge reduces the gap by at most a factor two. Choose a fixed rational $0 < \gamma \leq \delta/2$ and a fixed degree bound Δ for H. Trivial inputs, including empty formulas, can instead use fixed promise graphs, decreasing γ and increasing Δ if necessary. The graph is now nonempty, loopless, and has no isolated vertices. Keep the construction order on labels and count parallel edge occurrences separately.

For a sparse formula on N variables, the binary input length is $L_{\mathrm{enc}} = O(N \log(N+2))$. Dinur's starting PCP has size $L_{\mathrm{enc}}(\log L_{\mathrm{enc}})^{O(1)}$ and inverse-polylogarithmic gap. The $O(\log\log L_{\mathrm{enc}})$

amplification rounds each multiply size by a constant; the final degree reduction and both encodings have constant-factor cost [4, Theorem 8.1 and its proof; Lemma 1.7]. Hence

$$|V(H)|, |E(H)| \leq C' N (\log(N+2))^q$$

for fixed C', q. Taking $\max\{1, \lceil N/|E(H)| \rceil\}$ disjoint labelled copies gives a graph G with

$$N \leq |E(G)| \leq N + |E(H)|, \qquad |V(G)|/2 \leq |E(G)| \leq \Delta |V(G)|/2.$$

Copies preserve completeness, the gap, and the graph conventions. Enlarging a fixed integer p absorbs the constants and gives $|V(G)|, |E(G)| \leq N(\log(N+2))^p$. All steps are deterministic and polynomial-time. □

G.2 Balanced edges with distinct vertices

The balanced source counts vertex occurrences within an edge. The lookup construction needs distinct vertices, so we replace repeated occurrences by distinct labelled clones.

Theorem G.1 (Guruswami–Lee, [7, Theorem 1.1])**.** *For fixed integers $Q_{\mathrm{GL}}, k_{\mathrm{GL}} \geq 2$ and $\epsilon_{\mathrm{GL}} > 0$, it is NP-hard to distinguish $Q_{\mathrm{GL}} k_{\mathrm{GL}}$-uniform hypergraphs admitting a k_{GL}-colouring with at least $Q_{\mathrm{GL}} - 1$ occurrences of each colour in every edge from hypergraphs having no independent set of measure at least ϵ_{GL}.*

Proof of Proposition 8.2. Use the parameters

$$s_{\mathrm{a}} = 2^t, \qquad K_{\mathrm{a}} = s_{\mathrm{a}}^2, \qquad S_{\mathrm{a}} = 2^{s_{\mathrm{a}}}$$

and apply Theorem G.1 with $Q_{\mathrm{GL}} = k_{\mathrm{GL}} = s_{\mathrm{a}}$ and $\epsilon_{\mathrm{GL}} = 1/(2S_{\mathrm{a}})$. The YES case has the required balanced s_{a}-colouring. In the NO case a proper S_{a}-colouring would have an independent colour class of measure at least $1/S_{\mathrm{a}}$, contradicting the source promise.

Replace each vertex u by $K_{\mathrm{a}} S_{\mathrm{a}}$ labelled clones. For each ordered occurrence edge $(u_1, \ldots, u_{K_{\mathrm{a}}})$, include every tuple

$$(u_1^{(j_1)}, \ldots, u_{K_{\mathrm{a}}}^{(j_{K_{\mathrm{a}}})})$$

of pairwise distinct clones, in the displayed order. Retain repeated edge occurrences as separately labelled edges. Every occurrence edge has a lift, since there are at least K_{a} clones of each vertex. Giving clones their source vertex's colour preserves the colour multiset of every edge, and hence the YES promise.

Conversely, suppose the cloned hypergraph has a proper S_{a}-colouring. Give each source vertex a colour appearing on at least K_{a} of its clones. A monochromatic occurrence edge in this induced colouring would lift to a monochromatic edge of distinct clones: each source vertex occurs at most K_{a} times, so choose different clones of that colour for its positions. This contradicts properness. The induced colouring would therefore be proper, contradicting the NO promise.

Each vertex produces $K_{\mathrm{a}} S_{\mathrm{a}}$ clones and each edge produces at most $(K_{\mathrm{a}} S_{\mathrm{a}})^{K_{\mathrm{a}}}$ lifts. These are constants for fixed t, so enumeration is deterministic and polynomial-time, with degree allowed to depend on t. If the vertex set is empty, add one isolated labelled vertex. The resulting nonempty vertex set and ordered edges of distinct vertices give exactly the position map required by Definition 8.3. □

H Finite-field arrays

We use the polynomial-root construction of Alon, Goldreich, Håstad and Peralta [1, Construction 3 and Proposition 3]. The conversion from parity bounds to pattern probabilities is also recorded there, with credit to Vazirani [1, Lemma 1 and Corollary 1]. The following statement gives the exact interface needed by the lookup proof.

Lemma H.1. *For integers $n, r \geq 1$, a deterministic algorithm prints a nonempty labelled list of n-bit arrays whose uniform law satisfies*

$$\Pr[w|_I = z] \leq 2^{-|I|} + 2^{-r} \qquad (I \subseteq \{0, \dots, n-1\},\ z \in \{0,1\}^I).$$

Its running time and complete output length are polynomial in $n2^r$.

Proof. Let $R = 2^d$ be the least power of two at least $n2^r$. Then $d \geq 1$. In a fixed binary basis of $\mathbb{F}_R$, write $\mathrm{vec}(y)$ for the coordinate vector of y. For each seed $(u, y) \in \mathbb{F}_2^d \times \mathbb{F}_R$, define

$$w^{u,y}(j) = \langle u, \mathrm{vec}(y^j) \rangle_{\mathbb{F}_2}, \qquad 0 \leq j < n,$$

where $y^0 = 1$, also when $y = 0$. Give these R^2 seed labels the uniform law; equal arrays keep separate labels.

For a nonempty position set U, the polynomial

$$P_U(Z) = \sum_{j \in U} Z^j$$

is nonzero and has degree at most $n-1$. Conditional on y, the parity of the bits at U is $\langle u, \mathrm{vec}(P_U(y)) \rangle$. Its sign has mean zero over uniform u unless $P_U(y) = 0$, when its mean is one. The polynomial root bound therefore gives

$$\left| \mathbb{E}_{u,y} (-1)^{\sum_{j \in U} w^{u,y}(j)} \right| = \Pr_y[P_U(y) = 0] \leq \frac{n-1}{R} \leq 2^{-r}.$$

This includes $n = 1$, when the only polynomial is the constant one. Now expand the pattern indicator:

$$\mathbf{1}[w|_I = z] = 2^{-|I|} \sum_{U \subseteq I} (-1)^{\sum_{j \in U} (w(j) + z(j))}.$$

The empty-set term contributes $2^{-|I|}$, and the remaining terms contribute at most $(1 - 2^{-|I|})2^{-r}$, proving the pattern bound, also for empty I.

For constructibility, enumerate monic binary polynomials of degree d until finding an irreducible one. Test each by division by every monic polynomial of degrees 1 through $\lfloor d/2 \rfloor$: a reducible polynomial has a factor in that range, and an irreducible polynomial exists in every positive degree. For $d = 1$ the test range is empty. There are R candidates and fewer than $2R$ divisors, so the search takes polynomial time in R. The quotient field then supports arithmetic in polynomial time in d. Enumerate all R^2 seeds and compute successive powers and binary dot products to print each array. Since $R < 2n2^r$, this takes polynomial time and output length in $n2^r$, including the field representation and seed labels. □

Proof of Corollary 8.7. Number Ω_{a} by $0, \dots, n-1$ and apply Lemma H.1 with

$$n = K_{\mathrm{a}} H_{\mathrm{a}}, \qquad r = 4(b + 3t).$$

Here $2^{-r} = \epsilon_{\mathrm{a}}$ and $n2^r = K_{\mathrm{a}} 2^{12t} H_{\mathrm{a}}^5$, which is polynomial in H_{a} for fixed t. The integer parameters have $O_t(b+1)$ bits, and the bound includes $H_{\mathrm{a}} = 1$. □

References

[1] Noga Alon, Oded Goldreich, Johan Håstad, and René Peralta. Simple constructions of almost k-wise independent random variables. *Random Structures & Algorithms*, 3(3):289–304, 1992. doi: 10.1002/rsa.3240030308. URL `https://web.math.princeton.edu/~nalon/PDFS/aghp4.pdf`. Our theorem numbering refers to the author-hosted version dated 22 February 2002: Lemma 1, Corollary 1, Construction 3 and Proposition 3.

[2] Andrej Bogdanov, Kenji Obata, and Luca Trevisan. A lower bound for testing 3-colorability in bounded-degree graphs. In *Proceedings of the 43rd Annual IEEE Symposium on Foundations of Computer Science*, pages 93–102. IEEE Computer Society, 2002. doi: 10.1109/SFCS.2002.1181886. URL `https://lucatrevisan.github.io/pubs/bot.pdf`.

[3] Binh-Minh Bui-Xuan, Jan Arne Telle, and Martin Vatshelle. H-join decomposable graphs and algorithms with runtime single exponential in rankwidth. *Discrete Applied Mathematics*, 158 (7):809–819, 2010. doi: 10.1016/j.dam.2009.09.009. URL `https://perso.lip6.fr/binh-minh.bui-xuan/files/BTV.pdf`. Our Proposition 3.6 locator follows the author-hosted manuscript.

[4] Irit Dinur. The PCP theorem by gap amplification. *Journal of the ACM*, 54(3), June 2007. doi: 10.1145/1236457.1236459. URL `https://www.wisdom.weizmann.ac.il/~dinuri/mypapers/combpcp.pdf`. Article 12. Our theorem numbering refers to the author manuscript dated 13 February 2007.

[5] Serge Gaspers, Tao Zixu He, and Simon Mackenzie. NP-completeness of deterministic communication complexity via relaxed interlacing, 2026. URL `https://arxiv.org/abs/2508.05597v4`. Earlier version of the present paper, version 4, 13 April 2026.

[6] Serge Gaspers, Tao Zixu He, and Simon Mackenzie. NP-completeness and ETH-based additive inapproximability of deterministic communication complexity via relaxed interlacing. Earlier ETH stage of the present paper, prepared as version 5 of arXiv:2508.05597; version-specific reference to be updated before submission, 2026.

[7] Venkatesan Guruswami and Euiwoong Lee. Strong inapproximability results on balanced rainbow-colorable hypergraphs. *Combinatorica*, 38(3):547–599, 2018. doi: 10.1007/s00493-016-3383-0. URL `https://eccc.weizmann.ac.il/report/2014/043/`. Theorem numbering in our source discussion refers to ECCC TR14-043.

[8] Shuichi Hirahara, Rahul Ilango, and Bruno Loff. Hardness of constant-round communication complexity. In *36th Computational Complexity Conference (CCC 2021)*, volume 200 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 31:1–31:30, 2021. doi: 10.4230/LIPIcs.CCC.2021.31. URL `https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.CCC.2021.31`.

[9] Shuichi Hirahara, Rahul Ilango, and Bruno Loff. Communication complexity is NP-hard, 2025. URL `https://arxiv.org/abs/2507.10426v1`. Preprint, version 1, 14 July 2025.

[10] Pavel Hrubeš. Hard submatrices for non-negative rank and communication complexity. *Computational Complexity*, 34, 2025. doi: 10.1007/s00037-025-00269-4. URL `https://doi.org/10.1007/s00037-025-00269-4`. Article 9.

[11] Rahul Ilango, Bruno Loff, and Igor Carboni Oliveira. NP-hardness of circuit minimization for multi-output functions. In *35th Computational Complexity Conference (CCC 2020)*, volume

169 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 22:1–22:36, 2020. doi: 10.4230/LIPIcs.CCC.2020.22. URL https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.CCC.2020.22.

[12] Russell Impagliazzo and Ramamohan Paturi. On the complexity of k-SAT. *Journal of Computer and System Sciences*, 62(2):367–375, 2001. doi: 10.1006/jcss.2000.1727. URL https://cseweb.ucsd.edu/~paturi/myPapers/pubs/ImpagliazzoPaturi_2001_jcss.pdf.

[13] Russell Impagliazzo, Ramamohan Paturi, and Francis Zane. Which problems have strongly exponential complexity? *Journal of Computer and System Sciences*, 63(4):512–530, 2001. doi: 10.1006/jcss.2001.1774. URL https://cseweb.ucsd.edu/~paturi/myPapers/pubs/ImpagliazzoPaturiZane_2001_jcss.pdf.

[14] Daphne Koller and Nimrod Megiddo. Constructing small sample spaces satisfying given constraints. *SIAM Journal on Discrete Mathematics*, 7(2):260–274, 1994. doi: 10.1137/S0895480192228140. URL https://theory.stanford.edu/~megiddo/pdf/stoc93.pdf. First appeared in STOC 1993. Algorithm and theorem numbering here refers to the author-hosted conference version.

[15] Eyal Kushilevitz and Enav Weinreb. On the complexity of communication complexity. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing*, pages 465–474. Association for Computing Machinery, 2009. doi: 10.1145/1536414.1536479. URL https://doi.org/10.1145/1536414.1536479.

[16] Simon Mackenzie. Lossless hardness condensation in deterministic communication complexity, 2026. URL https://arxiv.org/abs/2609.28691v1. Preprint, version 1, 23 September 2026.

[17] Simon Mackenzie and Abdallah Saffidine. Refuting the direct sum conjecture for total functions in deterministic communication complexity, 2026. URL https://arxiv.org/abs/2411.19003v2. Corrected full version, 14 April 2026; conference version appeared in STOC 2025.

[18] Sang-il Oum, Sigve Hortemo Sæther, and Martin Vatshelle. Faster algorithms for vertex partitioning problems parameterized by clique-width. *Theoretical Computer Science*, 535:16–24, 2014. doi: 10.1016/j.tcs.2014.03.024. URL https://arxiv.org/abs/1311.0224v2. Our theorem numbering refers to arXiv version 2.

[19] Arshia Soltani Moakhar, Iman Gholami, Max Springer, Mahdi JafariRaviz, and Mohammad-Taghi Hajiaghayi. Beyond the library: An agentic framework for autoformalizing research mathematics, 2026. URL https://arxiv.org/abs/2606.31134v3. Version 3, 5 August 2026; direct-sum formalisation in Section 3.2.2 and Appendix C.3.

[20] Andrew Chi-Chih Yao. Some complexity questions related to distributive computing (preliminary report). In *Proceedings of the Eleventh Annual ACM Symposium on Theory of Computing*, pages 209–213. Association for Computing Machinery, 1979. doi: 10.1145/800135.804414. URL https://doi.org/10.1145/800135.804414.